

Hydrocarbons

Question1

The cycloalkene (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which C : Br ratio is 3:1. The percentage of bromine in the product (Y) is _____ %. (Nearest integer)

(Given : molar mass in g mol^{-1} H : 1, C : 12, O : 16, Br : 80)

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Answer: 66

Solution:

Bromination of a cycloalkene (one double bond) consumes **1 mole of Br_2 per mole of alkene**, which means **2 bromine atoms** add across the double bond.

So, product (Y) is a **dibromo** compound.

Given in (Y), the ratio C : Br = 3 : 1.

Let number of carbon atoms = n .

Number of bromine atoms = 2.

So,

$$\frac{n}{2} = \frac{3}{1} \Rightarrow n = 6$$

Hence (Y) has formula:



(since cyclohexene is C_6H_{10} and it adds Br_2)

Now molar mass of (Y):

- Carbon: $6 \times 12 = 72$
- Hydrogen: $10 \times 1 = 10$
- Bromine: $2 \times 80 = 160$

Total molar mass:

$$72 + 10 + 160 = 242 \text{ g mol}^{-1}$$

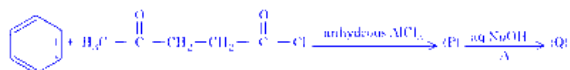
Percentage of bromine:

$$\% \text{Br} = \frac{160}{242} \times 100 \approx 66.12\%$$

Nearest integer = $\boxed{66\%}$.

Question2

Consider the following reaction of benzene.

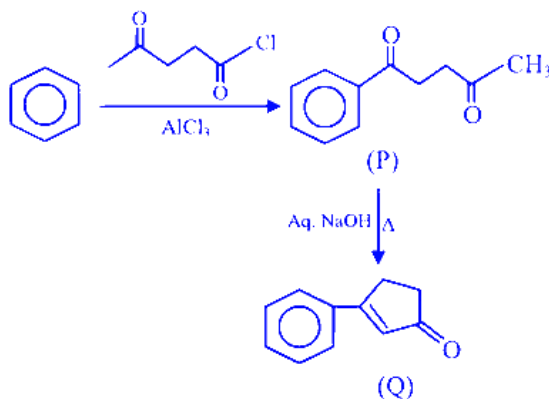


In compound (Q), the percentage of oxygen is ____ %. (Nearest integer)

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Answer: 10

Solution:



Molecular mass of compound Q is 157.

In Q, oxygen contributes mass 16 (for one oxygen atom).

So, percentage of oxygen in Q is % of oxygen in product 'Q' is $= \frac{16}{157} \times 100 = 10.19\%$

Nearest integer = 10%.

Question3

One mole of an alkane (x) requires 8 mole oxygen for complete combustion. Sum of number of carbon and hydrogen atoms in the alkane (x) is _____.

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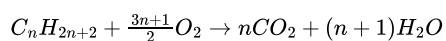
Answer: 17

Solution:

For an alkane, the general formula is

$$C_nH_{2n+2}$$

The combustion reaction of an alkane is



Given that 1 mole of the alkane requires 8 moles of oxygen, so

$$\frac{3n+1}{2} = 8$$

Now solve:

$$3n + 1 = 16$$

$$3n = 15$$

$$n = 5$$

So the alkane is



Now, total number of carbon and hydrogen atoms is

$$5 + 12 = 17$$

So, the required answer is

17

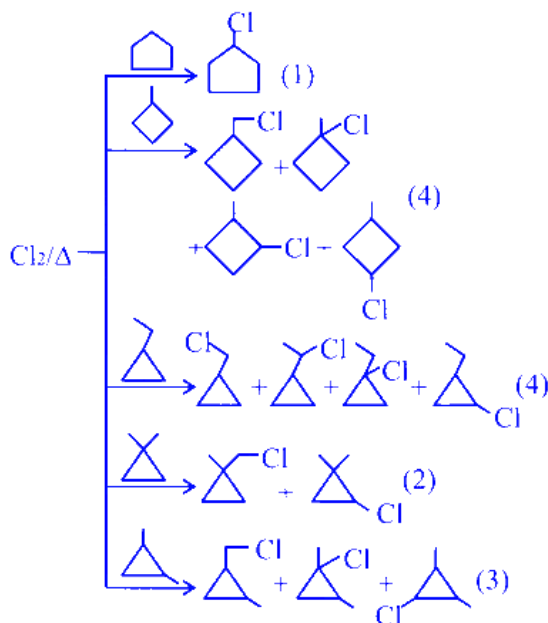
Question4

Consider the isomers of hydrocarbon with molecular formula C_5H_{10} . These isomers do not decolourise $KMnO_4$ solution. These isomers are subjected to chlorination with chlorine in presence of light to give monochloro compounds. The total number of monochloro compounds (structural isomers only) formed is _____.

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Answer: 14

Solution:



$$1 + 4 + 4 + 2 + 3 = 14$$

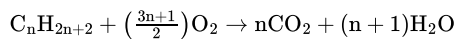
Question5

An alkane (Y) requires 8 moles of oxygen for complete combustion and on chlorination with $Cl_2/h\nu$, (Y) gives only one monochlorinated product (Z). The total number of primary carbon atoms in (Y) is ____.

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Answer: 4

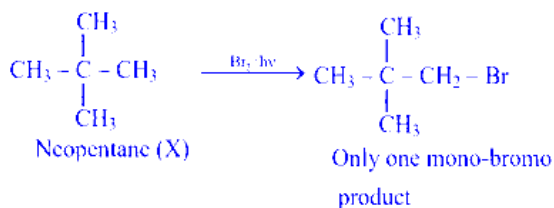
Solution:



$$\frac{3n+1}{2} = 8$$

$$3n+1 = 8 \times 2$$

$$n = 5 (\therefore C_5H_{12})$$



Number of primary carons = 4

Question6

80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL . When the system is treated with KOH solution, the volume decreases to 64 mL . The formula of the hydrocarbon is :

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Options:

A.



B.



C.

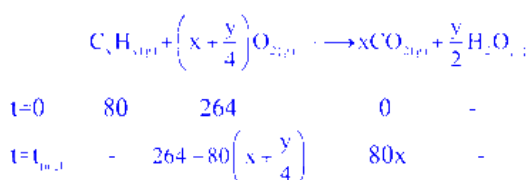


D.



Answer: A

Solution:



$$264 - 80 \left(x + \frac{y}{4} \right) + 80x = 224$$

$$264 - \frac{80y}{4} = 224$$

$$40 = \frac{80y}{4} \Rightarrow y = 2$$

$$264 - 80 \left(x + \frac{y}{4} \right) = 64$$

$$264 - 80 \left(x + \frac{1}{2} \right) = 64$$

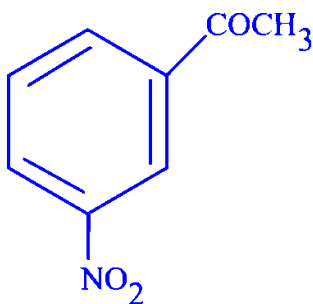
$$264 - 80x - 40 = 64$$

$$x = 2$$

Question7

Given below are two statements :

Statement I : Benzene is nitrated to give nitrobenzene, which on further treatment with $\text{CH}_3\text{COCl}/\text{AlCl}_3$ will give



Statement II : $-\text{NO}_2$ group is a *m*-directing, and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below

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Options:

A.

Statement I is incorrect but Statement II is correct

B.

Both Statement I and Statement II are correct

C.

Both Statement I and Statement II are incorrect

D.

Statement I is correct but Statement II is incorrect

Answer: A

Solution:

When benzene is nitrated, it forms nitrobenzene (benzene with a $-\text{NO}_2$ group).

Now, Friedel–Crafts acylation ($\text{CH}_3\text{COCl}/\text{AlCl}_3$) needs an aromatic ring that is sufficiently reactive to attack the electrophile.

The $-\text{NO}_2$ group is a strongly deactivating group. It pulls electron density away from the benzene ring, so the ring becomes much less reactive.

Because of this strong deactivation, nitrobenzene does **not** undergo Friedel–Crafts acylation with $\text{CH}_3\text{COCl}/\text{AlCl}_3$.

So, nitrobenzene will not give the shown acylated product under these conditions.

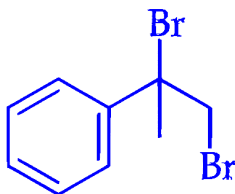
Question8

The dibromo compound [P] (molecular formula: $\text{C}_9\text{H}_{10}\text{Br}_2$) when heated with excess sodamide followed by treatment with dilute HCl gives [Q]. On warming [Q] with mercuric sulphate and dilute sulphuric acid yield [R] which gives positive Iodoform test but negative Tollen's test. The compound [P] is :

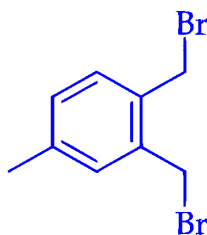
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Options:

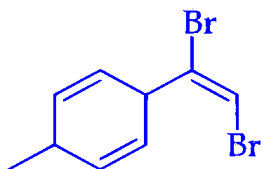
A.



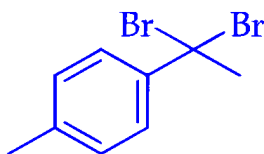
B.



C.

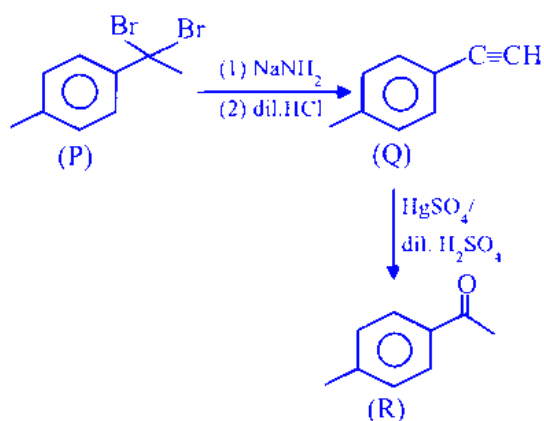


D.



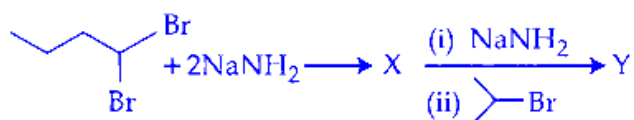
Answer: D

Solution:



Question9

Consider the following reaction :



The product Y formed is :

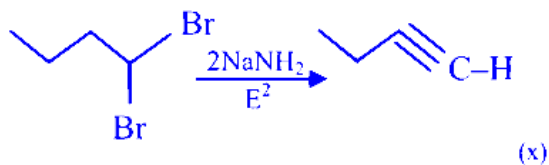
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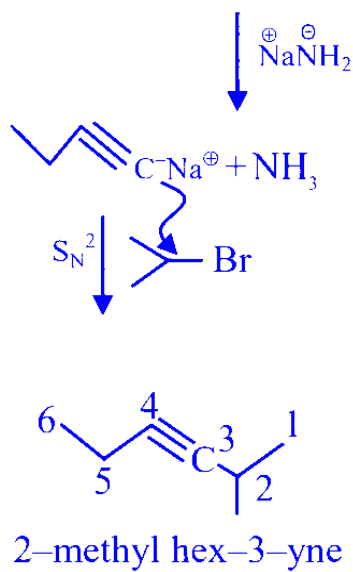
Options:

- A.
2-methylhex-3-yne
- B.
5-methylhex-2-yne
- C.
Isopropylbut-1-yne
- D.
2-methylhex-2-yne

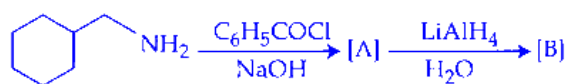
Answer: A

Solution:





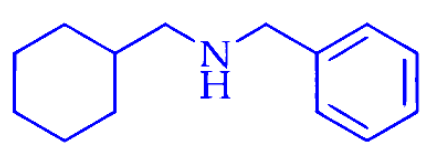
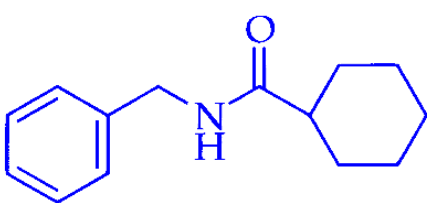
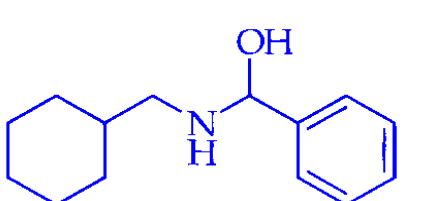
Question10



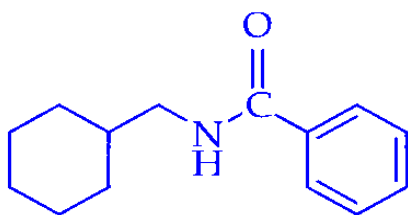
The final product [B] is :

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Options:

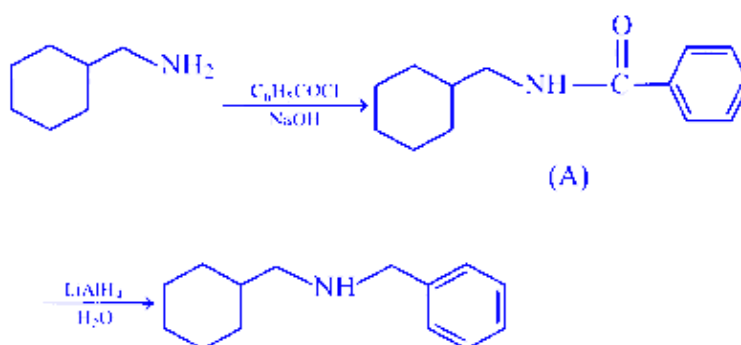
- A. 
- B. 
- C. 

D.



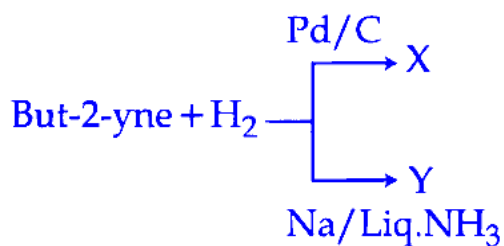
Answer: A

Solution:



Question11

But-2-yne and hydrogen (one mole each) are separately treated with (i) Pd/C and (ii) Na/ liq. NH₃ to give the products X and Y respectively.



Identify the incorrect statements.

A. X and Y are stereoisomers.

B. Dipole moment of X is zero.

C. Boiling point of X is higher than Y.

D. X and Y react with O₃/Zn + H₂O to give different products.

Choose the correct answer from the options given below :

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Options:

A.

B and D Only

B.

A and C Only

C.

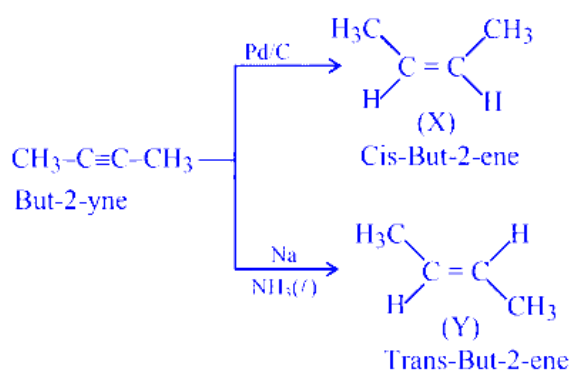
A and B Only

D.

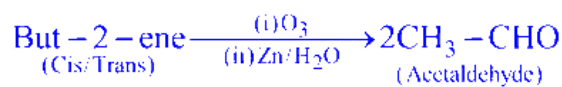
B and C Only

Answer: A

Solution:



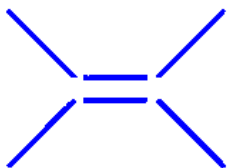
Dipole moment $\mu \neq 0$

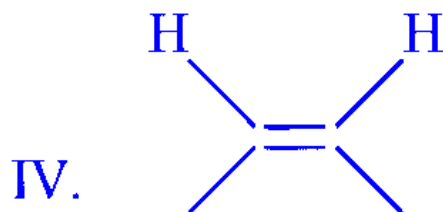
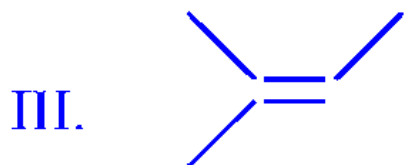
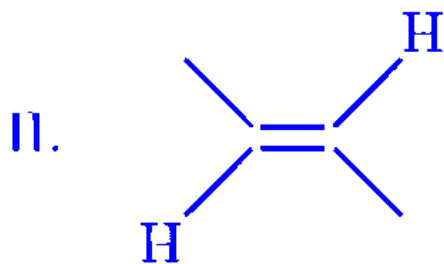


Question12

Arrange the following alkenes in decreasing order of stability.

I.





Choose the correct answer from the options given below:

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Options:

A.

III > I > II > IV

B.

III > II > I > IV

C.

I > III > II > IV

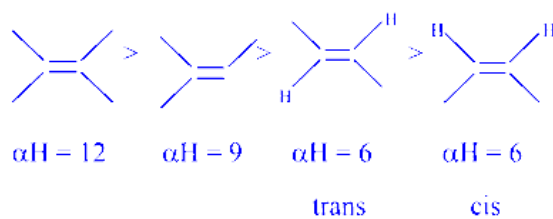
D.

I > III > IV > II

Answer: C

Solution:

Stability order :



Trans is more stable than cis.

Question13

The compound (X) on

(i) on heating in the presence of anhydrous AlCl_3 and HCl gas gives 2,4-dimethyl pentane

(ii) aromatization gives toluene and

(iii) cyclisation gives methyl cyclohexane

The correct name of compound (X) is:

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

Hept-2-ene

B.

Hept-1,3,5-triene

C.

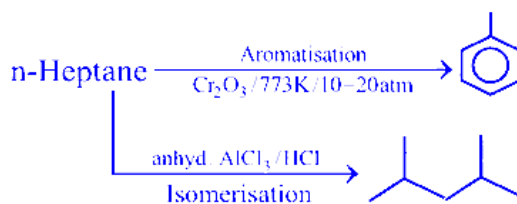
Heptane

D.

Hept-2,4,6-triene

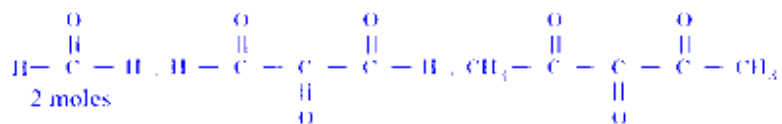
Answer: C

Solution:



Question14

An alkene (X) on ozonolysis followed by reduction gives following products.

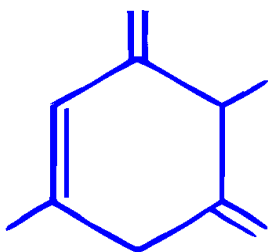


The alkene (X) is :

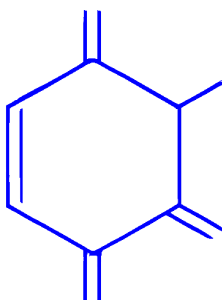
JEE Main 2026 (Online) 4th April Morning Shift

Options:

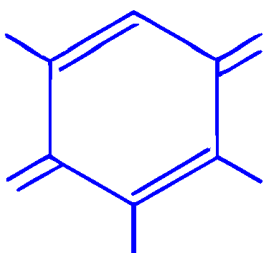
A.



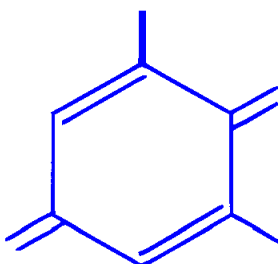
B.



C.

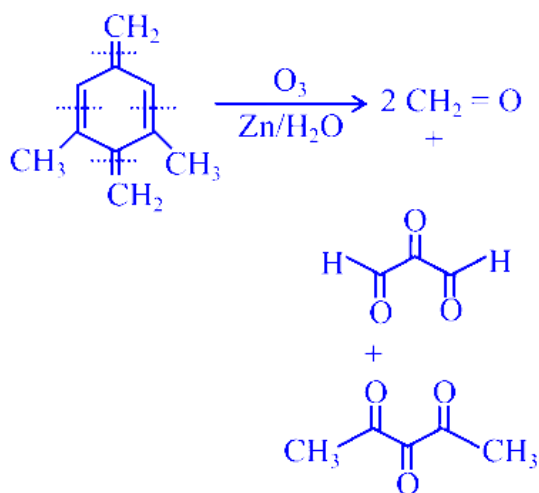


D.



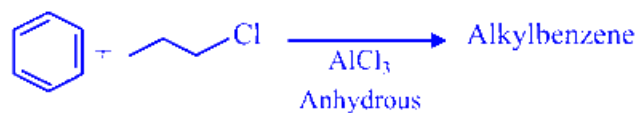
Answer: D

Solution:



Question15

For the following Friedel Craft's alkylation reaction, which of the statements are correct?



- A. Major product is n-propyl benzene.
- B. iso-propyl carbocation intermediate is also generated.
- C. Multiple substitution is inevitable.
- D. Introducing electron-donating substituent on benzene will not produce any alkyl benzene.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Morning Shift

Options:

- A.
- A and D only
- B.
- B and C only

C.

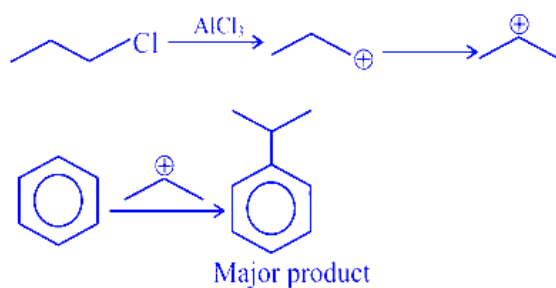
A and C only

D.

B and D only

Answer: B

Solution:



Question16

Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass 72 g mol^{-1} .

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Options:

A.

1, 1 - Dimethylcyclopropane

B.

2, 2 - Dimethylpropane

C.

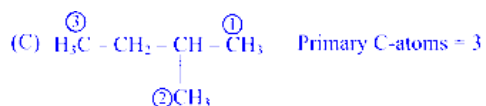
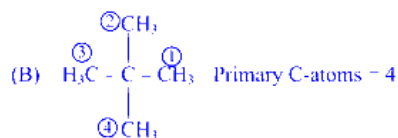
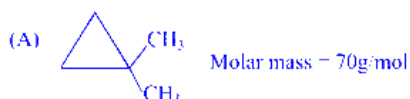
2 - Methylbutane

D.

n-pentane

Answer: C

Solution:



Question17

IUPAC name of the some alkenes are given below.

Find out the correct stability order.

A. 2-Methylbut-2-ene

B. cis-But-2-ene

C. 2,3-Dimethylbut-2-ene

D. Prop-1-ene

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

$C > A > B > D$

B.

$C > A > D > B$

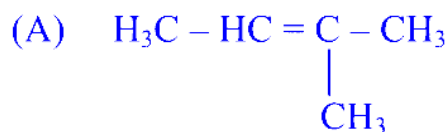
C.

$B > D > A > C$

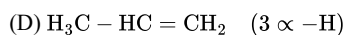
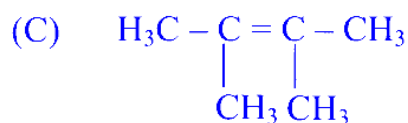
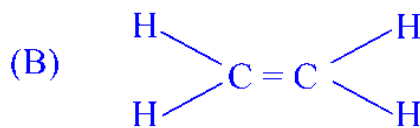
D.

$A > B > C > D$

Answer: A

Solution:

To compare the stability of alkenes, we mainly look at how many alkyl groups are attached to the double bond. More alkyl groups means more hyperconjugation, so the alkene becomes more stable.



More substituted alkenes are more stable because they have more $\alpha - \text{H}$ atoms. These $\alpha - \text{H}$ atoms take part in hyperconjugation, which stabilizes the double bond.

Now compare each alkene:

C. 2, 3-Dimethylbut-2-ene is the most substituted (tetra-substituted), so it is the most stable.

A. 2-Methylbut-2-ene is tri-substituted, so it is next in stability.

B. cis-But-2-ene is di-substituted, so it is less stable than A.

D. Prop-1-ene is mono-substituted, so it is the least stable.

Order: $C > A > B > D$.

Question18

Given below are two statements :

Statement I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water.

Statement II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: D

Solution:

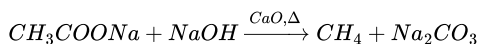
Analysis of Statement I

Statement I says: "Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water."

1. Decarboxylation of sodium ethanoate:

This reaction involves heating a sodium salt of a carboxylic acid with soda-lime (a mixture of NaOH and CaO). The general reaction is the removal of a carboxyl group.

For sodium ethanoate (CH_3COONa), the reaction is:



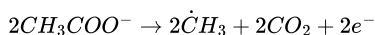
As you can see, methane (CH_4) is indeed the product. So, this part is correct.

2. Kolbe's electrolysis of sodium acetate:

Kolbe's electrolytic method involves the electrolysis of an aqueous solution of a sodium or potassium salt of a carboxylic acid.

For sodium acetate (CH_3COONa), at the anode, the acetate ions lose electrons and form methyl free radicals, which then combine to form ethane.

At Anode:

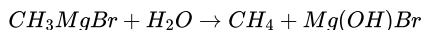


The product is ethane, not methane. As per our NCERT textbook, this method is used to prepare symmetrical alkanes with an even number of carbon atoms. Methane cannot be prepared by this method.

3. Reaction of CH_3MgBr with water:

CH_3MgBr is a Grignard reagent (methylmagnesium bromide). Grignard reagents are highly reactive and react with any source of a proton, such as water, to form hydrocarbons.

The reaction is:



Here, the methyl group from the Grignard reagent takes a hydrogen atom from water to form methane (CH_4). So, this part is also correct.

Conclusion for Statement I:

The statement claims that methane can be prepared by *all three* methods. Since Kolbe's electrolysis of sodium acetate produces ethane and **not** methane, the entire statement becomes **false**.

Analysis of Statement II

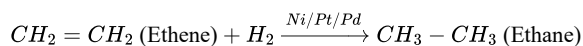
Statement II says: "Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction."

Let's check both parts.

1. Preparation from unsaturated hydrocarbons:

Alkanes are prepared from unsaturated hydrocarbons (alkenes and alkynes) by hydrogenation, which is the addition of hydrogen (H_2) across the double or triple bond.

For example:

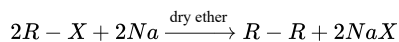


To prepare methane (CH_4), which has only one carbon atom, we would need an unsaturated hydrocarbon with one carbon atom. The smallest alkene (ethene) and alkyne (ethyne) have two carbon atoms. A one-carbon alkene or alkyne is not possible. Therefore, methane cannot be prepared by the hydrogenation of unsaturated hydrocarbons. This part is correct.

2. Preparation by Wurtz reaction:

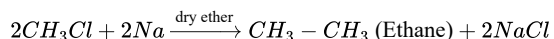
The Wurtz reaction involves the treatment of alkyl halides with sodium metal in dry ether to produce higher alkanes. It essentially joins two alkyl groups together.

The general reaction is:



This reaction always results in an alkane with an even number of carbon atoms, and it doubles the carbon chain. To prepare methane (CH_4), which has only one carbon atom, we would need to start with an alkyl halide having half a carbon atom, which is impossible.

If we use the smallest alkyl halide, methyl chloride (CH_3Cl), the Wurtz reaction would give ethane:



So, methane cannot be prepared by the Wurtz reaction. This part is also correct.

Conclusion for Statement II:

Both parts of the statement are correct. Methane indeed cannot be prepared from unsaturated hydrocarbons or by the Wurtz reaction. Therefore, Statement II is **true**.

Final Decision

- Statement I is **false**.
- Statement II is **true**.

Looking at the options, this corresponds to the choice where Statement I is false and Statement II is true.

The correct answer is **Option D**.

Question 19

n-Butane on monochlorination under photochemical condition gives an optically active compound " P ". " P " on further chlorination gives dichloro compounds.

The number of dichloro compounds obtained (ignore stereoisomers) is :

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Options:

A.

3

B.

4

C.

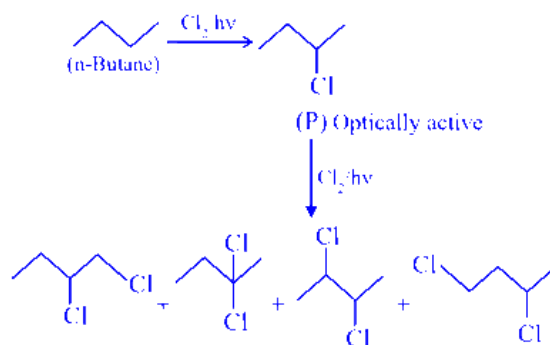
5

D.

6

Answer: B

Solution:



Total dichloro products (excluding stereoisomers) = 4

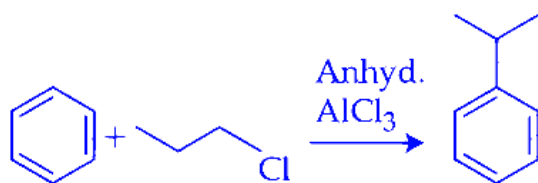
Question20

The major product of which of the following reaction is not obtained by rearrangement reaction?

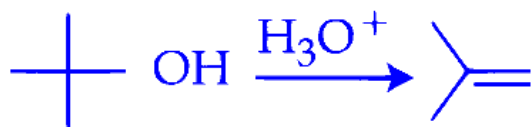
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Options:

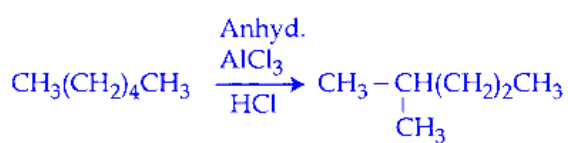
A.



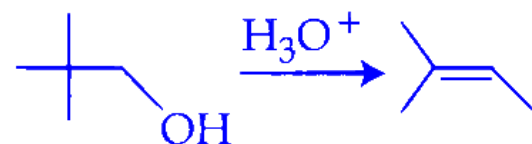
B.



C.

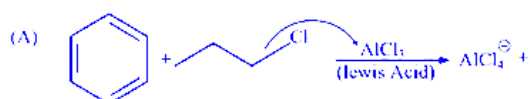


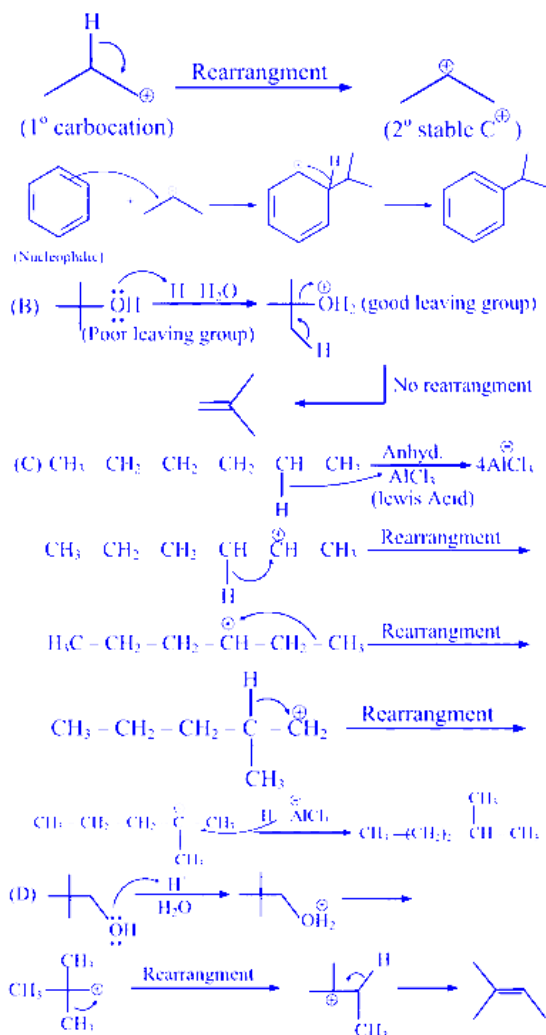
D.



Answer: B

Solution:





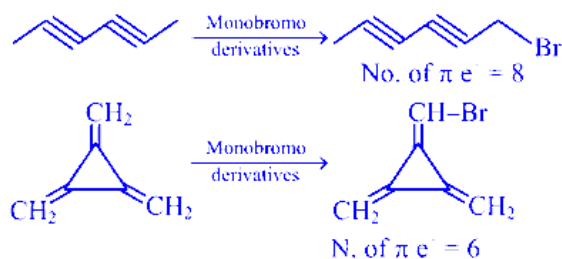
Question21

The compound with molecular formula C_6H_6 , which gives only one monobromo derivative and takes up four moles of hydrogen per mole for complete hydrogenation has _____ π electrons.

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Answer: 6

Solution:



Question22

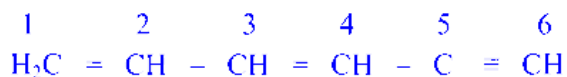
The sum of sigma (σ) and pi (π) bonds in Hex-1,3-dien-5-yne is _____.

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Answer: 15

Solution:

The compound given is hex-1,3-dien-5-yne structure is



A double bond has one π bond and one sigma bond.

A triple bond has one sigma bond and two π bonds.

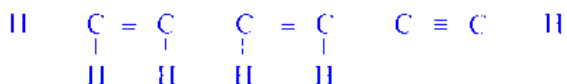
For the one triple bond, number of π bonds = 2

For one double bond, number of π bond = 1

For two double bonds, number of π bonds = 2

So, total number of π bonds = 2 + 2 = 4

For sigma bonds, count all carbon - carbon bonds and carbon - hydrogen bonds except the π bonds in double bonds and triple bond.



The total number of sigma bonds = 11

So, total number of bonds:

$$\sigma = 11$$

$$\pi = 4$$

Sum of σ and π bonds

$$= 11 + 4 = 15$$

Question 23

Isomeric hydrocarbons $\rightarrow$ negative Baeyer's test

(Molecular formula C_9H_{12})

The total number of isomers from above with four different non-aliphatic substitution sites is -

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Answer: 2

Solution:

Molecular formula of isomeric hydrocarbon $\rightarrow \text{C}_9\text{H}_{12}$ - Negative Baeyer's test

Baeyer's test is given by compounds that contain readily active carbon-carbon double bond.

So, in a negative Baeyer's test, there is no readily active $C=C$.

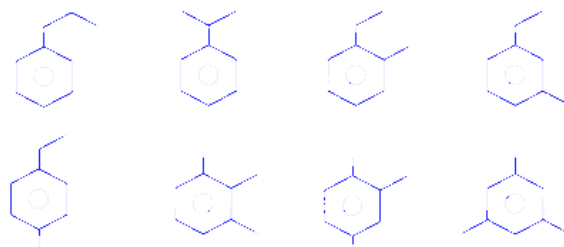
Compounds that give negative Baeyer's test are alkanes and aromatic compounds.

C_9H_{12} compound is not an alkane

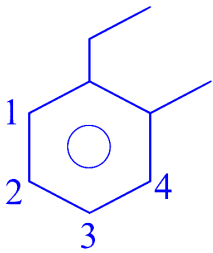
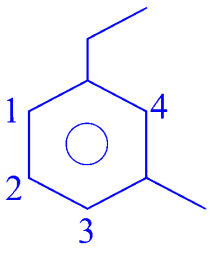
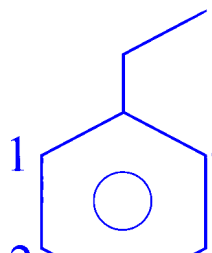
(Alkane general formula C_nH_{2n+2})

Hydrocarbons with the formula C_9H_{12} are considered as aromatic and isomers of substituted benzene rings.

The possible isomers of C_9H_{12} are



From these, total number of the isomers with four different non-aliphatic substitution sites are 2

		
Four positions (1,2,3,4) are different. So, it contain four different non-aliphatic substitution sites.	Four position (1,2,3,4) are different. So, it contain four different non-aliphatic substitution sites.	Four position (1,2,3,4) are not different. So, it contain four different non-aliphatic substitution sites.

Question24

Given below are two statements :

Statement I : One mole of propyne reacts with excess of sodium to liberate half a mole of H_2 gas.

Statement II : Four g of propyne reacts with $NaNH_2$ to liberate NH_3 gas which occupies 224 mL at STP.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

- A. Both Statement I and Statement II are incorrect
- B. Statement I is incorrect but Statement II is correct
- C. Both Statement I and Statement II are correct

D. Statement I is correct but Statement II is incorrect

Answer: D

Solution:

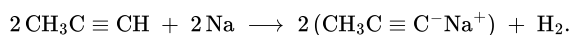
Statement I

“One mole of propyne reacts with excess of sodium to liberate half a mole of H_2 gas.”

Reaction and Stoichiometry

Propyne (terminal alkyne): $\text{CH}_3\text{C} \equiv \text{CH}$.

Reaction with sodium:



From this balanced equation, **2 moles** of propyne produce **1 mole** of H_2 .

Hence, **1 mole** of propyne will produce $\frac{1}{2}$ **mole** of H_2 .

Statement I is correct.

Statement II

“Four grams of propyne reacts with NaNH_2 to liberate NH_3 gas which occupies 224 mL at STP.”

Analysis

Moles of propyne

Molecular mass of propyne (C_3H_4):

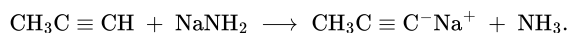
$$3 \times 12 + 4 \times 1 = 36 + 4 = 40 \text{ g/mol.}$$

Four grams of propyne is:

$$\frac{4 \text{ g}}{40 \text{ g/mol}} = 0.1 \text{ mol.}$$

Reaction with NaNH_2

For a **terminal alkyne**:



1 mole of propyne produces **1 mole** of NH_3 .

Moles of NH_3 produced

With 0.1 mol of propyne, we get 0.1 mol of NH_3 .

Volume of NH_3 at STP

1 mole of any ideal gas at STP $\approx 22.4 \text{ L} = 22400 \text{ mL}$.

0.1 mol of NH_3 occupies $0.1 \times 22.4 \text{ L} = 2.24 \text{ L} = 2240 \text{ mL}$.

However, **Statement II** says the liberated NH_3 occupies only **224 mL** at STP, which corresponds to 0.01 mol of NH_3 , not 0.1 mol. Therefore, the statement's volume is off by a factor of 10 and is thus **incorrect** if the reaction goes to completion in a typical way.

Statement II is incorrect.

Conclusion

Statement I is **correct**.

Statement II is **incorrect**.

Hence, the best choice is:

Option D: Statement I is correct but Statement II is incorrect.

Question25

The alkane from below having two secondary hydrogens is :

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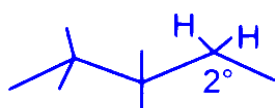
Options:

- A. 2,2,4,5-Tetramethylheptane
- B. 2,2,4,4-Tetramethylhexane
- C. 4-Ethyl-3,4-dimethyloctane
- D. 2,2,3,3-Tetramethylpentane

Answer: D

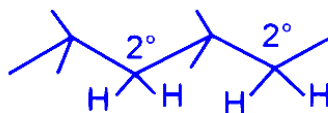
Solution:

2,2,3,3-Tetramethylpentane



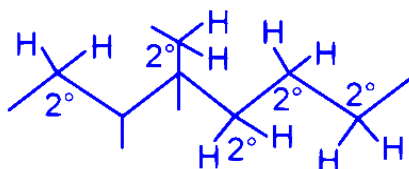
2 secondary Hydrogen

2,2,4,4-Tetramethylhexane



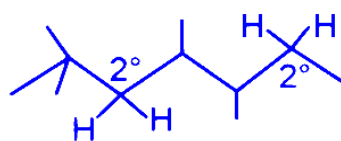
4 secondary Hydrogen

4-Ethyl-3,4-dimethyloctane



10 secondary Hydrogen

2,2,4,5-Tetramethylheptane



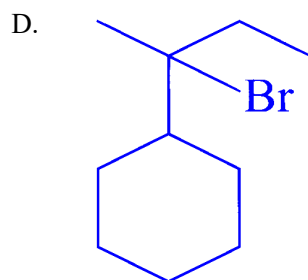
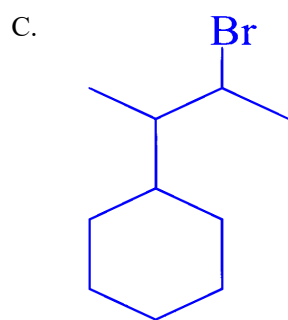
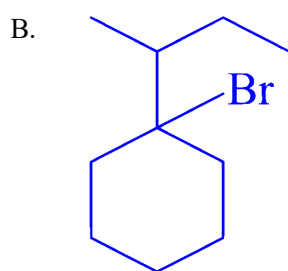
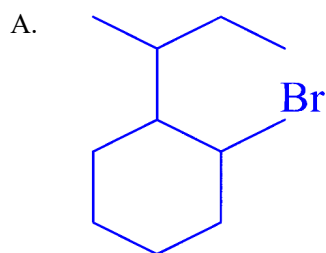
4 secondary Hydrogen

Question 26

When sec-butylcyclohexane reacts with bromine in the presence of sunlight, the major product is :

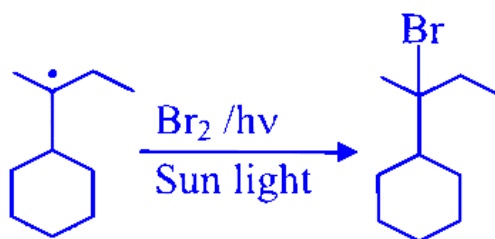
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Options:



Answer: D

Solution:



Formation of more stable free radical intermediate.

Question27

Match the List - I with List - II

	List - I Name reaction		List - II Product obtainable
(A)	Swarts reaction	(I)	Ethyl benzene
(B)	Sandmeyer's reaction	(II)	Ethyl iodide
(C)	Wurtz Fittig reaction	(III)	Cyanobenzene
(D)	Finkelstein reaction	(IV)	Ethyl fluoride

Choose the correct answer from the options given below:

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Options:

- A. A-II, B-III, C-I, D-IV
- B. A-II, B-I, C-III, D-IV
- C. A-IV, B-I, C-III, D-II
- D. A-IV, B-III, C-I, D-II

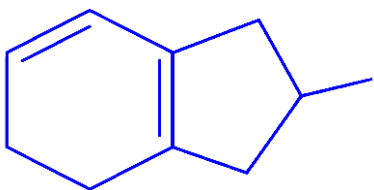
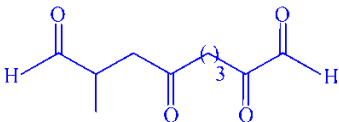
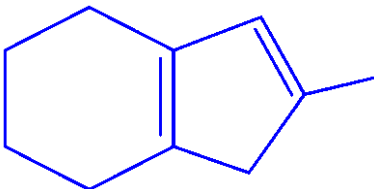
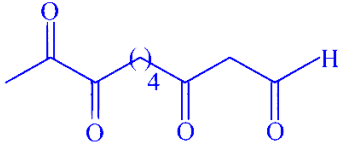
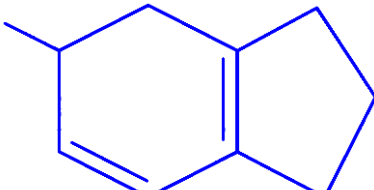
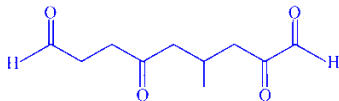
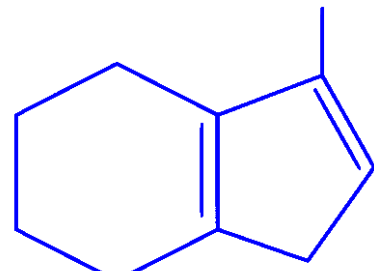
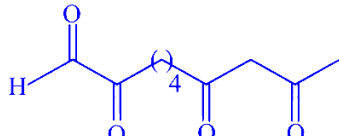
Answer: D

Solution:

	LIST-i Name reaction		LIST-II Product obtainable
(A)	Swarts reaction	(I)	$\text{Et} - \text{I} \xrightarrow[\text{DMF}]{\text{KF}} \text{Et} - \text{F}$
(B)	Sandmeyer's reaction	(II)	$\text{PhN}_2^{\oplus} \text{Cl}^{-} \xrightarrow{\text{CuCN/KCN}} \text{PhCN} + \text{N}_2$
(C)	Wurtz Fittig reaction	(III)	$\begin{array}{c} \text{Ph} - \text{Cl} + \text{EtCl} \xrightarrow[\text{ether}]{\text{Na}} \\ \text{Ph} - \text{Et} + \text{Ph} - \text{Ph} + \text{Et} - \text{Et} \end{array}$
(D)	Finkelstein reaction	(IV)	$\text{Et} - \text{Cl} \xrightarrow[\text{acetone}]{\text{NaI}} \text{Et} - \text{I} + \text{NaCl}$

Question28

Match List - I with List - II

	List - I (Isomers of $\text{C}_{10}\text{H}_{14}$)		List - II (Ozonolysis product)
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

Choose the correct answer from the options given below :

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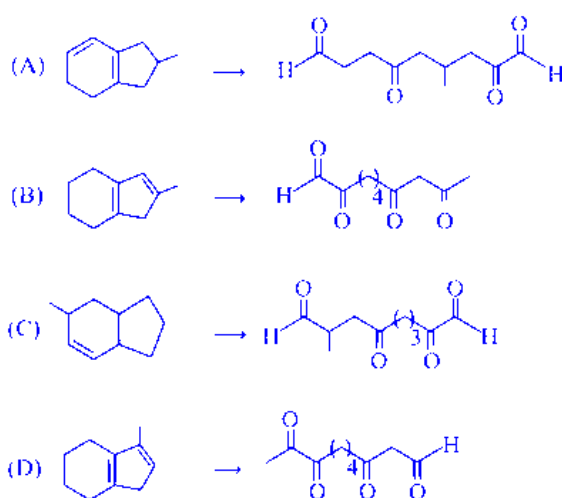
Options:

- A. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
B. (A) – (I), (B) – (IV), (C) – (III), (D) – (II)
C. (A) – (III), (B) – (II), (C) – (I), (D) – (IV)
D. (A) – (II), (B) – (II), (C) – (I), (D) – (IV)

Answer: A

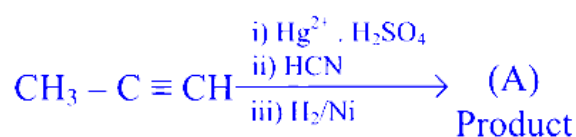
Solution:

Ozonolysis product



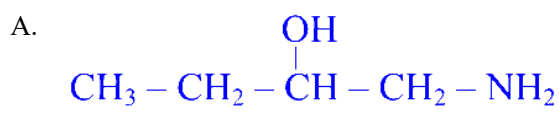
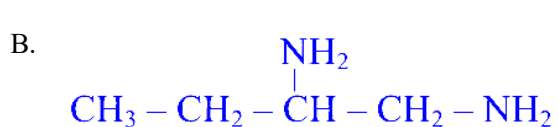
Question29

The product (A) formed in the following reaction sequence is

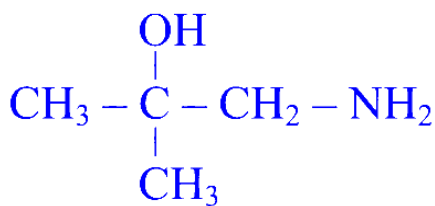


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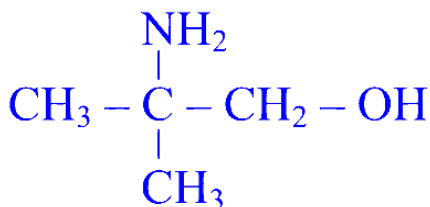
Options:

- A. 
- B. 

C.

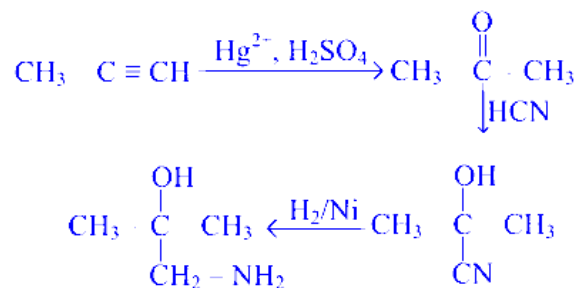


D.



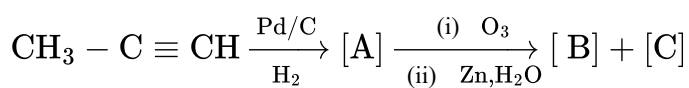
Answer: C

Solution:



Question30

Identify product [A], [B] and [C] in the following reaction sequence.



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Options:

A.

[A]: $\text{CH}_3\text{—CH=CH}_2$, [B]: CH_3CHO , [C]: $\text{CH}_3\text{CH}_2\text{OH}$

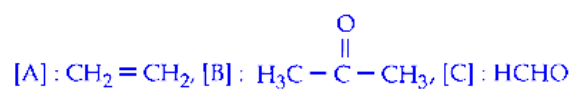
B.

[A]: $\text{CH}_3\text{—CH=CH}_2$, [B]: CH_3CHO , [C]: HCHO

C.

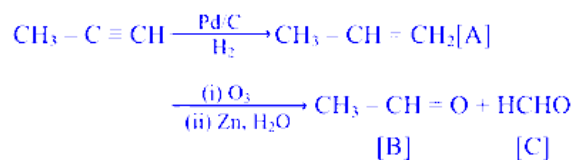
[A]: $\text{CH}_3\text{CH}_2\text{CH}_3$, [B]: CH_3CHO , [C]: HCHO

D.



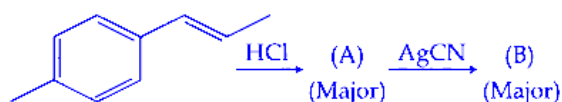
Answer: B

Solution:



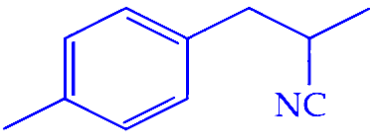
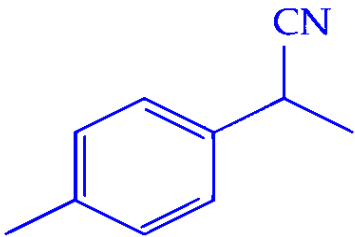
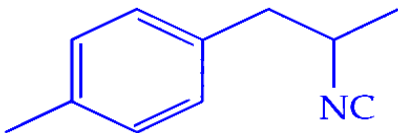
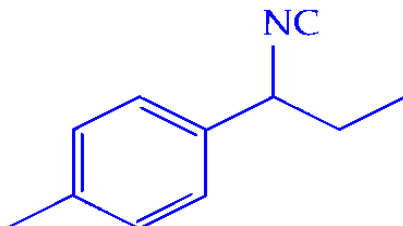
Question31

The product B formed in the following reaction sequence is:



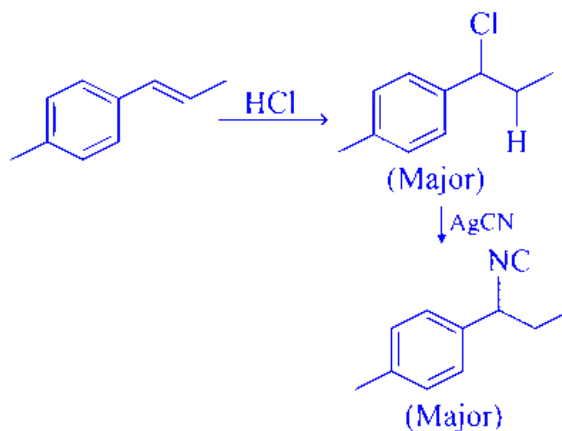
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Options:

- A. 
- B. 
- C. 
- D. 

Answer: D

Solution:



Question32

Total number of sigma (σ) _____ and pi(π) _____ bonds respectively present in hex-1-en-4-yne are:

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Options:

A.

14 and 3

B.

11 and 3

C.

13 and 3

D.

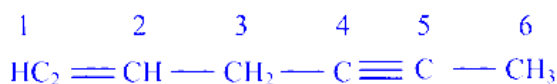
3 and 13

Answer: C

Solution:

The compound given is hex - 1 - en - 4 - yne

Structure is

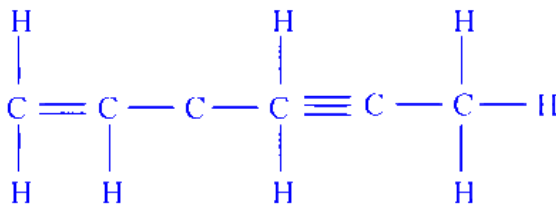


a double bond has one π bond and one sigma bond.

A triple bond has one σ bond and two π bonds.

So, the number of π bonds = $1 + 2 = 3$

For sigma bonds, count all carbon – carbon bonds and carbon – hydrogen bonds except the π bonds in double bond and triple bond.



The number of sigma bonds = 13

So, total number of bonds $\Rightarrow \sigma = 13 \quad \pi = 3$

Answer : option (C) 13 and 3

Question33

Match List - I with List - II.

List - I (Reaction)		List - II (Name of reaction)	
(A)	$2 \text{C}_6\text{H}_5\text{X} + 2\text{Na} \xrightarrow[\text{Ether}]{\text{Dry}} \text{C}_6\text{H}_5\text{C}_6\text{H}_5 + 2\text{NaX}$	(I)	Lucas reaction
(B)	$\text{ArN}_2^+\text{X}^- \xrightarrow[\text{HCl}]{\text{Cu}} \text{ArCl} + \text{N}_2 \uparrow + \text{CuX}$	(II)	Finkelstein reaction
(C)	$\text{C}_2\text{H}_5\text{Br} + \text{NaI} \xrightarrow[\text{Acetone}]{\text{Dry}} \text{C}_2\text{H}_5\text{I} + \text{NaBr}$	(III)	Fittig reaction
(D)	$\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)\text{CH}_3 \xrightarrow[\text{ZnCl}_2]{\text{HCl}} \text{CH}_3\text{C}(\text{Cl})(\text{CH}_3)\text{CH}_3$	(IV)	Gatterman reaction

Choose the correct answer from the options given below :

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Options:

A. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

B. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

D. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Answer: D

Solution:

	List - I (Reaction)		List - II (Name of reaction)
A.	$2 \text{C}_6\text{H}_5\text{X} + 2\text{Na} \xrightarrow[\text{Ether}]{\text{Dry}} \text{C}_6\text{H}_5\text{C}_6\text{H}_5 + 2\text{NaX}$	III.	Fittig reaction
B.	$\text{ArN}_2^+\text{X}^- \xrightarrow[\text{HCl}]{\text{Cu}} \text{ArCl} + \text{N}_2 \uparrow + \text{CuX}$	IV.	Gatterman reaction
C.	$\text{C}_2\text{H}_5\text{Br} + \text{NaI} \xrightarrow[\text{Acetone}]{\text{Dry}} \text{C}_2\text{H}_5\text{I} + \text{NaBr}$	II.	Finkelstein reaction
D.	$\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)\text{CH}_3 \xrightarrow[\text{ZnCl}_2]{\text{HCl}} \text{CH}_3\text{C}(\text{Cl})(\text{CH}_3)\text{CH}_3$	IV.	Lucas reaction

Question 34

Given below are two statements :

Statement (I) : Neopentane forms only one monosubstituted derivative.

Statement (II) : Melting point of neopentane is higher than n-pentane.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 2nd April Evening Shift

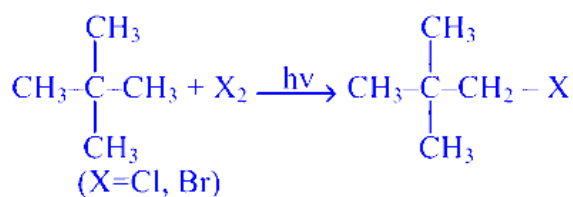
Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Statement I is correct but Statement II is incorrect
- D. Both Statement I and Statement II are incorrect

Answer: B

Solution:

Both Statement I and Statement II are correct.



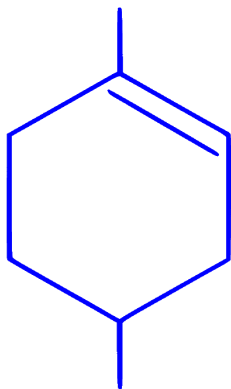
Question35

Which compound would give 3-methyl-6-oxoheptanal upon ozonolysis?

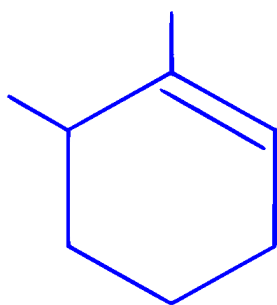
JEE Main 2025 (Online) 3rd April Morning Shift

Options:

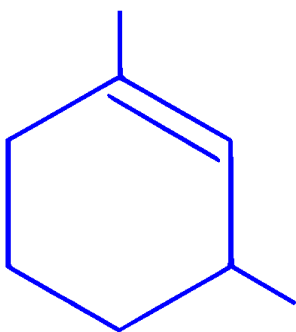
A.



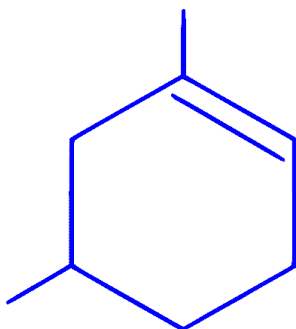
B.



C.

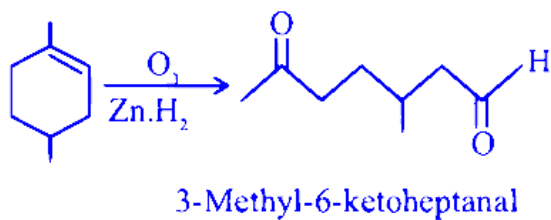


D.



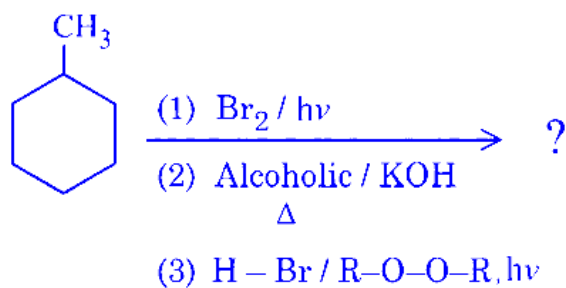
Answer: A

Solution:



Question36

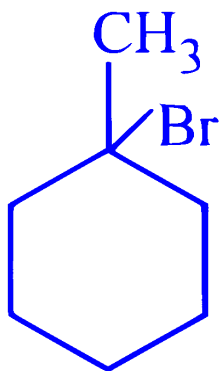
Predict the major product of the following reaction sequence:-



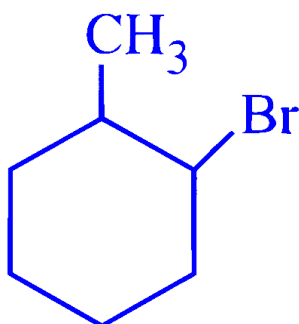
JEE Main 2025 (Online) 4th April Morning Shift

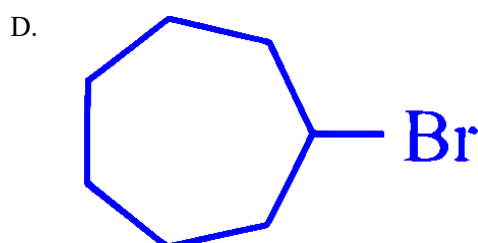
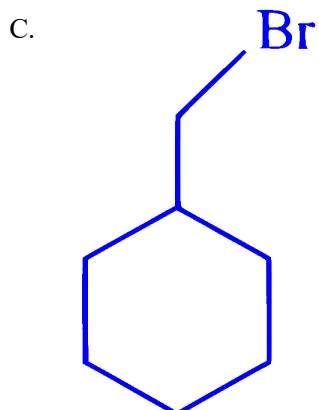
Options:

A.



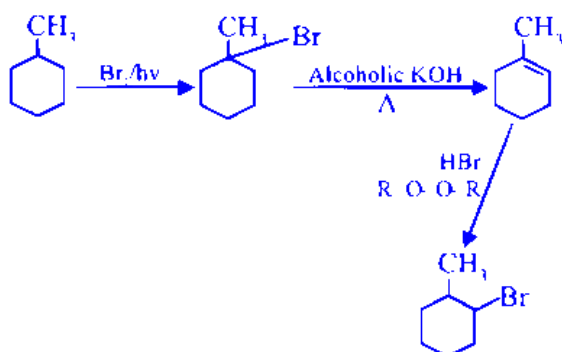
B.





Answer: B

Solution:



Question37

Given below are two statements:

Statement I : Ozonolysis followed by treatment with $\text{Zn}, \text{H}_2\text{O}$ of cis-2-butene gives ethanal.

Statement II : The product obtained by ozonolysis followed by treatment with $\text{Zn}, \text{H}_2\text{O}$ of 3, 6-dimethyloct-4-ene has no chiral carbon atom.

In the light of the above statements, choose the correct answer from the options given below

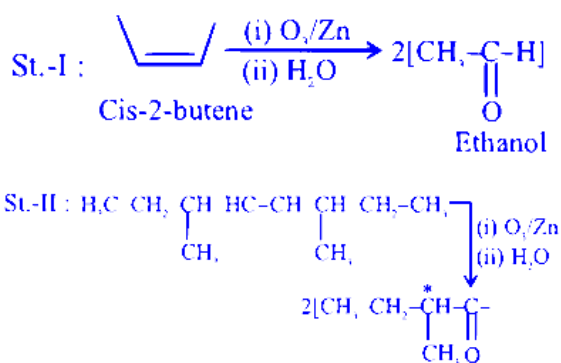
JEE Main 2025 (Online) 7th April Morning Shift

Options:

- A. Statement I is false but Statement II are true
- B. Both Statement I and Statement II are False
- C. Statement I is true but Statement II is false
- D. Both Statement I and Statement II are true

Answer: C

Solution:

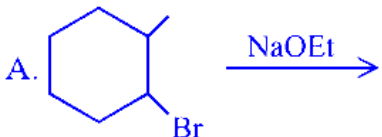
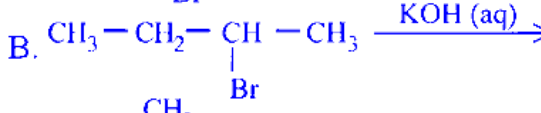
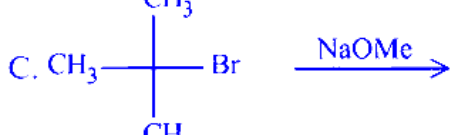
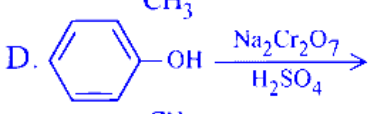
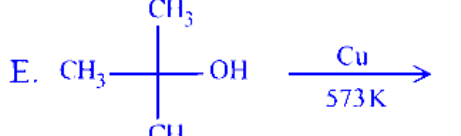


St-I : Correct statement

St-II : In correct statement because product has chiral centre.

Question38

The reactions which cannot be applied to prepare an alkene by elimination, are

- A. 
- B. 
- C. 
- D. 
- E. 

Choose the correct answer from the options given below:

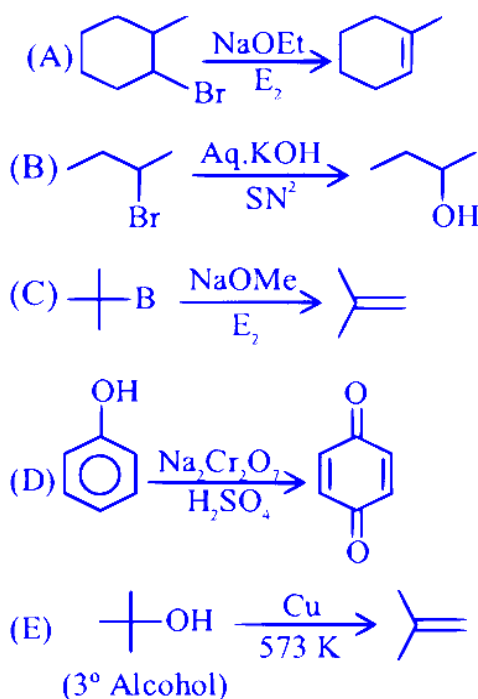
JEE Main 2025 (Online) 7th April Morning Shift

Options:

- A. B & E Only
- B. B, C & D Only
- C. A, C & D Only
- D. B & D Only

Answer: D

Solution:



Option (B) and (D) reaction are not able to form alkene as a product.

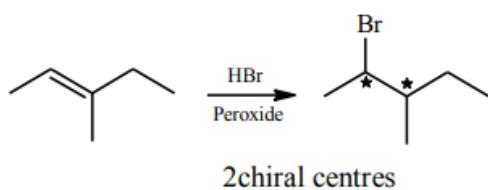
Question39

3-Methylhex-2-ene on reaction with HBr in presence of peroxide forms an addition product (A). The number of possible stereoisomers for 'A' is _____

[27-Jan-2024 Shift 1]

Answer: 4

Solution:



No. of stereoisomers = 4

Question40

Among the following, total number of meta directing functional groups is (Integer based)

$-\text{OCH}_3$, $-\text{NO}_2$, $-\text{CN}$, $-\text{CH}_3$, $-\text{NHCOCH}_3$,
 $-\text{COR}$, $-\text{OH}$, $-\text{COOH}$, $-\text{Cl}$

[27-Jan-2024 Shift 1]

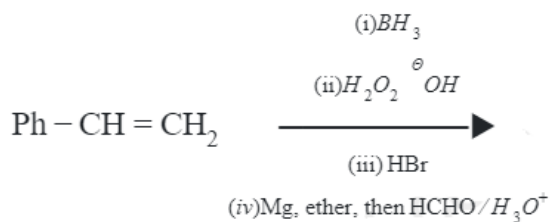
Answer: 4

Solution:

$-\text{NO}_2$, $-\text{C} \equiv \text{N}$, $-\text{COR}$, $-\text{COOH}$
are meta directing.

Question41

The final product A, formed in the following reaction sequence is:



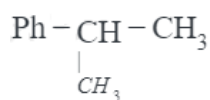
[27-Jan-2024 Shift 2]

Options:

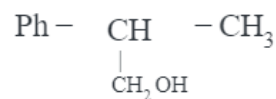
A.

$\text{Ph}-\text{CH}_2-\text{CH}_2-\text{CH}_3$

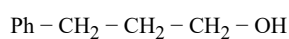
B.



C.

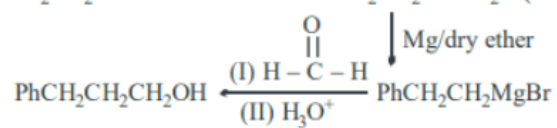
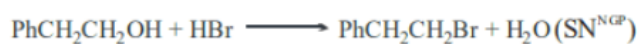
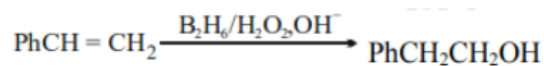


D.



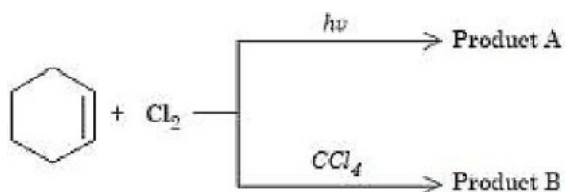
Answer: D

Solution:



Question42

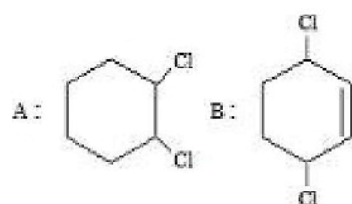
Identify product A and product B :



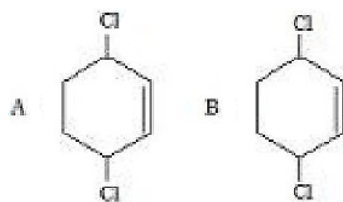
[29-Jan-2024 Shift 1]

Options:

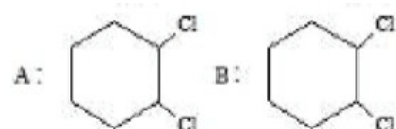
A.



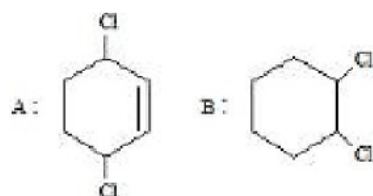
B.



C.

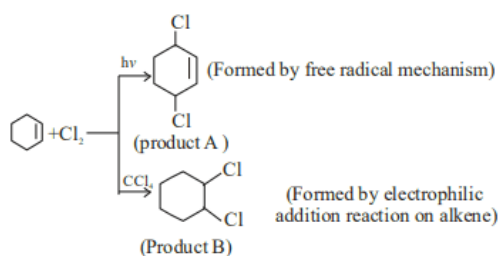


D.



Answer: D

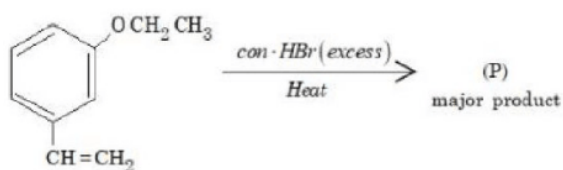
Solution:



Hence correct Ans. (4)

Question43

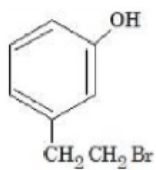
The major product (P) in the following reaction is



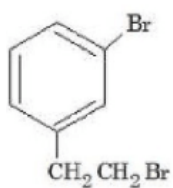
[29-Jan-2024 Shift 1]

Options:

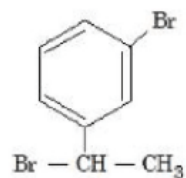
A.



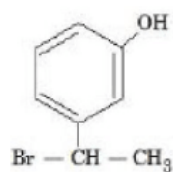
B.



C.

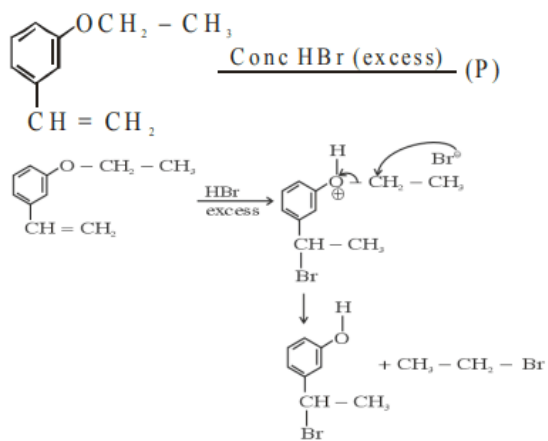


D.



Answer: D

Solution:



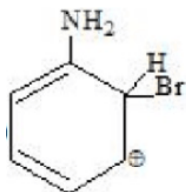
Question44

The arenium ion which is not involved in the bromination of Aniline is .

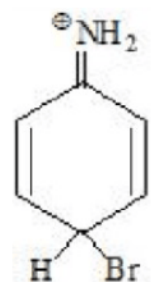
[29-Jan-2024 Shift 1]

Options:

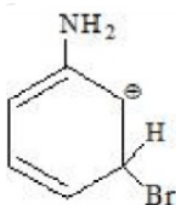
A.



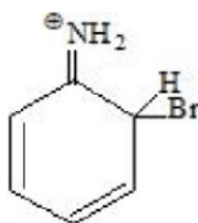
B.



C.



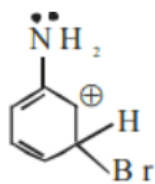
D.



Answer: C

Solution:

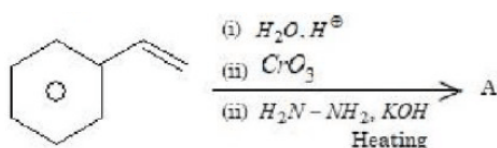
Since NH_2 group is o/p directing hence arenium ion will not be formed by attack at meta position i.e.



Hence Answer is (3)

Question45

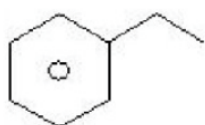
The final product A formed in the following multistep reaction sequence is



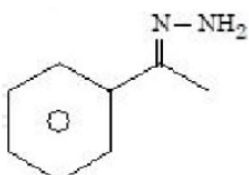
[29-Jan-2024 Shift 1]

Options:

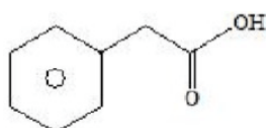
A.



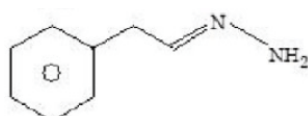
B.



C.

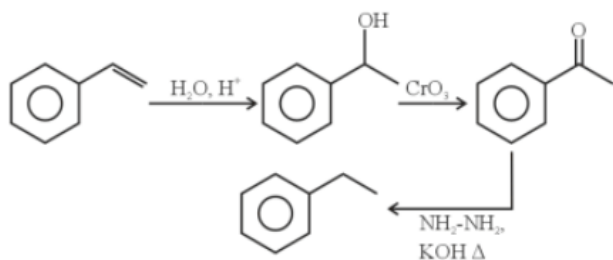


D.

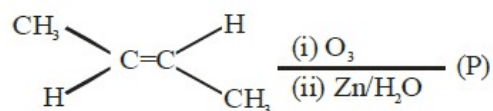


Answer: A

Solution:



Question46

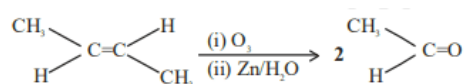


Consider the given reaction. The total number of oxygen atoms present per molecule of the product (P) is ____

[29-Jan-2024 Shift 1]

Answer: 1

Solution:

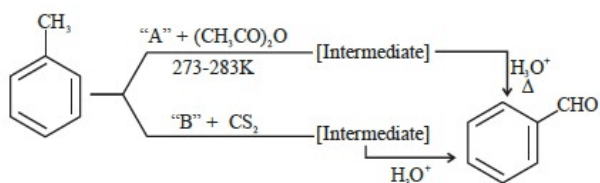


Hence total number of oxygen atom present per

molecule CC=O is 1.

Question47

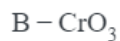
In the given reactions identify the reagent A and reagent B



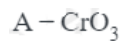
[30-Jan-2024 Shift 1]

Options:

A.



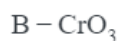
B.



C.

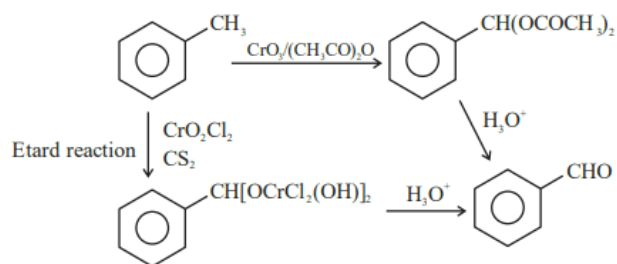


D.



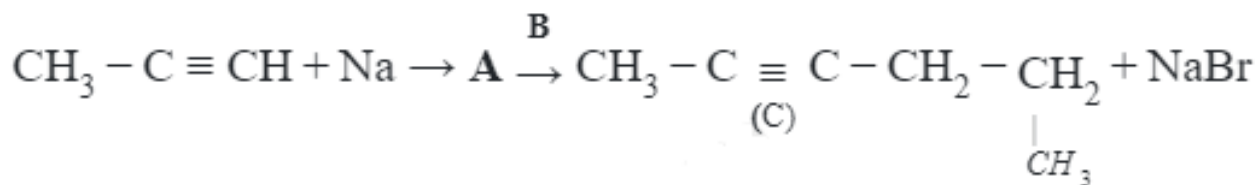
Answer: B

Solution:



Question48

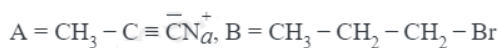
Compound A formed in the following reaction reacts with B gives the product C. Find out A and B.



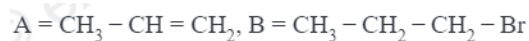
[30-Jan-2024 Shift 1]

Options:

A.



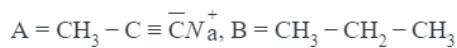
B.



C.

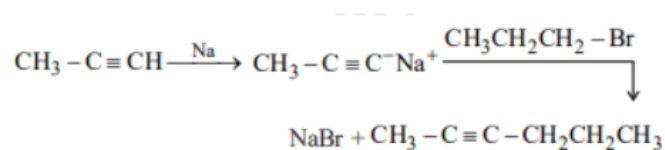


D.



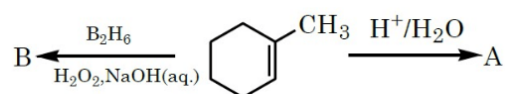
Answer: A

Solution:



Question49

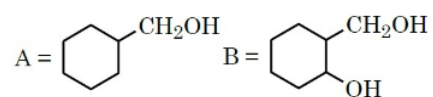
Product A and B formed in the following set of reactions are:



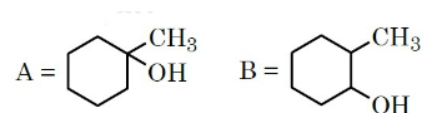
[30-Jan-2024 Shift 2]

Options:

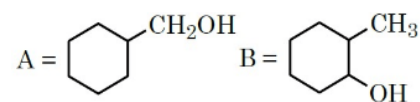
A.



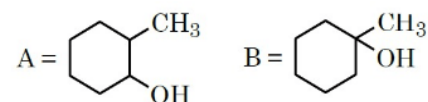
B.



C.

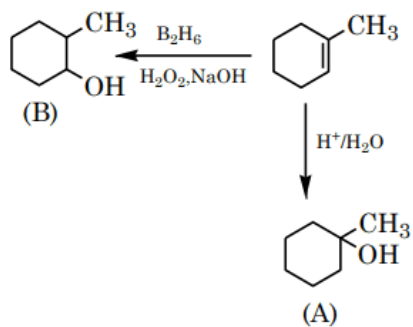


D.



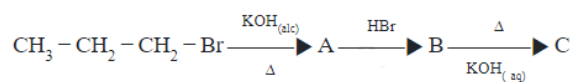
Answer: B

Solution:



Question50

The product (C) in the below mentioned reaction is:



[31-Jan-2024 Shift 1]

Options:

A.

Propan-1-ol

B.

Propene

C.

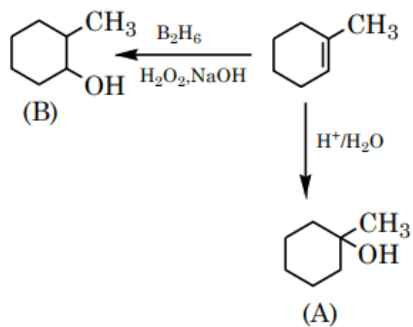
Propyne

D.

Propan-2-ol

Answer: D

Solution:



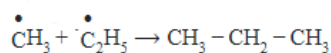
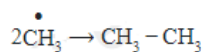
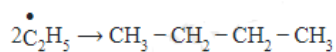
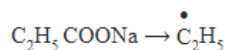
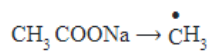
Question51

Number of alkanes obtained on electrolysis of a mixture of CH_3COONa and $\text{C}_2\text{H}_5\text{COONa}$ is ____

[31-Jan-2024 Shift 1]

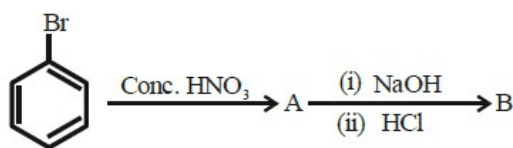
Answer: 3

Solution:



Question52

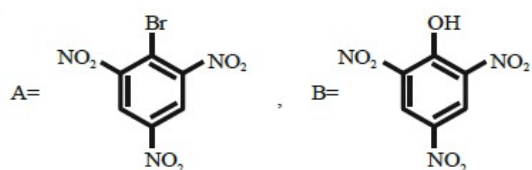
Identify A and B in the following reaction sequence.



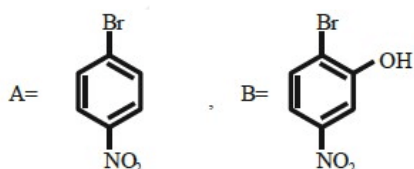
[31-Jan-2024 Shift 2]

Options:

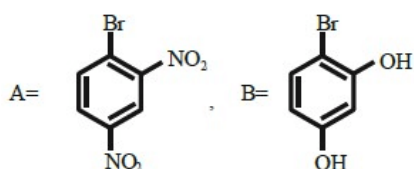
A.



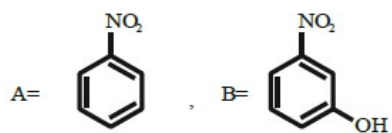
B.



C.

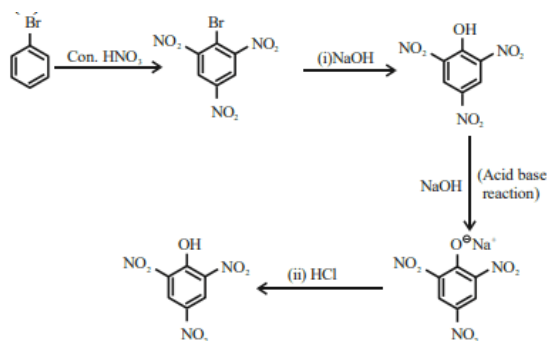


D.



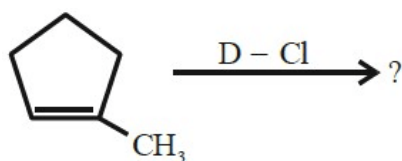
Answer: A

Solution:



Question53

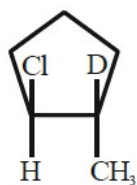
Major product of the following reaction is -



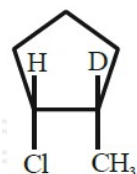
[31-Jan-2024 Shift 2]

Options:

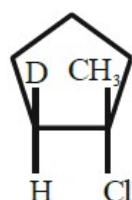
A.



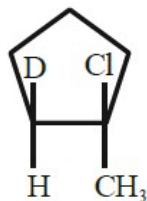
B.



C.

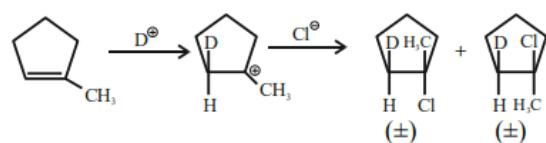


D.



Answer: 0

Solution:



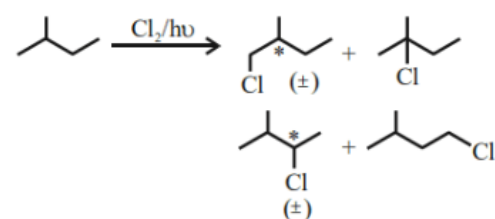
Question54

Number of isomeric products formed by monochlorination of 2-methylbutane in presence of sunlight is _____

[31-Jan-2024 Shift 2]

Answer: 6

Solution:



∴ Number of isomeric products = 6

Question55

Given below are two statements:

Statement (I) : The NH_2 group in Aniline is ortho and para directing and a powerful activating group.

Statement (II) : Aniline does not undergo FriedelCraft's reaction (alkylation and acylation).

In the light of the above statements, choose the most appropriate answer from the options given below :

[1-Feb-2024 Shift 1]

Options:

A.

Both Statement I and Statement II are correct

B.

Both Statement I and Statement II are incorrect

C.

Statement I is incorrect but Statement II is correct

D.

Statement I is correct but Statement II is incorrect

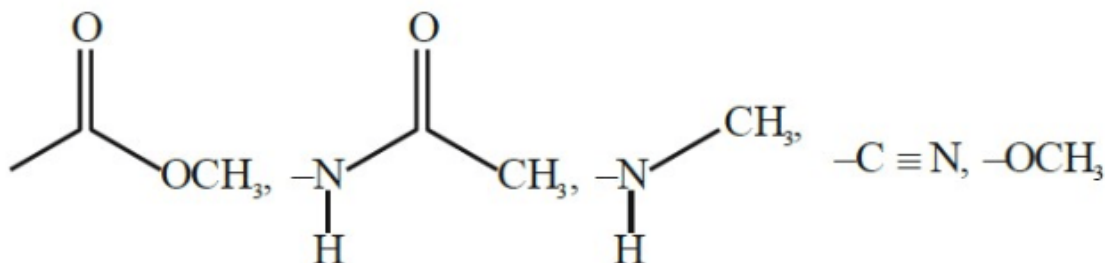
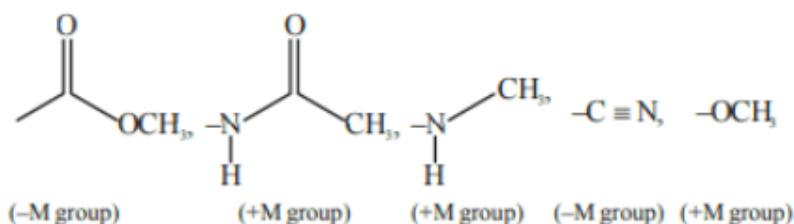
Answer: A**Solution:**

The NH_2 group in Aniline is ortho and para directing and a powerful activating group as NH_2 has strong +M effect.

Aniline does not undergo Friedel-Craft's reaction (alkylation and acylation) as Aniline will form complex with AlCl_3 which will deactivate the benzene ring.

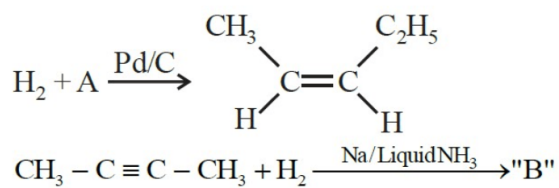
Question 56

Total number of deactivating groups in aromatic electrophilic substitution reaction among the following is

**[1-Feb-2024 Shift 1]****Answer: 2****Solution:**

Question57

In the given reactions identify



[1-Feb-2024 Shift 2]

Options:

A.

A : 2-Pentyne

B : trans -2- butene

B.

A : n-Pentane

B : trans -2 - butene

C.

A : 2 - Pentyne

B : Cis -2 - butene

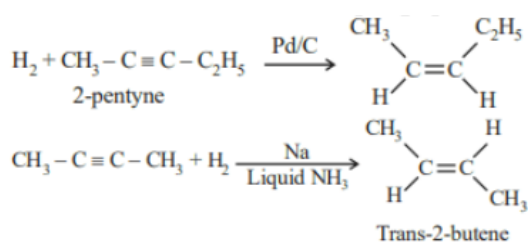
D.

A : n-Pentane

B : Cis -2- butene

Answer: A

Solution:



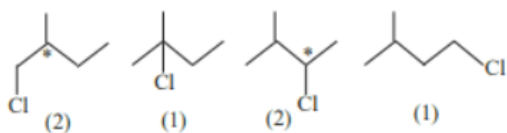
Question58

Total number of isomeric compounds (including stereoisomers) formed by monochlorination of 2-methylbutane is _____

[1-Feb-2024 Shift 2]

Answer: 6

Solution:



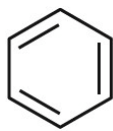
Question59

Which of the following compound will most easily be attacked by an electrophile?

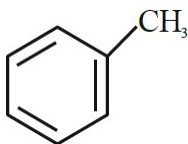
[1-Feb-2024 Shift 1]

Options:

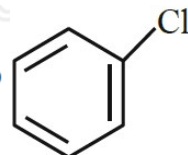
A.



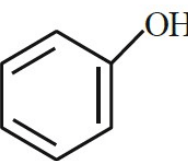
B.



C.



D.



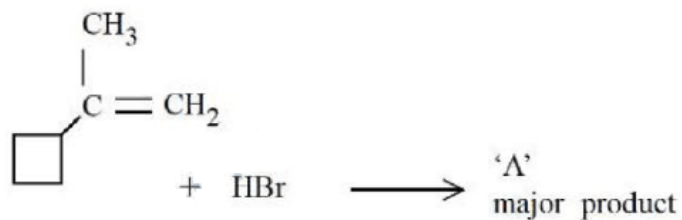
Answer: D

Solution:

Higher the electron density in the benzene ring more easily it will be attacked by an electrophile. Phenol has the highest electron density amongst all the given compound.

Question60

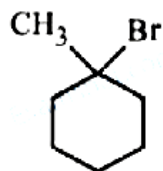
In the following given reaction 'A' is



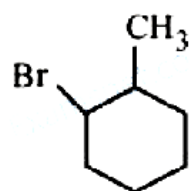
[24-Jan-2023 Shift 1]

Options:

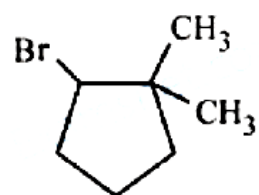
A.



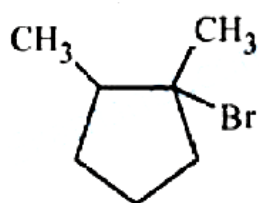
B.



C.

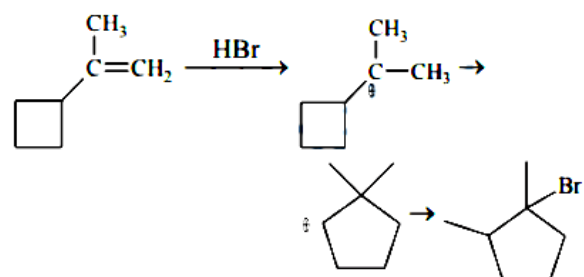


D.



Answer: D

Solution:



Question61

Given below are two statements :

Statement I :

under Clemmensen reduction conditions will give $\text{HOOC} \rightarrow$

Statement II :

under Wolff-Kishner reduction condition will give

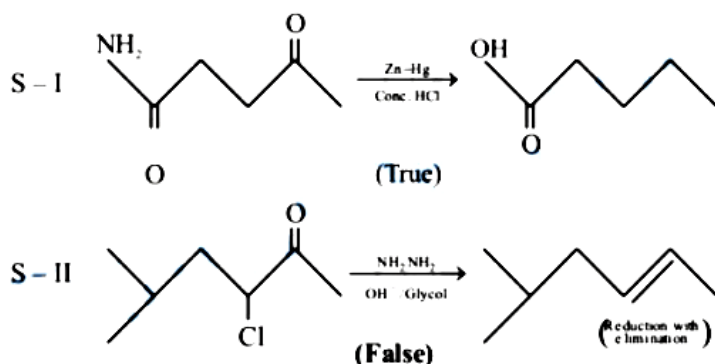
In the light of the above statements, choose the correct answer from the options given below :
[24-Jan-2023 Shift 2]

Options:

- A. Statement I is false but Statement II is true
- B. Both Statement I and Statement II are false
- C. Statement I is true but Statement II is false
- D. Both Statement I and Statement II are true

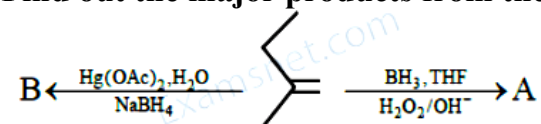
Answer: C

Solution:



Question62

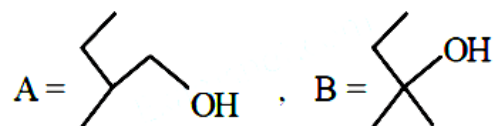
Find out the major products from the following reactions.



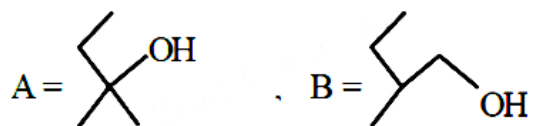
[24-Jan-2023 Shift 2]

Options:

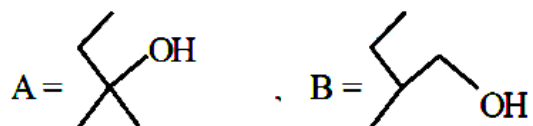
A.



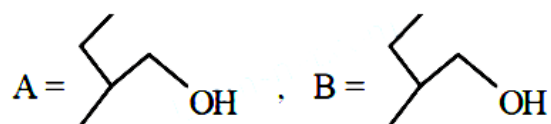
B.



C.

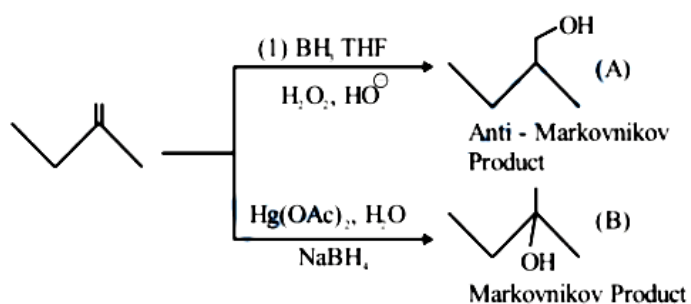


D.



Answer: A

Solution:

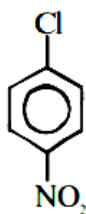


Question63

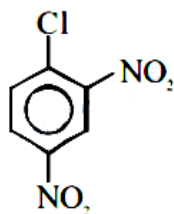
The compound which will have the lowest rate towards nucleophilic aromatic substitution on treatment with OH^- is
 [25-Jan-2023 Shift 1]

Options:

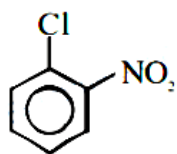
A.



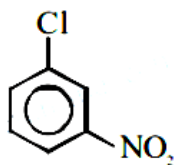
B.



C.



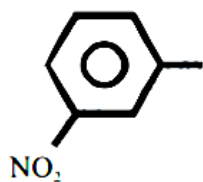
D.



Answer: D

Solution:

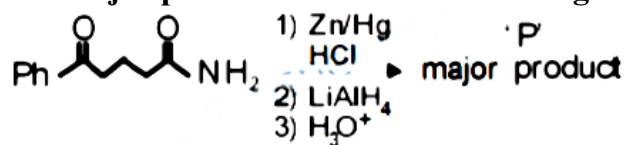
Electron withdrawing groups are highly ineffective at meta position in nucleophilic aromatic substitution reactions.
Hence compound



rate in nucleophilic aromatic substitution.

Question64

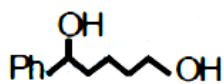
The major product ' P ' for the following sequence| of reactions is:



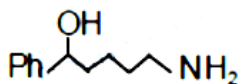
[29-Jan-2023 Shift 1]

Options:

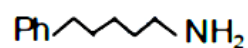
A.



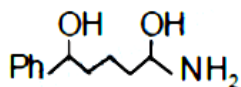
B.



C.

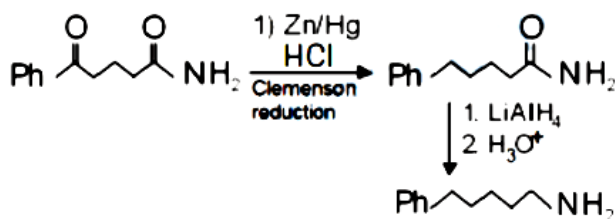


D.



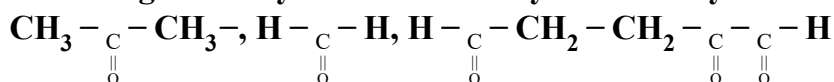
Answer: C

Solution:



Question65

17 mg of a hydrocarbon (M.F. $C_{10}H_{16}$) takes up 8.40 mL of the H_2 gas measured at $0^\circ C$ and 760 mm of Hg. Ozonolysis of the same hydrocarbon yields



The number of double bond/s present in the hydrocarbon is _____.
[29-Jan-2023 Shift 1]

Answer: 3

Solution:

$$\text{Moles of hydrocarbon} = \frac{17 \times 10^{-3}}{136} = 1.25 \times 10^{-4}$$

Mole of H_2 gas

$$\Rightarrow 1 \times \frac{8.40}{1000} = n \times 0.0821 \times 273$$

$$\Rightarrow n = 3.75 \times 10^{-4}$$

Hydrogen molecule used for 1 molecule of hydrocarbon is 3

$$= \frac{3.75 \times 10^{-4}}{1.25 \times 10^{-4}} = 3$$

Question66

The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement)
[29-Jan-2023 Shift 2]

Options:

A. 1-Bromo-2-methylbutane

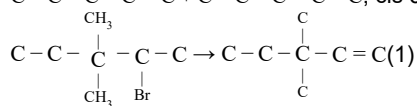
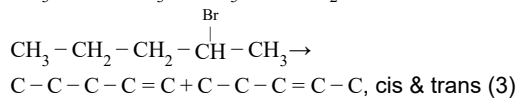
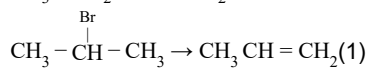
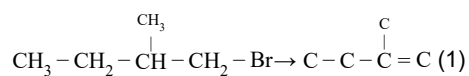
B. 2-Bromopropane

C. 2-Bromopentane

D. 2-Bromo-3,3-dimethylpentane

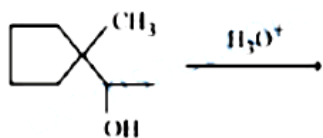
Answer: C

Solution:



Question67

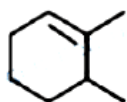
Find out the major product for the following reaction.



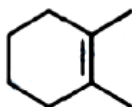
[29-Jan-2023 Shift 2]

Options:

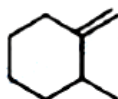
A.



B.



C.

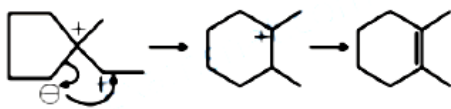


D.



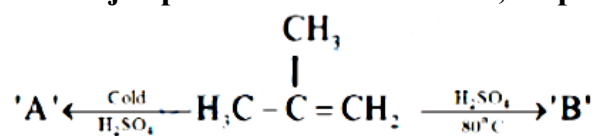
Answer: B

Solution:



Question68

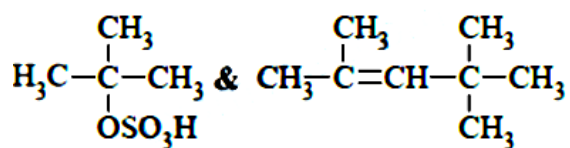
The major products 'A' and 'B', respectively, are



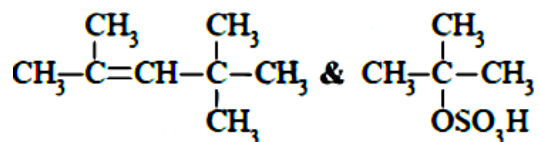
[30-Jan-2023 Shift 1]

Options:

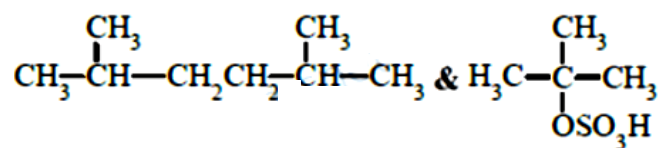
A.



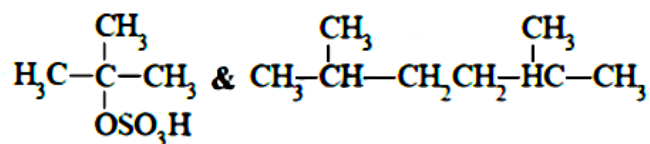
B.



C.

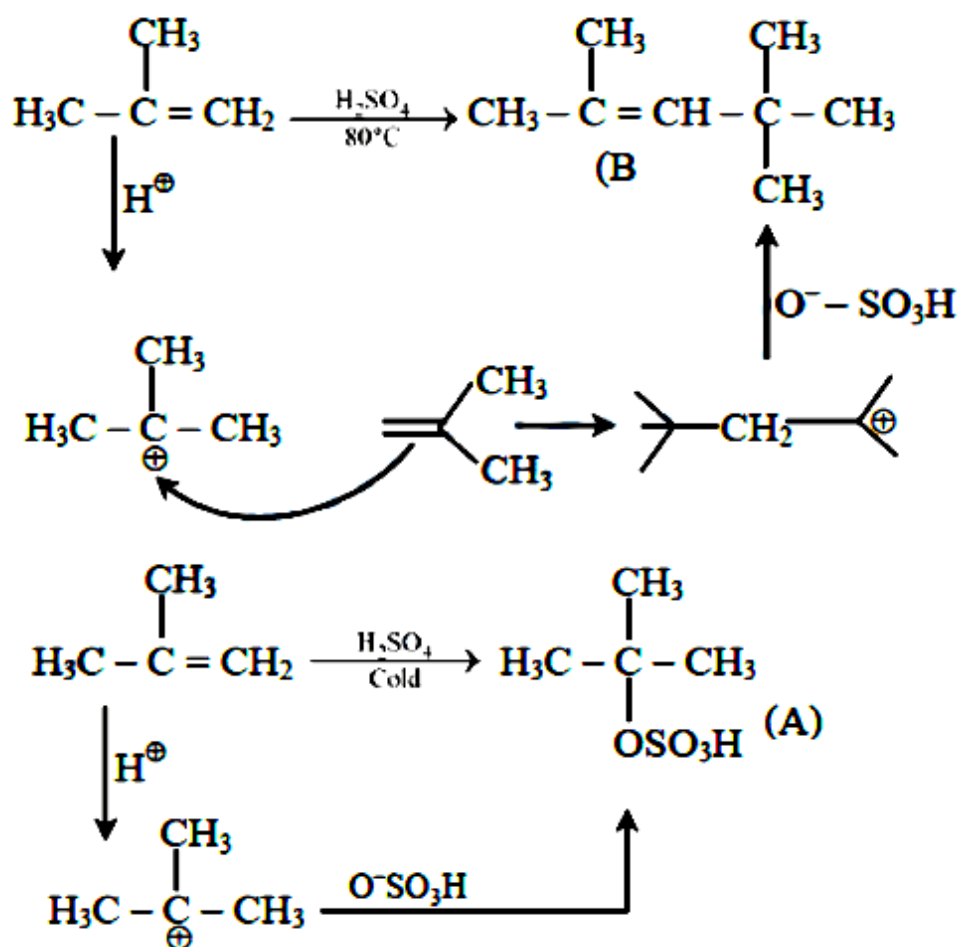


D.



Answer: A

Solution:



Question69

Choose the correct set of reagents for the following conversion trans (Ph - CH = CH - CH₃) → cis(Ph - CH = CH - CH₃)

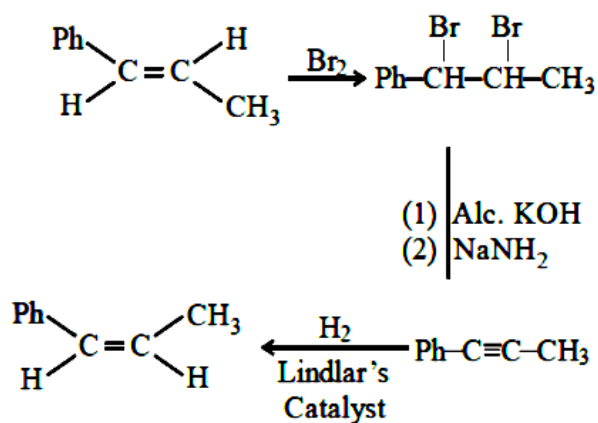
[31-Jan-2023 Shift 1]

Options:

- A. Br₂, alc KOH, NaNH₂, Na(Liq NH₃)
- B. Br₂, alc KOH, NaNH₂, H₂ Lindlar Catalyst
- C. Br₂, aq KOH, NaNH₂, H₂ Lindlar Catalyst
- D. Br₂, aq KOH, NaNH₂, Na(Liq NH₃)

Answer: B

Solution:



Question70

A hydrocarbon 'X' with formula C_6H_8 uses two moles of H_2 on catalytic hydrogenation of its one mole. On ozonolysis, 'X' yields two moles of methane dicarbaldehyde. The hydrocarbon 'X' is:

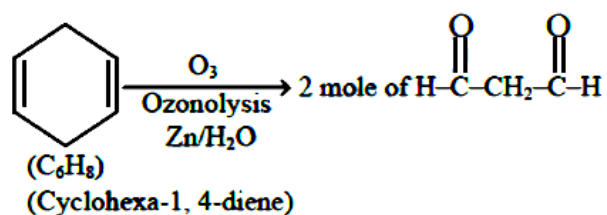
[31-Jan-2023 Shift 2]

Options:

- A. hexa-1, 3, 5-triene
- B. 1-methylcyclopenta-1, 4-diene
- C. cyclohexa-1, 3-diene
- D. cyclohexa-1, 4-diene

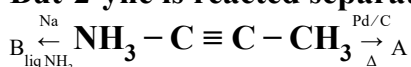
Answer: D

Solution:



Question71

But-2-yne is reacted separately with one mole of Hydrogen as shown below:



Identify the incorrect statements from the options given below:

- A. A is more soluble than B.
- B. The boiling point & melting point of A are higher and lower than B respectively.
- C. A is more polar than B because dipole moment of A is zero.

D. Br₂ adds easily to B than A.

[1-Feb-2023 Shift 1]

Options:

A. B and C only

B. B, C and D only

C. A, C and D only

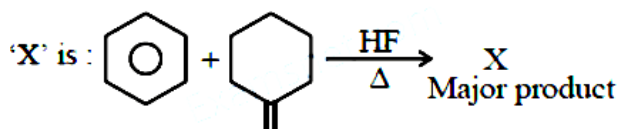
D. C and D only

Answer: D

Solution:

Incorrect statements are C and D only

Question72



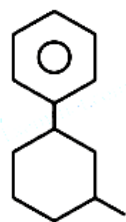
[1-Feb-2023 Shift 2]

Options:

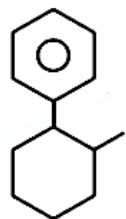
A.



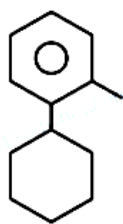
B.



C.

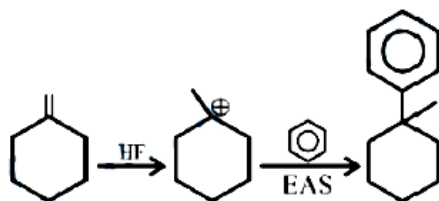


D.



Answer: A

Solution:



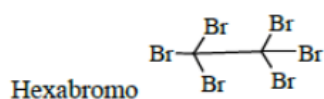
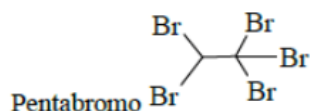
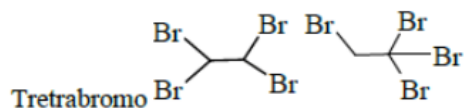
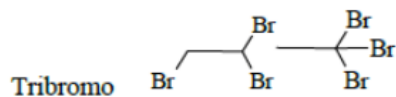
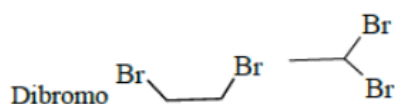
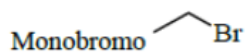
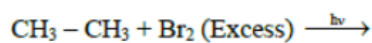
Question73

Number of bromo derivatives obtained on treating ethane with excess of Br_2 , in diffused sunlight is...

[6-Apr-2023 shift 1]

Answer: 9

Solution:

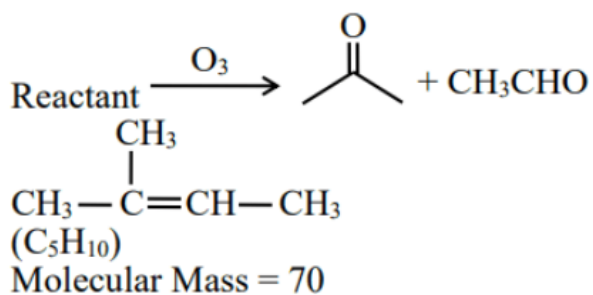


Question74

Molar mass of the hydrocarbon (X) which on ozonolysis consumes one mole of O_3 per mole of (X) and gives one mole each of ethanol and propanone is _____ $gmol^{-1}$ (Molar mass of C : $12gmol^{-1}$, H : $1gmol^{-1}$)
[8-Apr-2023 shift 1]

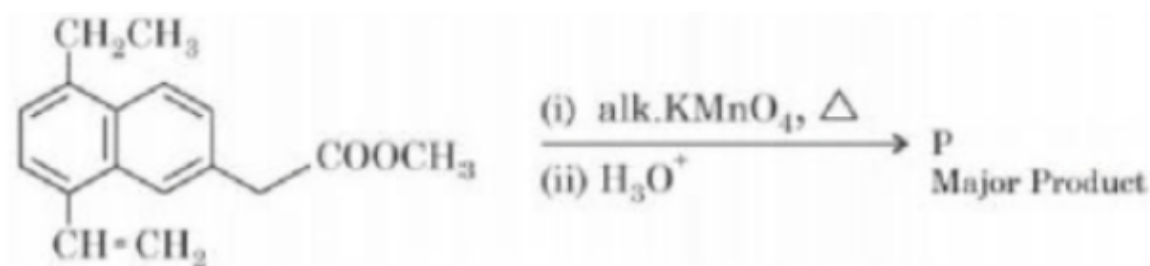
Answer: 70

Solution:



Question 75

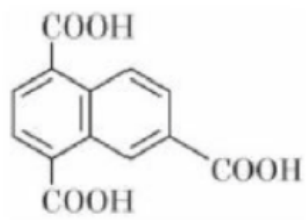
The major product 'P' formed in the given reaction is



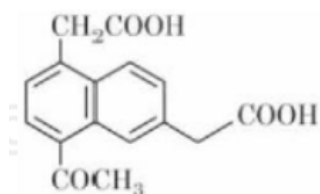
[10-Apr-2023 shift 1]

Options:

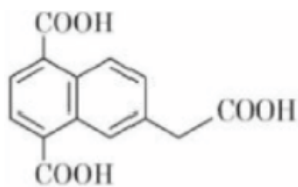
A.



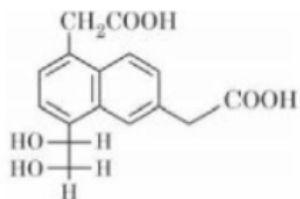
B.



C.

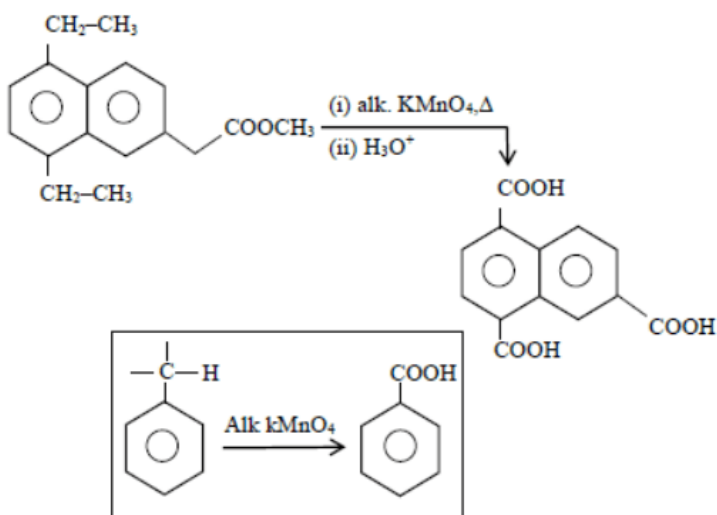


D.



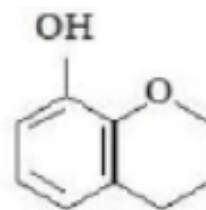
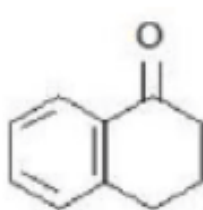
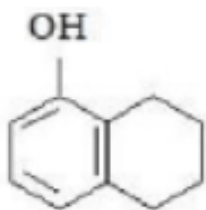
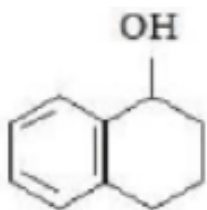
Answer: A

Solution:



Question76

Arrange the following compounds in increasing order of rate of aromatic electrophilic substitution reaction



[11-Apr-2023 shift 1]

Options:

A. c, a, b, d

B. d, b, c, a

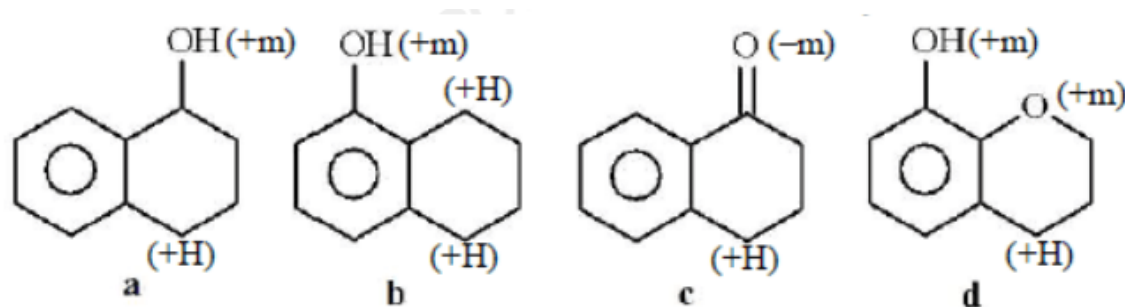
C. d, b, a, c

D. b, c, a, d

Answer: A

Solution:

Benzene becomes more reactive towards EAS when any substituent raises the electron density.

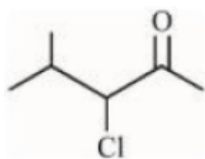


Correct order $c < a < b < d$

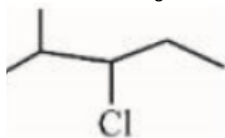
Question 77

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.

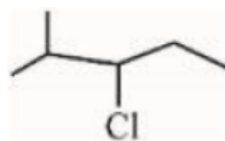
Assertion A



can be subjected to Wolff-Kishner reduction to give



Reason R: Wolff-Kishner reduction is used to convert



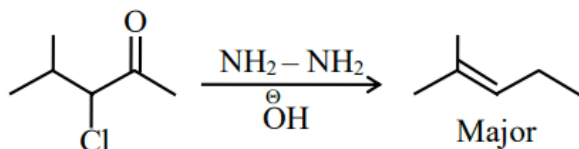
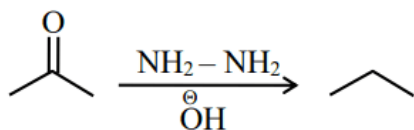
In the light of the above statements, choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

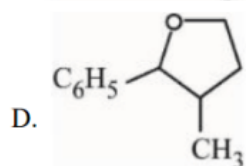
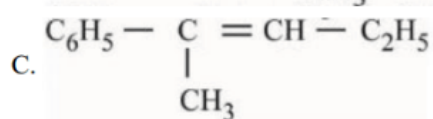
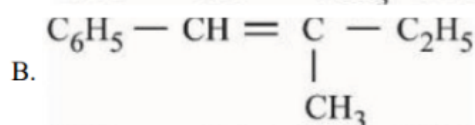
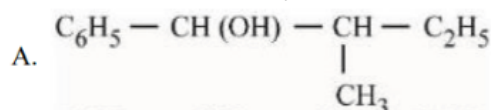
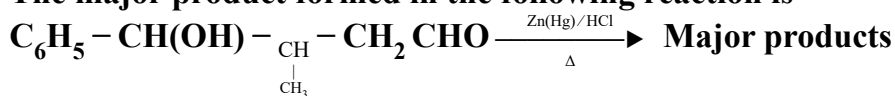
- A. Both A and R are true and R is the correct explanation of A
- B. A is true but R is false
- C. Both A and R are true but R is NOT the correct explanation of A
- D. A is false but R is true

Answer: D

Solution:

Assertion (A)**Reason (R)****Question78**

The major product formed in the following reaction is



Choose the correct answer from the options given below:

[11-Apr-2023 shift 2]

Options:

A. C only

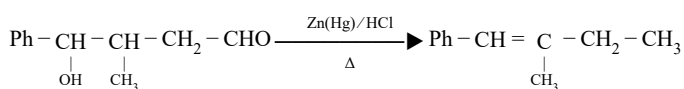
B. A only

C. B only

D. D only

Answer: C

Solution:

**Question79**

2-hexene $\xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) O}_3}$ **Products**

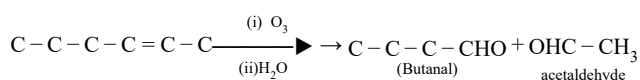
The two products formed in above reaction are -
[12-Apr-2023 shift 1]

Options:

- A. Butanal and acetaldehyde
- B. Butanal and acetic acid
- C. Butanoic acid and acetaldehyde
- D. Butanoic acid and acetic acid

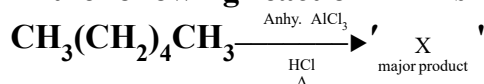
Answer: A

Solution:



Question80

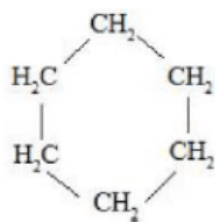
In the following reaction ' X ' is



[13-Apr-2023 shift 1]

Options:

- A. $\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{Cl}$
- B.

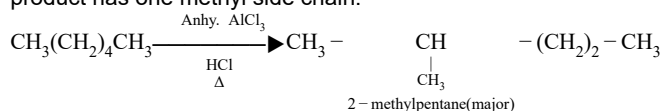


- C. $\text{Cl}-\text{CH}_2-(\text{CH}_2)_4-\text{CH}_2-\text{Cl}$
- D. $\text{CH}_3\text{CH}(\text{CH}_3)-(\text{CH}_2)_2\text{CH}_3$

Answer: D

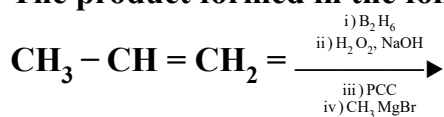
Solution:

n-alkanes on heating in this presence of anhydrous AlCl_3 and hydrogen chloride gas isomerise to branched chain alkanes. The major product has one methyl side chain.



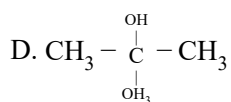
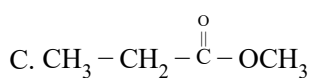
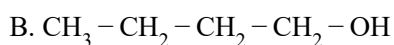
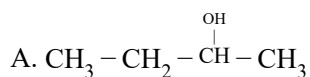
Question81

The product formed in the following multistep reaction is:



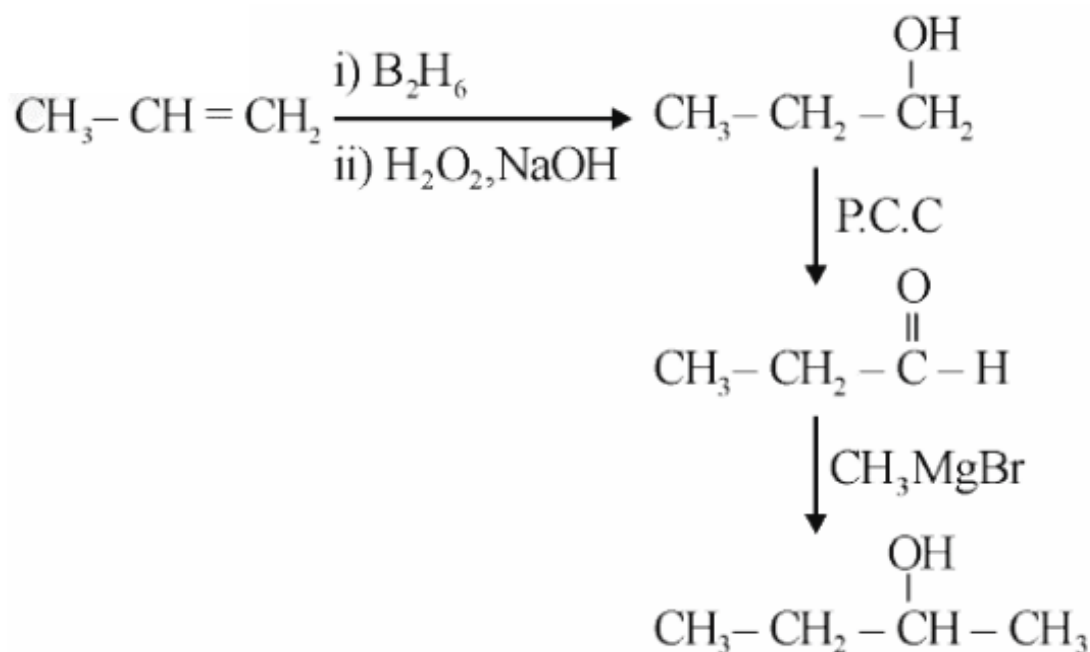
[15-Apr-2023 shift 1]

Options:



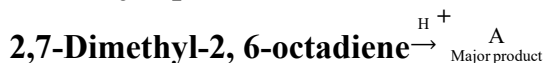
Answer: A

Solution:



Question82

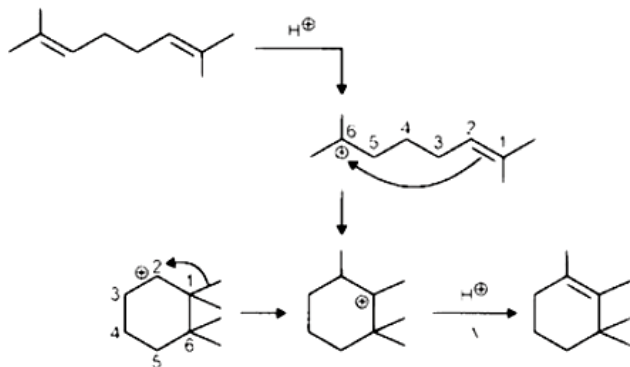
The major product 'A' of the following given reaction has _____ sp^2 hybridized carbon atoms.



[24-Jun-2022-Shift-1]

Answer: 2

Solution:



Number of sp^2 hybridised carbon atoms = 2

Question83

Given below are two statements:

Statement I : The presence of weaker π -bonds make alkenes less stable than alkanes.

Statement II : The strength of the double bond is greater than that of carbon-carbon single bond.

In the light of the above statements, choose the correct answer from the options given below:

[24-Jun-2022-Shift-2]

Options:

- A. Both Statement I and Statement II are correct.
- B. Both Statement I and Statement II are incorrect.
- C. Statement I is correct but Statement II is incorrect.
- D. Statement I is incorrect but Statement II is correct.

Answer: A

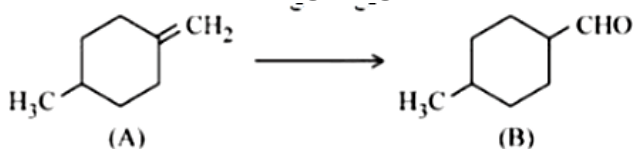
Solution:

The π -bond present in alkenes is weaker than σ bond present in alkanes. That makes alkenes less stable than alkanes. Therefore, statement-I is correct.

Carbon-carbon double bond is stronger than Carbon-carbon single bond because more energy is required to break 1 sigma and 1 pi bond than to break 1 sigma bond only. Therefore, statement-II is also correct.

Question84

Which of the following reagents / reactions will convert 'A' to 'B' ?



[24-Jun-2022-Shift-2]

Options:

A. PCC oxidation

B. Ozonolysis

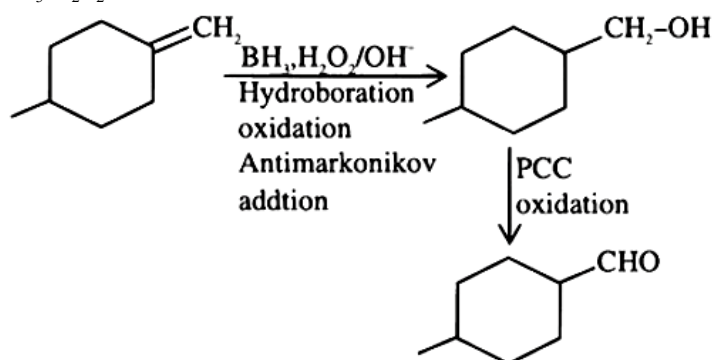
C. BH_3 , $\text{H}_2\text{O}_2/\text{OH}^-$ followed by PCC oxidation

D. HBr, hydrolysis followed by oxidation by $\text{K}_2\text{Cr}_2\text{O}_7$.

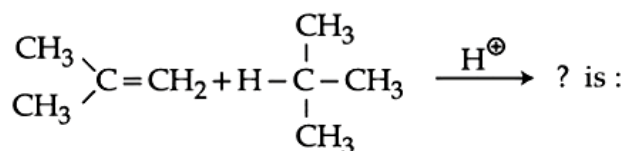
Answer: C

Solution:

BH_3 , $\text{H}_2\text{O}_2/\text{OH}^-$ followed by PCC oxidation.



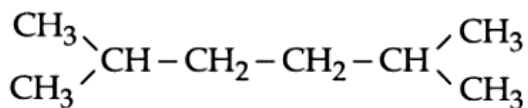
Question85



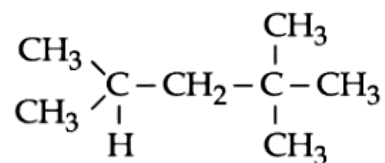
[25-Jun-2022-Shift-1]

Options:

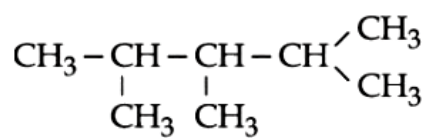
A.



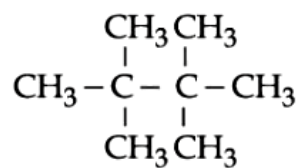
B.



C.

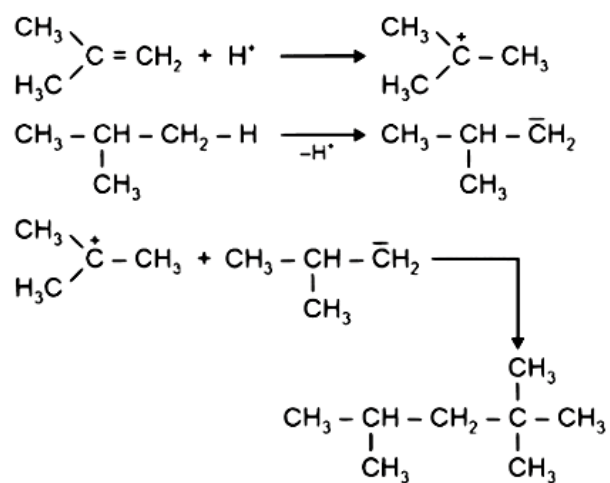


D.



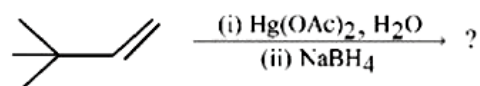
Answer: B

Solution:



Question 86

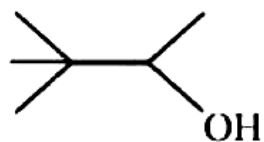
The major product in the following reaction



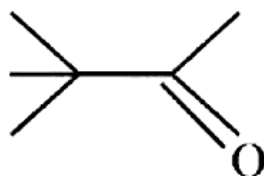
[26-Jun-2022-Shift-2]

Options:

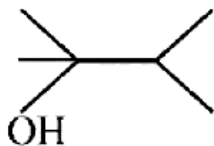
A.



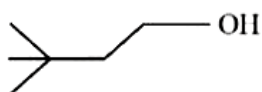
B.



C.

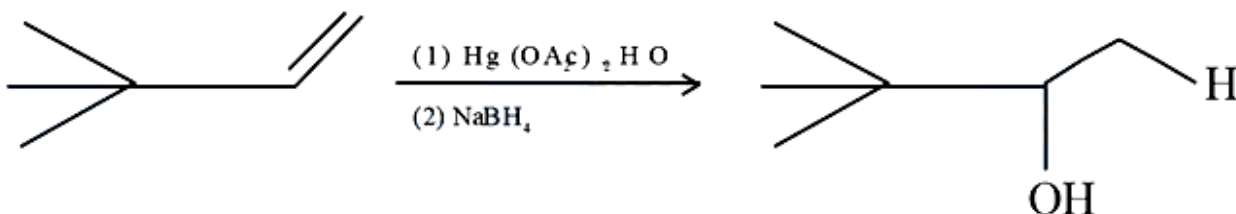


D.



Answer: A

Solution:



Oxymercuration-Demercuration

Addition of H_2O

Markovnikov's addition without rearrangement

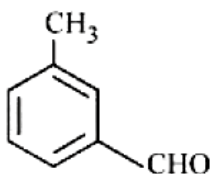
Question87

Halogenation of which one of the following will yield m-substituted product with respect to methyl group as a major product?

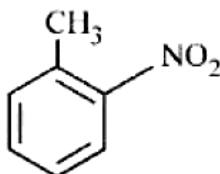
[26-Jun-2022-Shift-2]

Options:

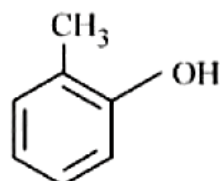
A.



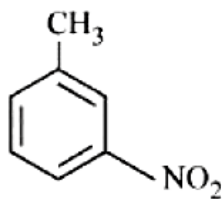
B.



C.



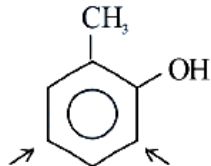
D.



Answer: C

Solution:

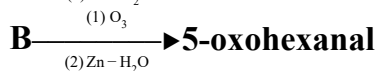
Electrophile will attack at ortho and para position with respect to better electron releasing group (ERG)



Para position with respect to $-\text{OH}$ ($+\text{R}$) group and it will be meta position with respect to $-\text{CH}_3$ group.

Question88

'A' and 'B' respectively are:



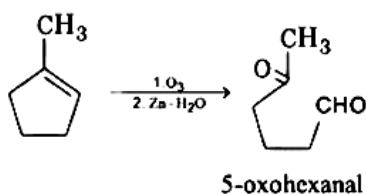
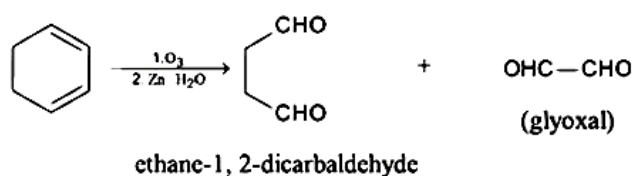
[27-Jun-2022-Shift-1]

Options:

- A. 1-methylcyclohex-1,3-diene & cyclopentene.
- B. Cyclohex-1,3-diene & cyclopentene
- C. 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene
- D. Cyclohex-1,3-diene & 1-methylcyclopent-1-ene

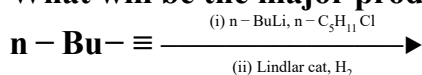
Answer: D

Solution:



Question89

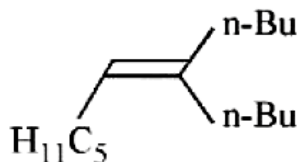
What will be the major product of following sequence of reactions?



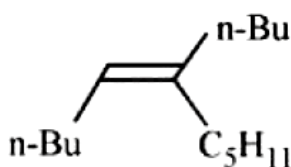
[27-Jun-2022-Shift-2]

Options:

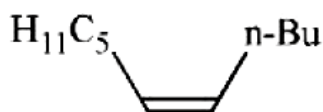
A.



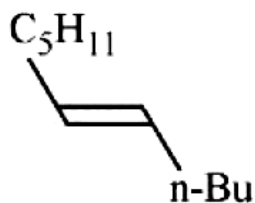
B.



C.

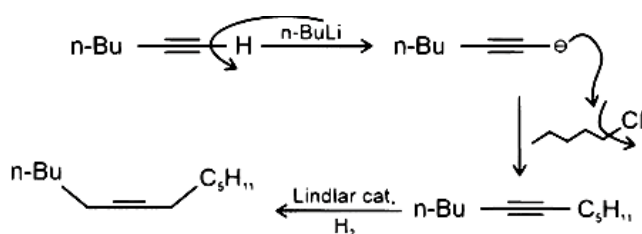


D.



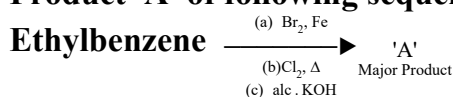
Answer: C

Solution:



Question90

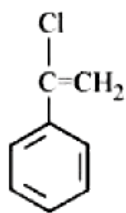
Product 'A' of following sequence of reactions is



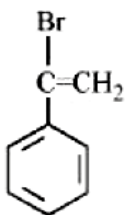
[27-Jun-2022-Shift-2]

Options:

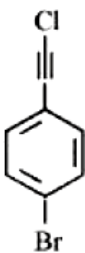
A.



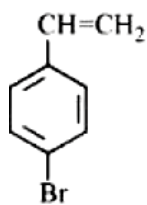
B.



C.

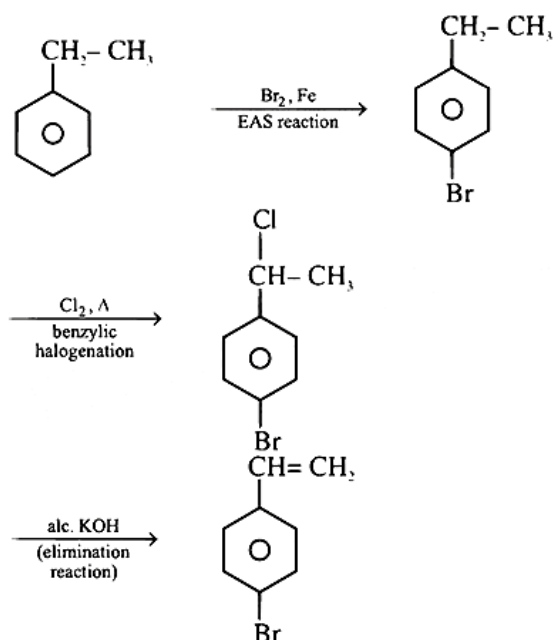


D.



Answer: D

Solution:



Question91

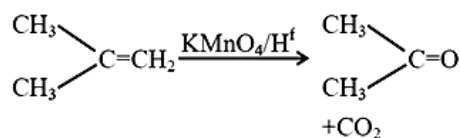
Two isomers ' A ' and ' B ' with molecular formula C_4H_8 give different products on oxidation with KMnO_4 in acidic medium. Isomer ' A ' on reaction with KMnO_4/H^+ results in effervescence of a gas and gives ketone. The compound ' A ' is
[29-Jun-2022-Shift-1]

Options:

- A. But-1-ene.
- B. cis-But-2-ene.
- C. trans-But-2-ene.
- D. 2-methyl propene.

Answer: D

Solution:



Question92

In Friedel-Crafts alkylation of aniline, one gets
[29-Jun-2022-Shift-2]

Options:

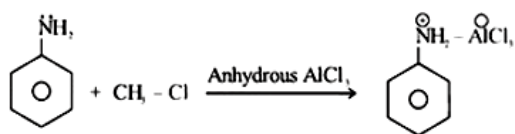
- A. alkylated product with ortho and para substitution.
- B. secondary amine after acidic treatment.

C. an amide product.

D. positively charged nitrogen at benzene ring.

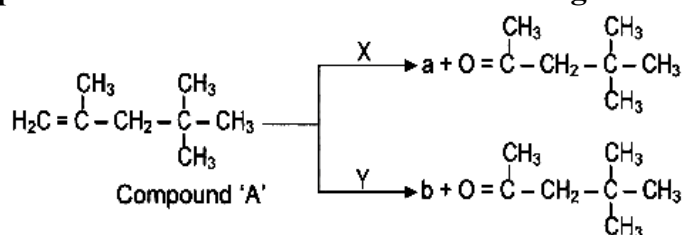
Answer: D

Solution:



Question93

A compound 'A' on reaction with 'X' and 'Y' produces the same major product but different by product 'a' and 'b'. Oxidation of 'a' gives a substance produced by ants.



'X' and 'Y' respectively are
[25-Jul-2022-Shift-1]

Options:

A. KMnO_4/H^+ and dil. $\text{KMnO}_4/273\text{K}$

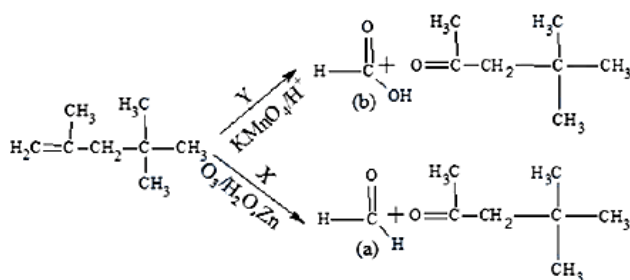
B. KMnO_4 (dilute), 273K and KMnO_4/H^+

C. KMnO_4/H^+ and $\text{O}_3, \text{H}_2\text{O}/\text{Zn}$

D. $\text{O}_3, \text{H}_2\text{O}/\text{Zn}$ and KMnO_4/H^+

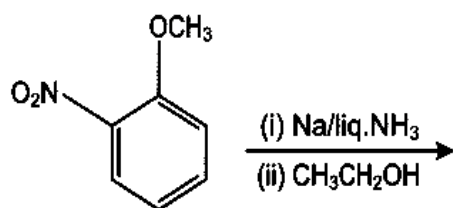
Answer: D

Solution:



Question94

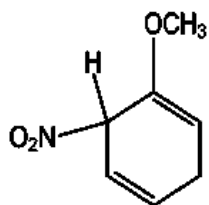
The major product of the following reaction is



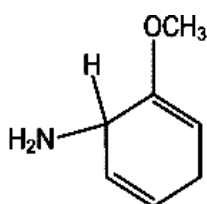
[26-Jul-2022-Shift-1]

Options:

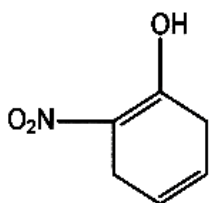
A.



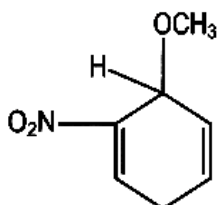
B.



C.

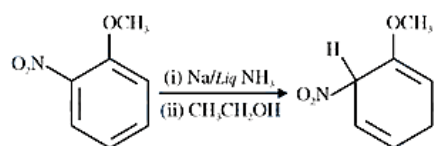


D.



Answer: A

Solution:



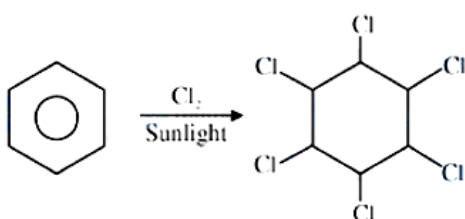
Given reaction is an example of birch reduction.

Question95

In the presence of sunlight, benzene reacts with Cl_2 to give product, X. The number of hydrogens in X is _____.
[26-Jul-2022-Shift-1]

Answer: 6

Solution:

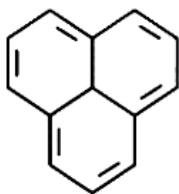


Question96

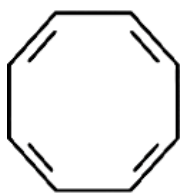
Which of the following is not an example of benzenoid compound?
[26-Jul-2022-Shift-2]

Options:

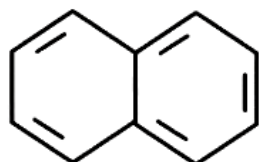
A.



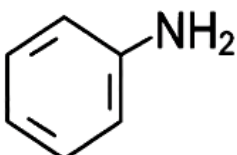
B.



C.



D.



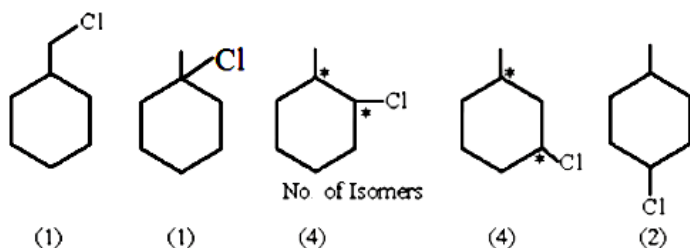
Answer: B

Question97

Total number of isomers (including stereoisomers) obtained on monochlorination of methylcyclohexane is _____.
[26-Jul-2022-Shift-2]

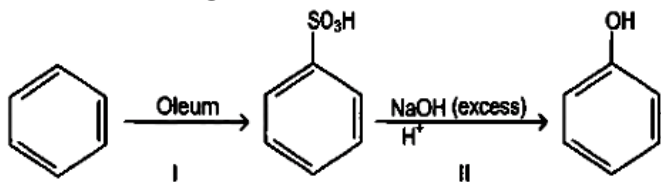
Answer: 12

Solution:



Question98

In the following reaction

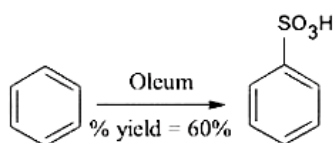


The % yield for reaction I is 60% and that of reaction II is 50%. The overall yield of the complete reaction is _____. [nearest integer]
[27-Jul-2022-Shift-1]

Answer: 30

Solution:

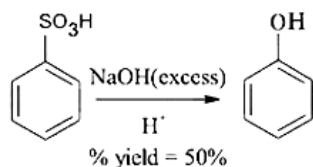
(I)



Let initial moles of reactant taken = n

Total moles obtained for benzene sulphonic acid (with % yield = 60%) = $0.6n$

(II)



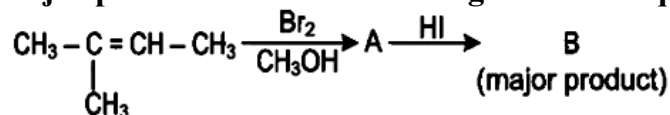
Moles of benzene sulphonic acid before reaction II = $0.6n$

Moles obtained for phenol (with % yield = 50%) = $0.6 \times 0.5n = 0.3n$

So over all % yield of complete reaction = $\frac{0.3n}{n} \times 100 = 30$

Question99

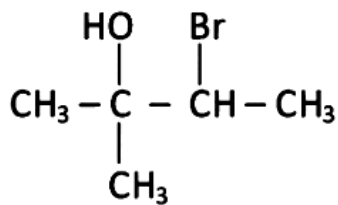
Major product B' of the following reaction sequence is :



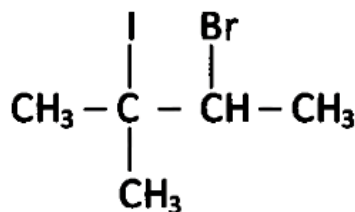
[27-Jul-2022-Shift-2]

Options:

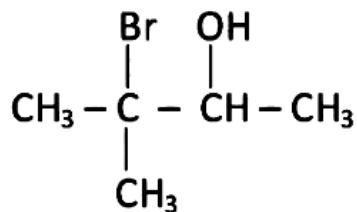
A.



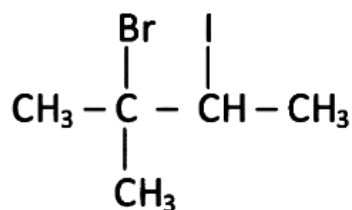
B.



C.

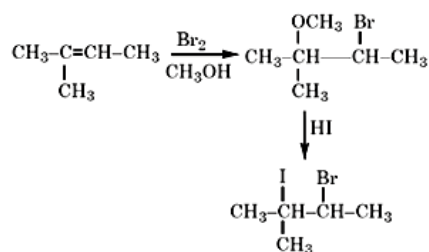


D.



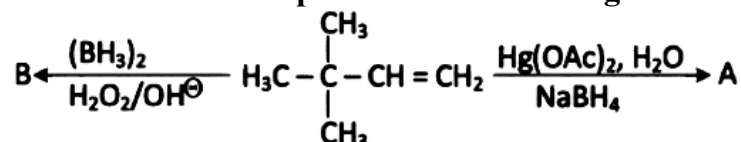
Answer: B

Solution:



Question100

Choose the correct option for the following reactions.



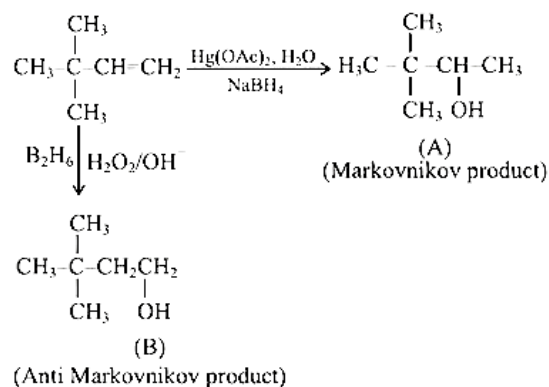
[28-Jul-2022-Shift-1]

Options:

- A. 'A' and 'B' are both Markovnikov addition products.
- B. 'A' is Markovnikov product and 'B' is anti-Markovnikov product.
- C. 'A' and 'B' are both anti-Markovnikov products.
- D. 'B' is Markovnikov and 'A' is anti-Markovnikov product.

Answer: B

Solution:



Question101

Arrange the following in increasing order of reactivity towards nitration

- A. p-xylene
- B. bromobenzene
- C. mesitylene
- D. nitrobenzene
- E. benzene

Choose the correct answer from the options given below

[28-Jul-2022-Shift-2]

Options:

A. $C < D < E < A < B$

B. $D < B < E < A < C$

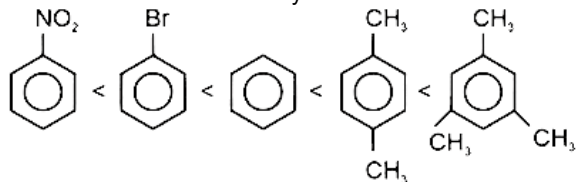
C. $D < C < E < A < B$

D. $C < D < E < B < A$

Answer: B

Solution:

The correct order of reactivity towards nitration is



as electron releasing groups on benzene ring facilitate the nitration at benzene ring.

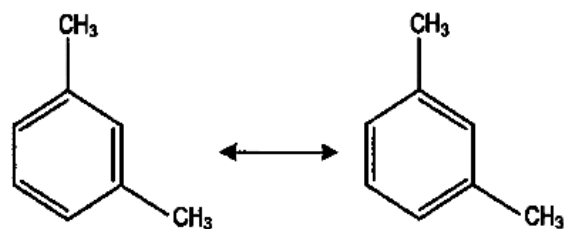
Question102

Which among the following pairs of the structures will give different products on ozonolysis?
(Consider the double bonds in the structures are rigid and not delocalized.)

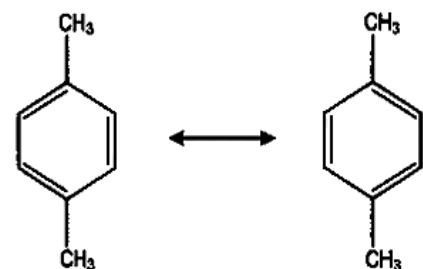
[29-Jul-2022-Shift-1]

Options:

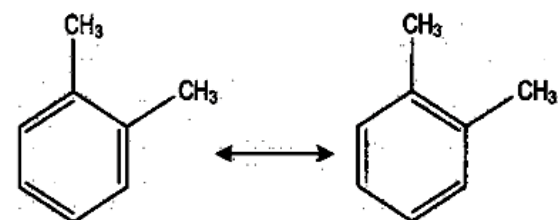
A.



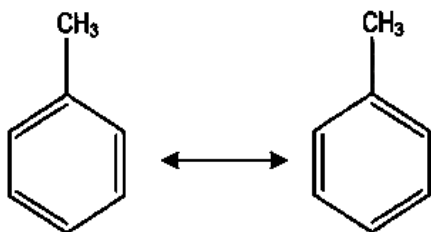
B.



C.



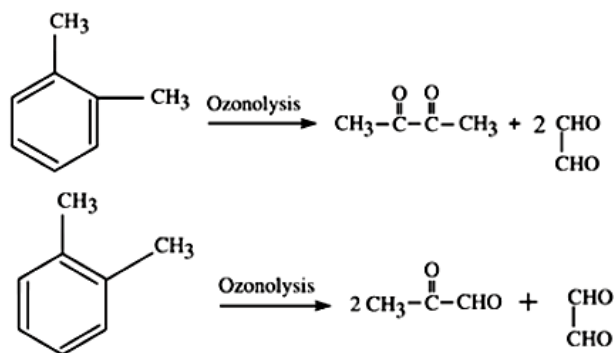
D.



Answer: C

Solution:

O-xylene has different resonating structures which will produce different ozonolysis products.
m-xylene p-xylene and toluene have identical resonating structures which will give identical ozonolysis products.

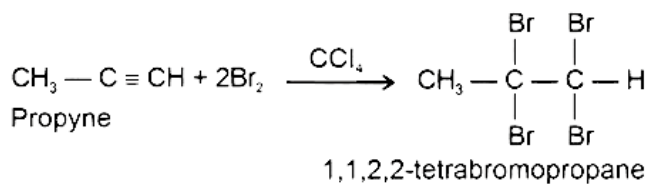


Question103

In bromination of Propyne, with Bromine, 1, 1, 2, 2-tetrabromopropane is obtained in 27\% yield. The amount of 1, 1, 2, 2-tetrabromopropane obtained from 1g of Bromine in this reaction is _____ x 10⁻¹ g. (Nearest integer)
(Molar Mass : Bromine = 80g/mol)
[29-Jul-2022-Shift-1]

Answer: 3

Solution:



2 moles Br₂ ≡ 1 mole 1,1,2,2-tetrabromopropane $\frac{1}{160}$ mole Br₂ ≡ $\frac{1}{2} \times \frac{1}{160}$ mole 1,1,2,2-tetrabromopropane

But yield of reaction is only 27%

Moles of 1,1,2,2-tetrabromopropane

$$= \frac{1}{2} \times \frac{1}{160} \times \frac{27}{100}$$

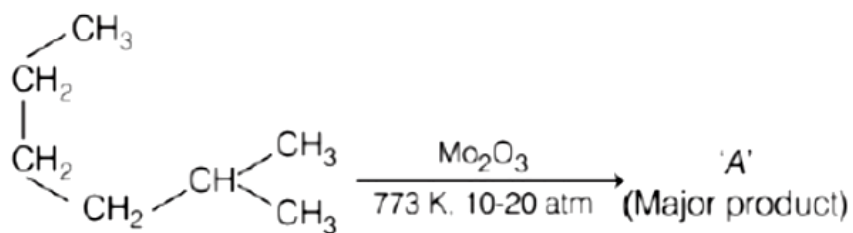
Molar mass of 1,1,2,2-tetrabromopropane = 360g

$$= \frac{1}{2} \times \frac{1}{160} \times \frac{27}{100} \times 360g$$

$$\approx 3 \times 10^{-1}g$$

Question 104

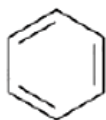
Identify A in the given chemical reaction.



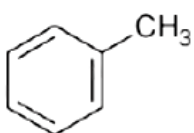
[25 Feb 2021 Shift 1]

Options:

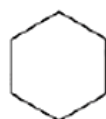
A.



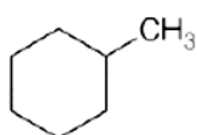
B.



C.

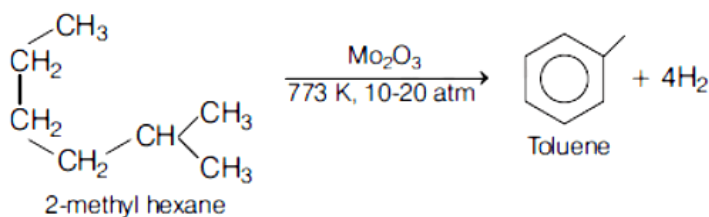


D.



Answer: B

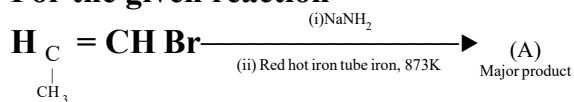
Solution:



Oxides of molybdenum, vanadium and chromium, i.e. Mo_2O_3 , V_2O_5 and Cr_2O_3 work as catalyst. When they are operated under certain temperature and pressure with n-heptane then they form toluene which is aromatic compound. Mo_2O_3 behave as aromatising agent.

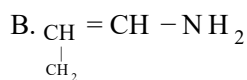
Question105

For the given reaction



[26 Feb 2021 Shift 1]

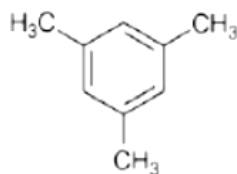
Options:



C.



D.



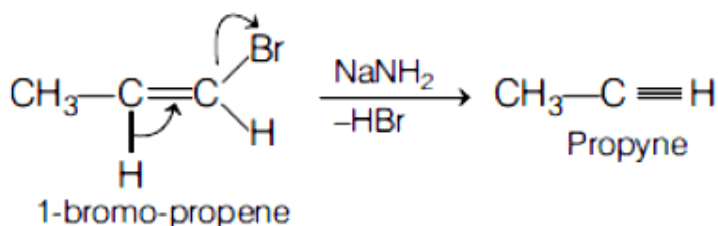
Answer: D

Solution:

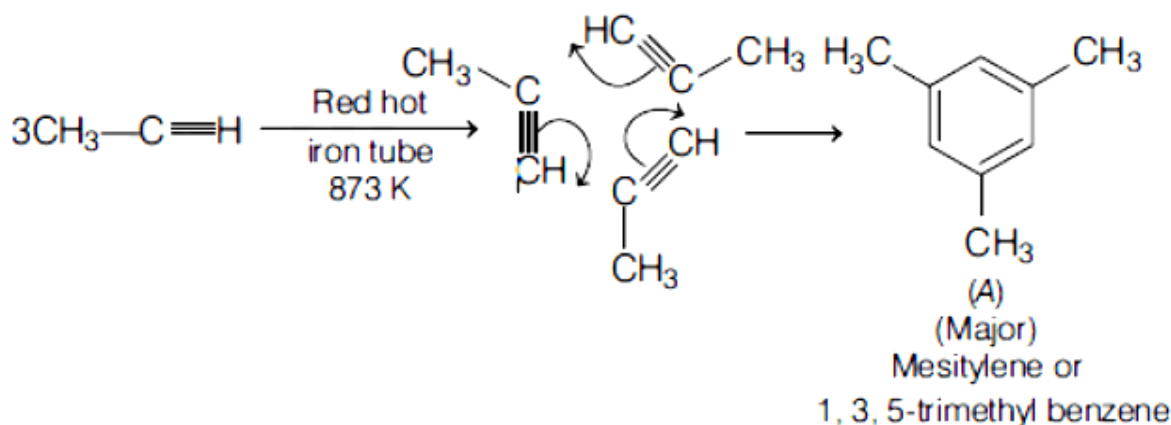
When 1-bromo-prop-1-ene reacts with NaNH_2 (a strong base), β -elimination or dehydrobromination, $(-\text{HBr})$ takes place to give propyne in next step. When propyne is passed through red hot iron tube, cyclic trimerisation (aromatisation also) occurs to produce mesitylene (A) as

major product.

Step I



Step II



Question106

The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is
[25 Feb 2021 Shift 2]

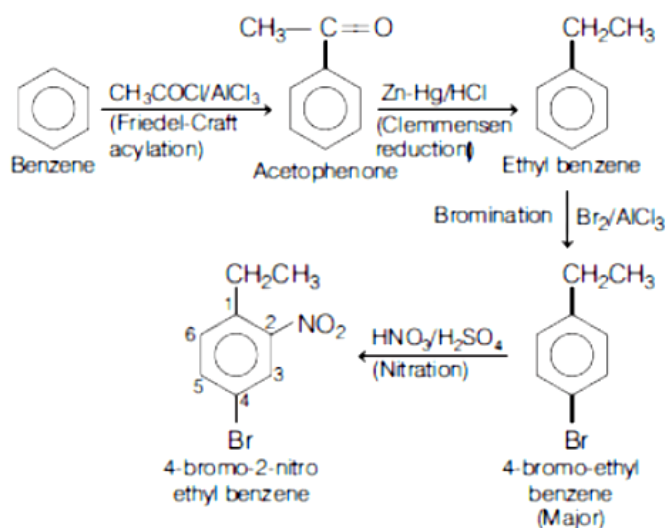
Options:

- A. $\text{CH}_3\text{COCl} / \text{AlCl}_3$, $\text{Br}_2 / \text{AlBr}_3$, $\text{HNO}_3 / \text{H}_2\text{SO}_4$, Zn / HCl
- B. $\text{CH}_3\text{COCl} / \text{AlCl}_3$, $\text{Zn} - \text{Hg} / \text{HCl}$, $\text{Br}_2 / \text{AlBr}_3$, $\text{HNO}_3 / \text{H}_2\text{SO}_4$
- C. $\text{Br}_2 / \text{AlBr}_3$, $\text{CH}_3\text{COCl} / \text{AlCl}_3$, $\text{HNO}_3 / \text{H}_2\text{SO}_4$, Zn / HCl
- D. $\text{HNO}_3 / \text{H}_2\text{SO}_4$, $\text{Br}_2 / \text{AlCl}_3$, $\text{CH}_3\text{COCl} / \text{AlCl}_3$, $\text{Zn} - \text{Hg} / \text{HCl}$

Answer: B

Solution:

Benzene on reaction with $\text{CH}_3\text{COCl} / \text{AlCl}_3$ gives acetophenone which on reduction with $\text{Zn} - \text{Hg} / \text{HCl}$ gives ethyl benzene. Bromination of ethyl benzene gives 4-bromo-ethylbenzene which upon nitration gives 4-bromo-2-nitroethyl benzene.

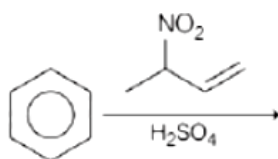


[I-effect of Br directs NO_2^- at its meta-position]

So, option (b) is the correct sequence of reagents used.

Question107

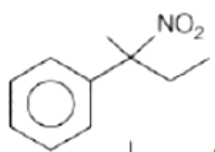
The major product of the following reaction is



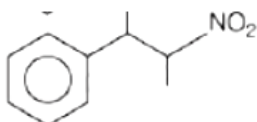
[25 Feb 2021 Shift 2]

Options:

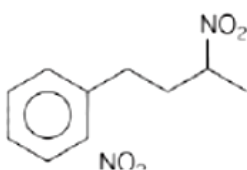
A.



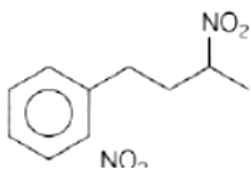
B.



C.



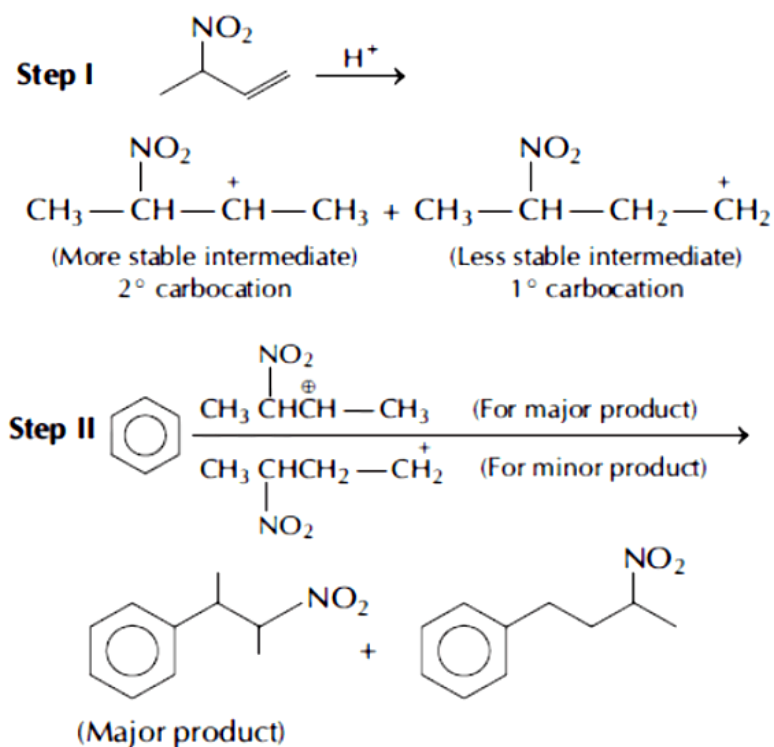
D.



Answer: B

Solution:

The reaction take place as follows:



Question108

Consider the following chemical reaction.

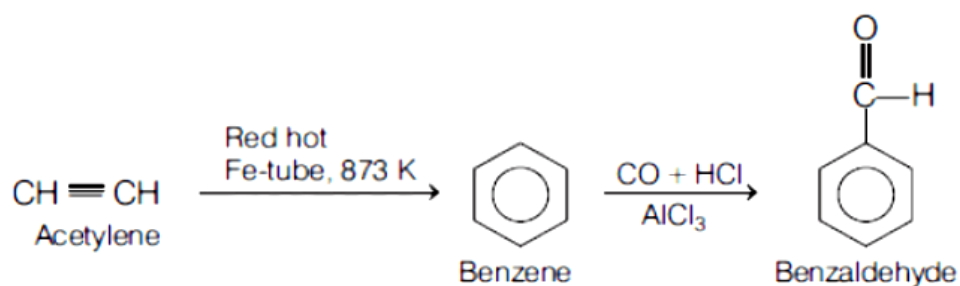
$\text{CH} \equiv \text{CH} \xrightarrow[(2) \text{ CO, HCl, AlCl}_3]{(1) \text{ Red hot Fe tube, } 873\text{K}} \text{Product}$ The number of sp^2 hybridised carbon atom(s) present in the product is

[25 Feb 2021 Shift 1]

Answer: 7

Solution:

The reaction take place as follows



Every carbon atom found in benzaldehyde (6 carbon in benzene ring +1 in aldehyde group), forms three bonds with neighbouring atoms- (one double bond + two single bond). Therefore, in benzaldehyde total number of $\text{sp}^2\text{C}'$ are 7 .

Question109

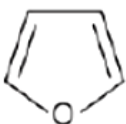
Which one of the following compounds is non-aromatic ?
 [24 Feb 2021 Shift 2]

Options:

A.



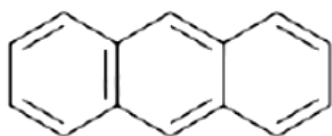
B.



C.



D.

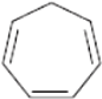
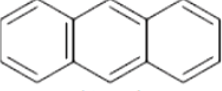


Answer: A

Solution:

Compound, which obey Huckel rule, $(4n + 2)\pi$ is called aromatic compound.
 Compound, which obey $4n\pi$ rule, is called anti-aromatic compound.

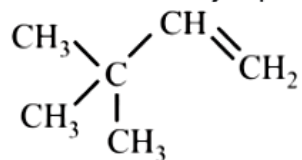
Compound which contain one or more sp^3 carbon in its structure is called non-aromatic compound. The nature of following ions/compounds are as follows :

- (a)  sp^3 Carbon $\rightarrow$ Non-aromatic
- (b)  ($6e^-$) $\rightarrow$ Aromatic (follow Huckel's rule)
- (c)  ($2e^-$) $\rightarrow$ Aromatic (follow Huckel's rule)
- (d)  $\rightarrow$ Anthracene $\rightarrow$ Aromatic ($14e^-$)

Only (a) is non-aromatic and rest all are aromatic.

Question110

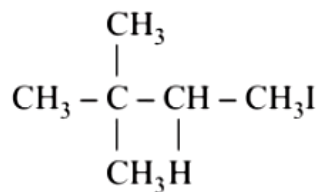
What is the major product formed by HI on reaction with



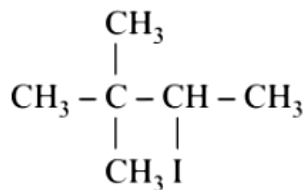
[24feb2021shift1]

Options:

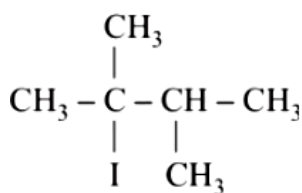
A.



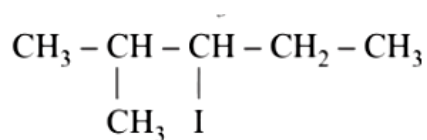
B.



C.

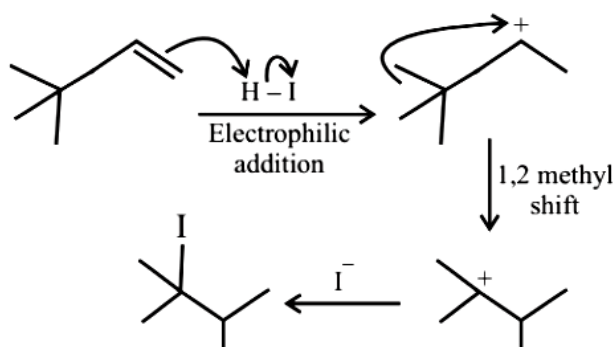


D.



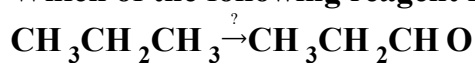
Answer: C

Solution:



Question111

Which of the following reagent is used for the following reaction?



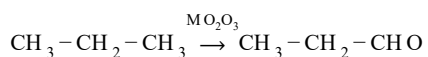
[24feb2021shift1]

Options:

- A. Manganese acetate
- B. Copper at high temperature and pressure
- C. Molybdenum oxide
- D. Potassium permanganate

Answer: C

Solution:



Question112

Given below are two statements.

Statement-I 2-methylbutane on oxidation with KMnO_4 gives 2-methylbutan-2-ol.

Statement-II n-alkanes can be easily oxidised to corresponding alcohols with KMnO_4 .

Choose the correct option.

[17 Mar 2021 Shift 2]

Options:

- A. Both statements I and II are correct.
- B. Both statements I and II are incorrect.
- C. Statement I is correct but statement II is incorrect.
- D. Statement I is incorrect but statement II is correct.

Answer: C

Solution:

Statement I is correct, whereas statement II is incorrect. Corrected statement is as follows: Tertiary alkanes can be oxidised to alcohol by treating with KMnO_4 . n-alkanes are very less reactive and have no reaction with KMnO_4 . n-alkanes $\xrightarrow{\text{KMnO}_4}$ No reaction

Question113

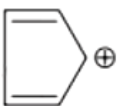
Which of the following is an aromatic compound?
[17 Mar 2021 Shift 1]

Options:

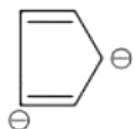
A.



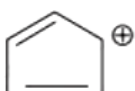
B.



C.

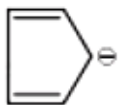


D.



Answer: A

Solution:



is an aromatic compound ($6\pi e^-$).

Aromatic compound possess following characteristics:

(i) Planarity

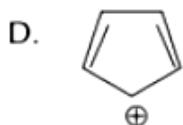
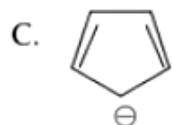
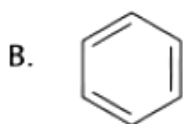
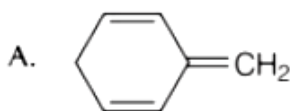
(ii) Complete delocalisation of the π -electrons in the ring.

(iii) Presence of $(4n + 2)\pi$ -electrons in the ring,

where n is an integer ($n = 0, 1, 2, \dots$). That compound follow all aromatic characteristics.

Question114

Among the following, the aromatic compounds are



Choose the correct answer from the following options.
[16 Mar 2021 Shift 1]

Options:

A. Only (A) and (B)

B. Only (B) and (C)

C. (B),(C) and (D)

D. (A),(B) and (C)

Answer: B

Solution:

(b) For aromatic compounds following conditions should be satisfied

(i) planarity

(ii) complete delocalisation of π -electrons in the ring.

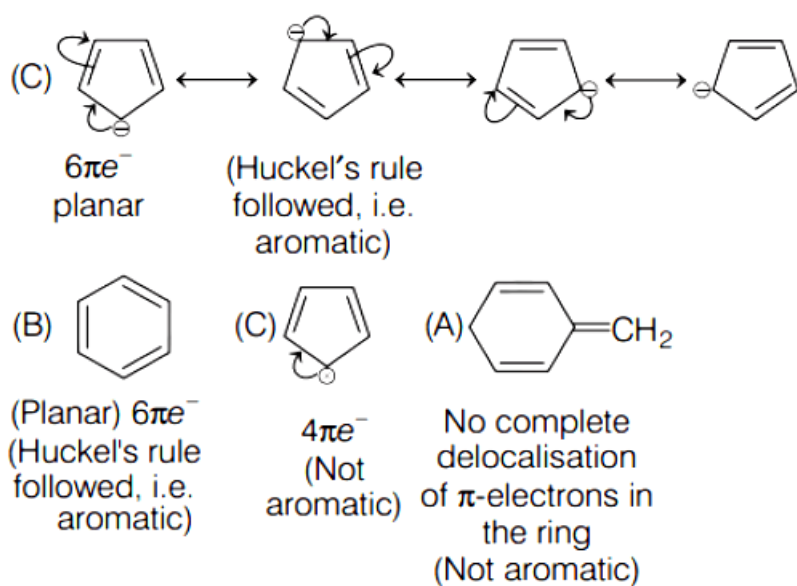
(iii) presence of $(4n + 2)\pi$ -electrons in the ring, where $n = 0, 1, 2, \dots$

This is called Huckel rule.

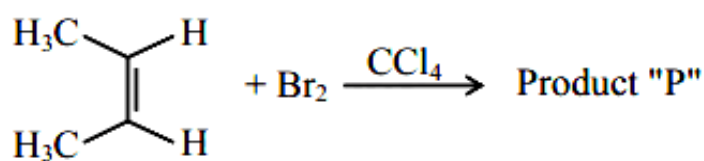
e.g. $n = 0, 2\pi e^-$

$n = 1, 6\pi e^-$

$n = 2, 10\pi e^-$



Question115

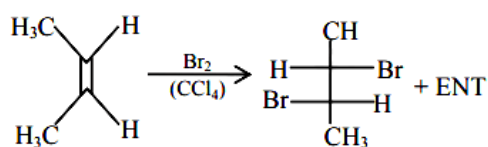


Consider the above chemical reaction. The total number of stereoisomers possible for Product 'P' is _____ .

[25 Jul 2021 Shift 2]

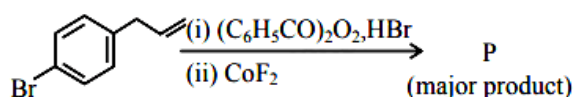
Answer: 2

Solution:



The total number of products possible = 2

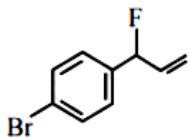
Question116



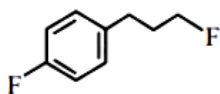
Major product P of above reaction, is :
[20 Jul 2021 Shift 2]

Options:

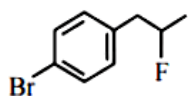
A.



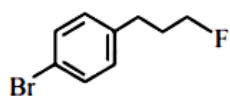
B.



C.

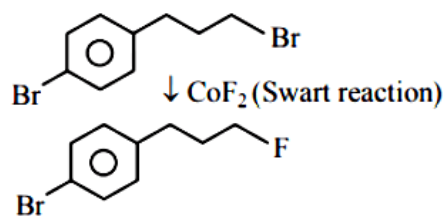
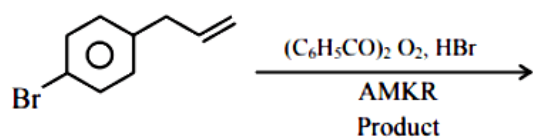


D.



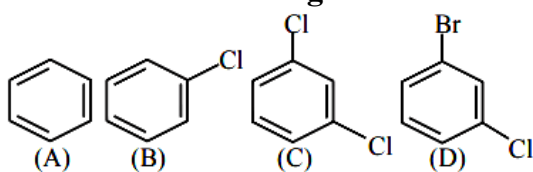
Answer: D

Solution:



Question117

The correct decreasing order of densities of the following compounds is :



[25 Jul 2021 Shift 2]

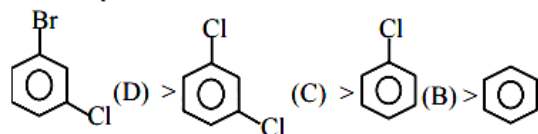
Options:

- A. (D) > (C) > (B) > (A)
- B. (C) > (D) > (A) > (B)
- C. (C) > (B) > (A) > (D)
- D. (A) > (B) > (C) > (D)

Answer: A

Solution:

The density order



Question118

**Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where :
[20 Jul 2021 Shift 2]**

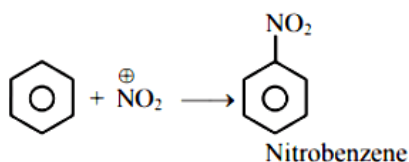
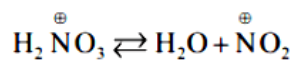
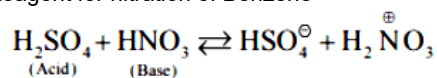
Options:

- A. both H_2SO_4 and HNO_3 act as a bases
- B. HNO_3 acts as an acid and H_2SO_4 acts as a base
- C. both H_2SO_4 and HNO_3 act as an acids
- D. HNO_3 acts as a base and H_2SO_4 acts as an acid

Answer: D

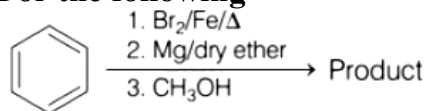
Solution:

Reagent for nitration of Benzene



Question119

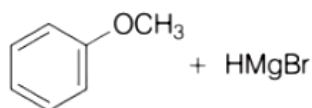
For the following



[31 Aug 2021 Shift 2]

Options:

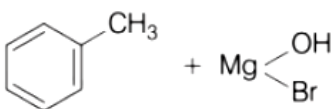
A.



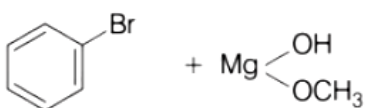
B.



C.



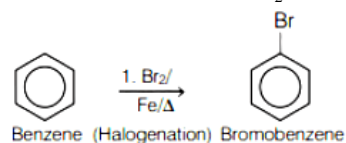
D.



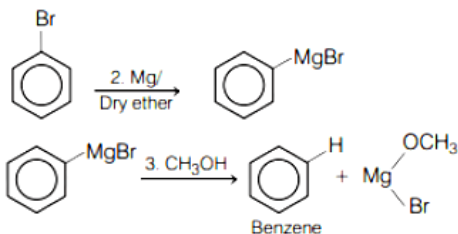
Answer: B

Solution:

Benzene on reaction with Br_2/Fe give 5 bromo benzene as follows



Bromobenzene on reaction with Mg / dry ether and CH_3OH gives benzene as follows



Question120

Which one of the following compounds is not aromatic?

[26 Aug 2021 Shift 2]

Options:

A.



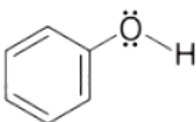
B.



C.



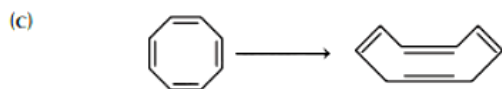
D.



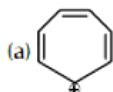
Answer: C

Solution:

The compound which follows Huckel's rule has planar structure is aromatic in nature.

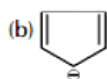


It exist in a non-planar tube shape, so it is non-aromatic.

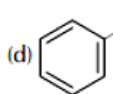


It is aromatic in nature.

Number of $\pi e^- = 4n + 2$ ($n = 2$) = $6 \pi e^-$



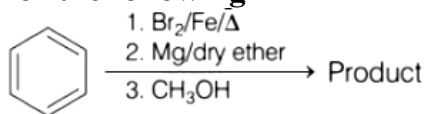
It is also aromatic in nature as the lone pair is in conjugation with πe^- .



It is aromatic in nature as it contains benzene ring.

Question121

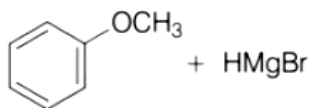
For the following



[31 Aug 2021 Shift 2]

Options:

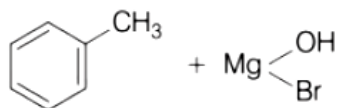
A.



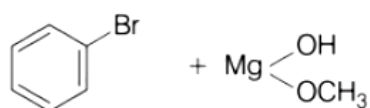
B.



C.



D.



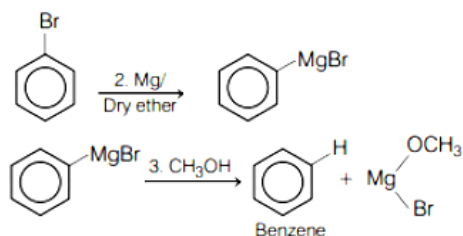
Answer: B

Solution:

Benzene on reaction with Br_2/Fe give 5 bromo benzene as follows



Bromobenzene on reaction with Mg / dry ether and CH_3OH gives benzene as follows



Question122

Which one of the following compounds is not aromatic?
[26 Aug 2021 Shift 2]

Options:

A.



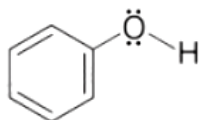
B.



C.



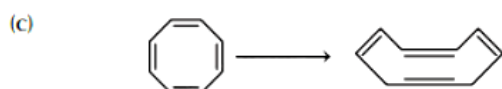
D.



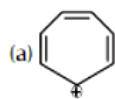
Answer: C

Solution:

The compound which follows Huckel's rule has planar structure is aromatic in nature.

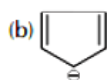


It exist in a non-planar tube shape, so it is non-aromatic.

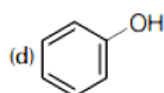


It is aromatic in nature.

Number of $\pi e^- = 4n + 2$ ($n = 2$) = 6 πe^-



It is also aromatic in nature as the lone pair is in conjugation with πe^- .



It is aromatic in nature as it contains benzene ring.

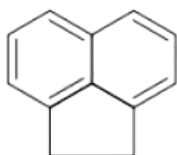
Question123

Which one of the following compounds is aromatic in nature ?

[1 Sep 2021 Shift 2]

Options:

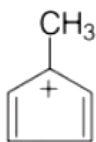
A.



B.



C.



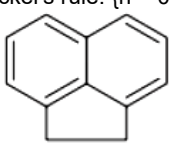
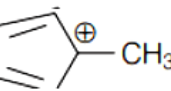
D.



Answer: D

Solution:

Aromatic compounds are conjugated planar ring system having delocalised π - electron clouds and follows $(4n + 2)$ π - electron rule, i.e. Huckel's rule. $\{n = 0, 1, 2, \dots\}$

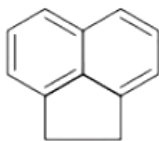
- (a)  $\rightarrow 10\pi e^-$ in cyclic conjugation place.
It is aromatic.
- (b)  $\rightarrow 4\pi e^-$ in cyclic conjugation
It is anti-aromatic.
- (c)  $\rightarrow 4\pi e^-$ in cyclic conjugation.
It is anti-aromatic.
- (d)  $\rightarrow 6\pi e^-$ in cyclic conjugation.
It is aromatic.

Question 124

Which one of the following compounds is aromatic in nature ?
[1 Sep 2021 Shift 2]

Options:

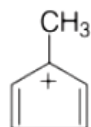
A.



B.



C.



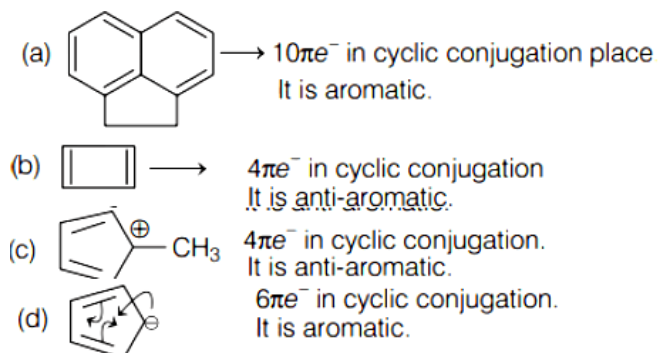
D.



Answer: D

Solution:

Aromatic compounds are conjugated planar ring system having delocalised π - electron clouds and follows $(4n + 2)$ π - electron rule, i.e. Huckel's rule. $\{n = 0, 1, 2, \dots\}$



Question125

Experimentally reducing a functional group cannot be done by which one of the following reagents ?

[1 Sep 2021 Shift 2]

Options:

- A. Pt - C/H₂
- B. Na/H₂
- C. Pd - C/H₂
- D. Zn/H₂O

Answer: B

Solution:

A functional group cannot be reduced by Na/H₂ as it is not a reducing agent.

Question126

The correct order of heat of combustion for following alkadienes is:



[Jan. 09, 2020(I)]

Options:

- A. (A) < (B) < (C)
- B. (A) < (C) < (B)
- C. (C) < (B) < (A)
- D. (B) < (C) < (A)

Answer: A

Solution:

In isomers of hydrocarbon heat of combustion is inversely proportional to the stability.

Stability order : A > B > C

Order of heat of combustion : A < B < C

Question127

Which of the following has the shortest C-Cl bond?

[Jan. 09,2020 (II)]

Options:

- A. $\text{Cl} - \text{CH} = \text{CH}_2$
- B. $\text{Cl} - \text{CH} = \text{CHNO}_2$
- C. $\text{Cl} - \text{CH} = \text{CH} - \text{CH}_3$
- D. $\text{Cl} - \text{CH} = \text{CH} - \text{OCH}_3$

Answer: B

Solution:

In $\text{Cl}-\text{CH}=\text{CH}-\text{NO}_2$ double bond character in carbonchlorine bond is maximum due to resonance and so the bond length is shortest.

Question128

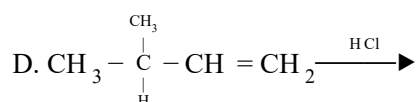
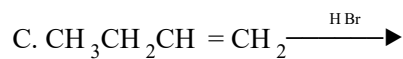
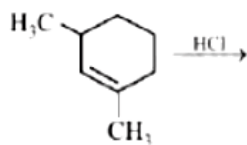
Which of the following reactions will not produce a racemic product?

[Jan. 09, 2020 (II)]

Options:

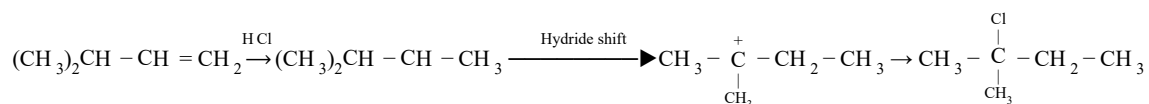


B.



Answer: D

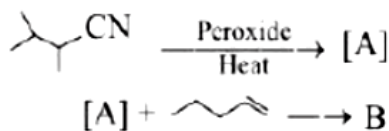
Solution:



There is no chiral carbon in the product of this reaction.

Question129

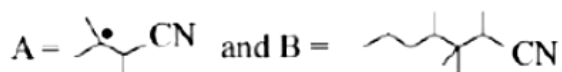
The major products A and B in the following reactions are:



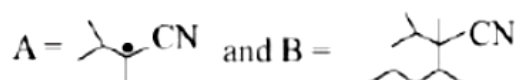
[Jan. 08,2020 (I)]

Options:

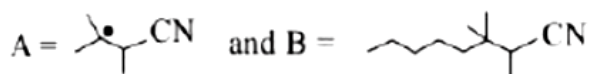
A.



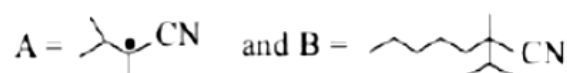
B.



C.

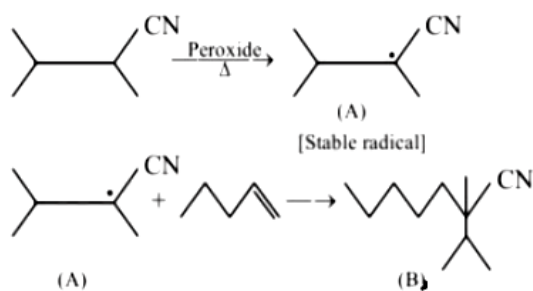


D.



Answer: A

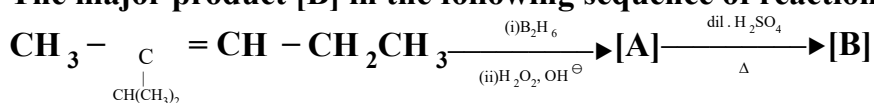
Solution:



(A) is more stable radical and undergoes Markovnikov addition to form (B).

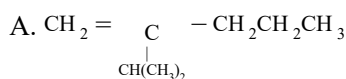
Question130

The major product [B] in the following sequence of reactions is:

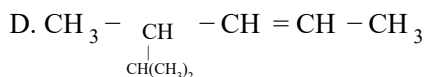
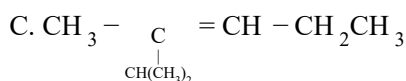


[Jan. 08,2020 (II)]

Options:

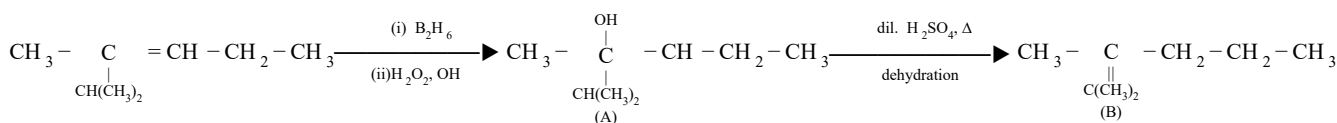


B.



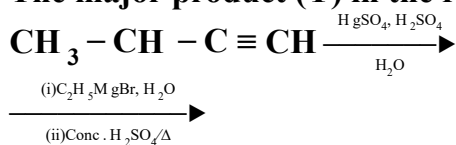
Answer: B

Solution:



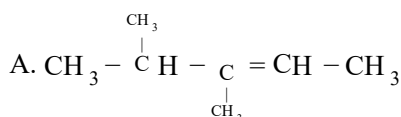
Question131

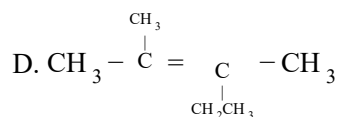
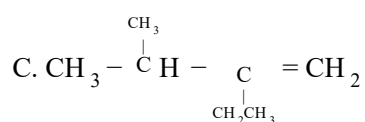
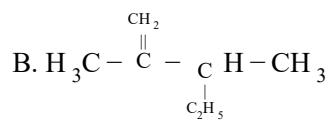
The major product (Y) in the following reactions is:



[Jan. 09, 2020 (I)]

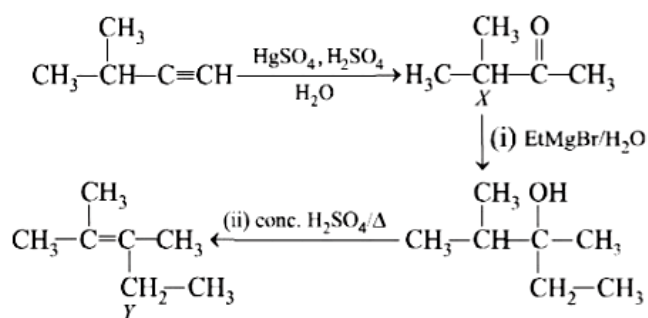
Options:





Answer: D

Solution:



Question132

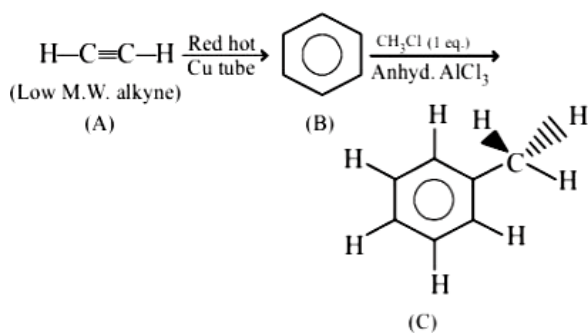
In the following sequence of reactions the maximum number of atoms present in molecule 'C' in one plane is _____. A $\xrightarrow[\text{Cu tube}]{\text{Red hot}}$ B $\xrightarrow[\text{Anhydrous AlCl}_3]{\text{CH}_3\text{Cl (1 eq.)}}$

(A is a lowest molecular weight alkyne)

[NV. Jan. 08,2020 (II)]

Answer: 13

Solution:



Question133

The number of sp^2 hybrid orbitals in a molecule of benzene is:
[Jan. 09,2020 (II)]

Options:

- A. 24
- B. 6
- C. 18
- D. 12

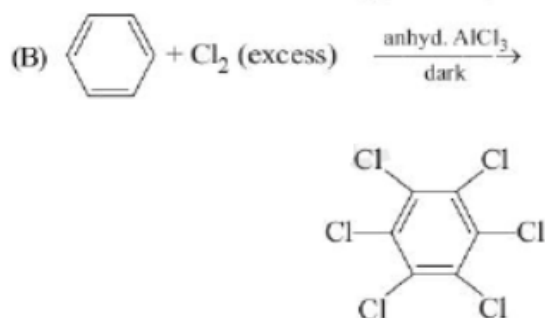
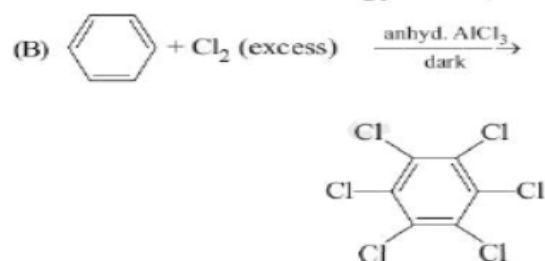
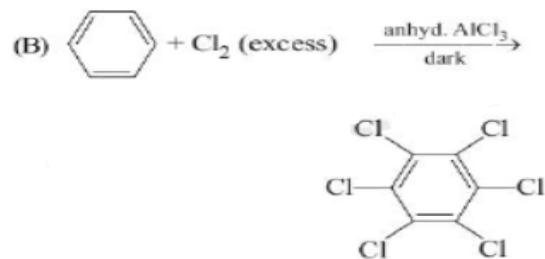
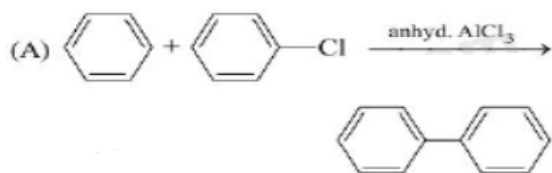
Answer: C

Solution:

In benzene each carbon atom is sp^2 hybridised. Therefore total $18sp^2$ hybrid orbitals are present in benzene.

Question134

Consider the following reactions:



Which of these reactions are possible?
[Jan. 07, 2020(II)]

Options:

- A. (A) and (B)
- B. (A) and (D)
- C. (B), (C) and (D)
- D. (B) and (D)

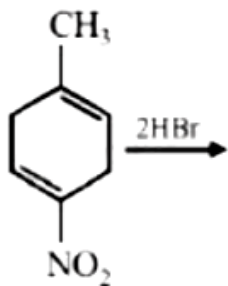
Answer: D

Solution:

Vinyl halides and aryl halides are unreactive towards Friedel Craft's reaction. Therefore reactions (A) and (C) are not possible.

Question135

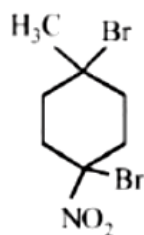
The major product of the following reaction is :



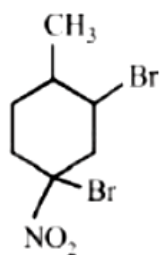
[Sep. 06, 2020(I)]

Options:

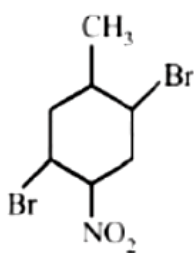
A.



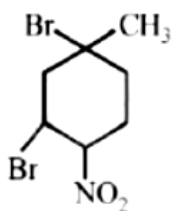
B.



C.

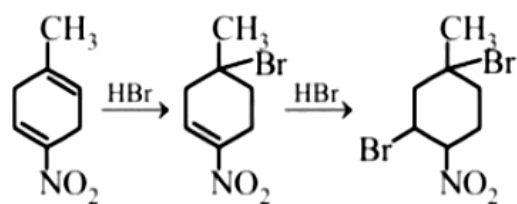


D.



Answer: D

Solution:



Question136

Which of the following compounds shows geometrical isomerism?
[Sep. 06,2020(I)]

Options:

A. 2 -methylpent-2-ene

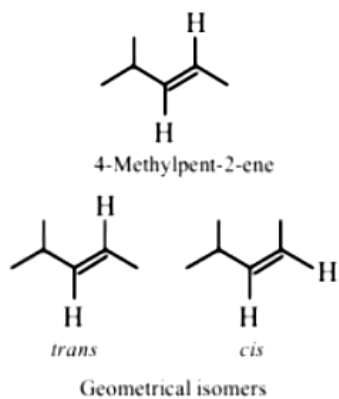
B. 4 -methylpent-2-ene

C. 4-methylpent-1-ene

D. 2-methylpent-1-ene

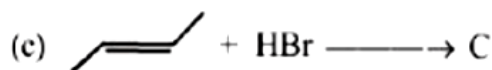
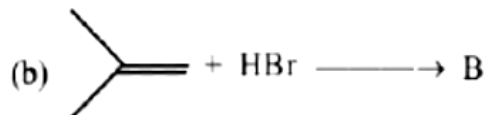
Answer: B

Solution:



Question137

The increasing order of the boiling point of the major products A, B and C of the following reactions will be:



[Sep .06,2020(II)]

Options:

A. $B < C < A$

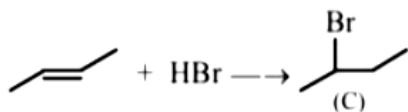
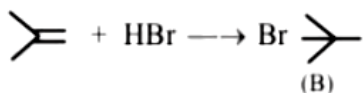
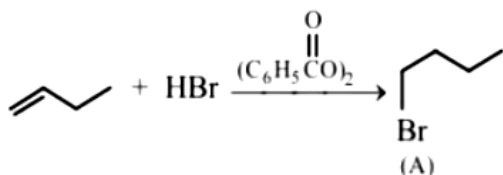
B. $C < A < B$

C. $A < B < C$

D. $A < C < B$

Answer: A

Solution:



The boiling points of isomeric haloalkanes decrease with increase in branching.
So order of B.P. is $A > C > B$.

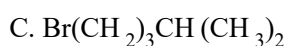
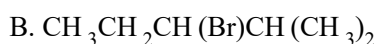
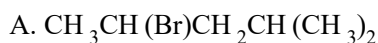
Question 138

The major product formed in the following reaction is:



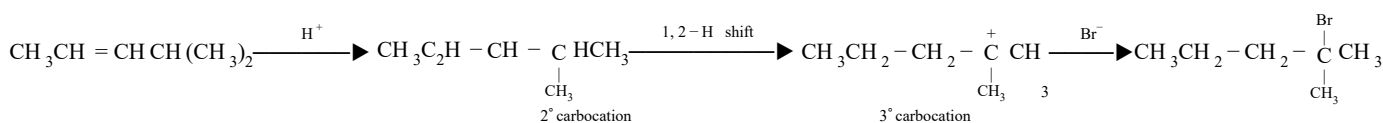
[Sep. 05, 2020(II)]

Options:



Answer: D

Solution:



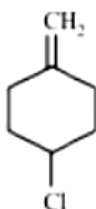
Question 139

Among the following compounds, geometrical isomerism is exhibited by:

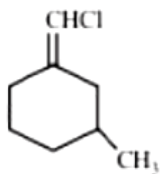
[Sep. 05, 2020 (II)]

Options:

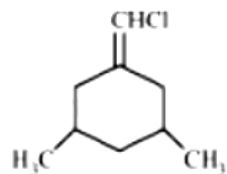
A.



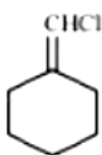
B.



C.

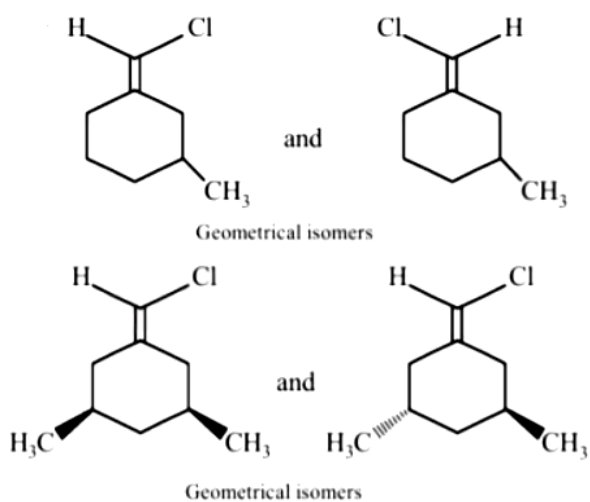


D.



Answer: C

Solution:



Question140

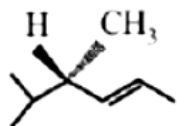
Which of the following compounds produces an optically inactive compound on hydrogenation?
[Sep. 03,2020 (I)]

Options:

A.



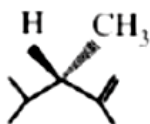
B.



C.

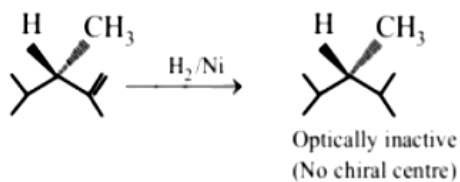


D.



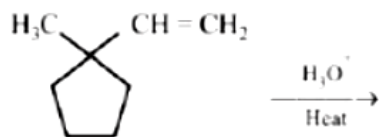
Answer: D

Solution:



Question141

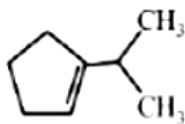
The major product in the following reaction is:



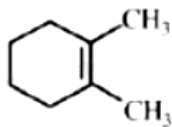
[Sep. 02,2020(I)]

Options:

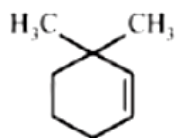
A.



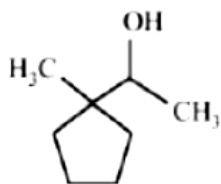
B.



C.

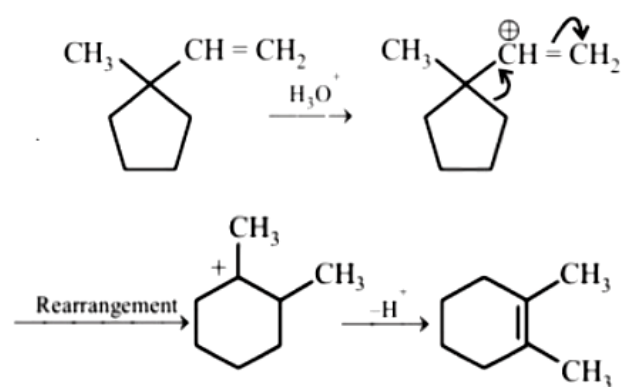


D.



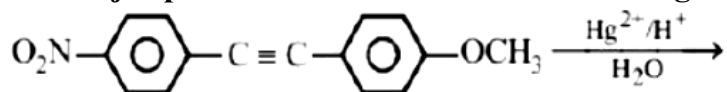
Answer: B

Solution:



Question142

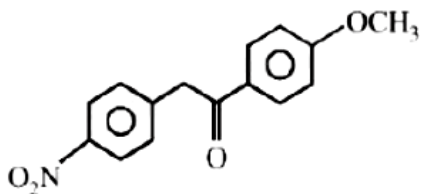
The major product obtained from the following reaction is:



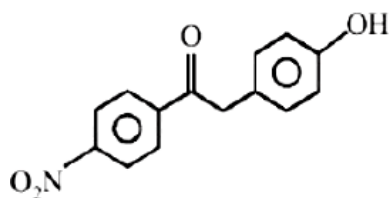
[Sep. 06,2020 (I)]

Options:

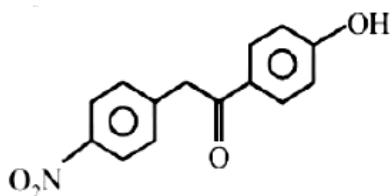
A.



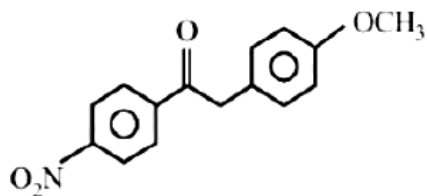
B.



C.

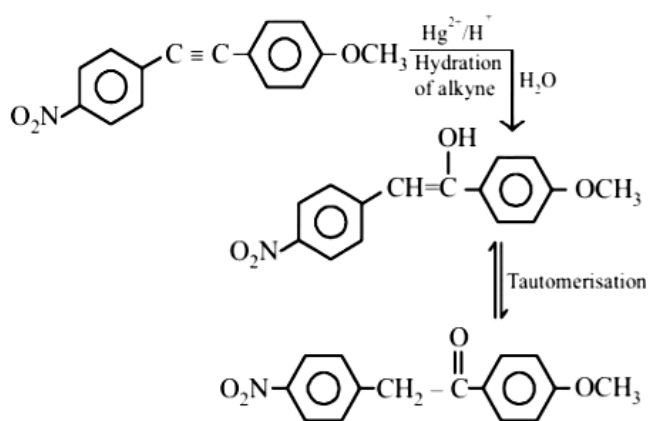


D.



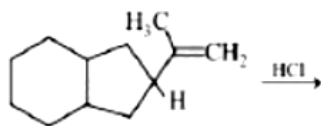
Answer: A

Solution:



Question143

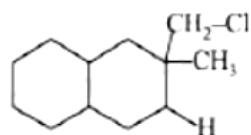
The major product of the following reaction is :



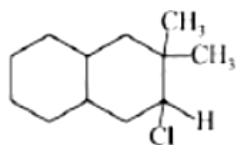
[Jan. 12, 2019 (II)]

Options:

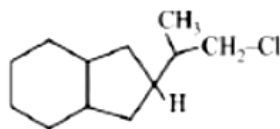
A.



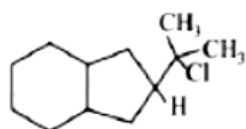
B.



C.

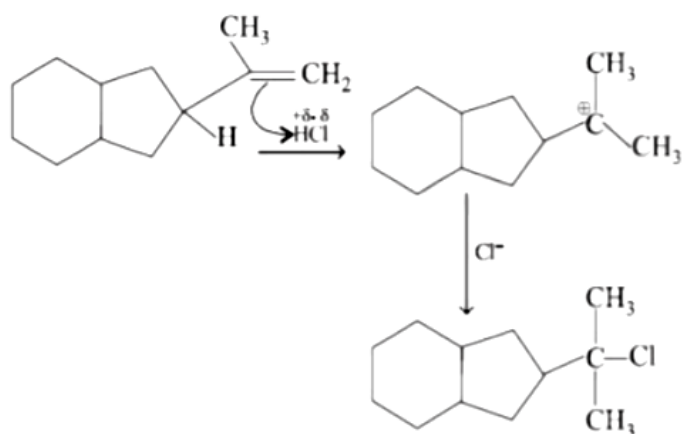


D.



Answer: D

Solution:



Question144

The correct order for acid strength of compounds $\text{CH} \equiv \text{CH}$, $\text{CH}_3 - \text{C} \equiv \text{CH}$ and $\text{CH}_2 = \text{CH}_2$ is as follows:

[Jan. 12, 2019 (I)]

Options:

A. $\text{CH} = \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH}$

B. $\text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2$

C. $\text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{H} \text{C} = \text{CH}$

D. $\text{H} \text{C} = \text{CH} > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2$

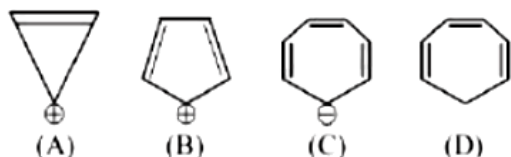
Answer: D

Solution:

Alkynes having sp -hybridised carbon has max. s character and hence max. electronegativity with the result it attracts the electron pair of the bond $C-H$ to a greater extent and makes the removal of proton easier than the alkenes having sp^2 hybridised carbon. Now presence of electron donating group ($-CH_3$) in alkyne $CH_3-C \equiv CH$ decreases acidic strength of the compound. Hence the correct order of acidic strength is
 $H-C \equiv CH > H_3C-C \equiv CH > CH_2=CH_2$

Question145

Which compound (s) out of the following is\are not aromatic?



[Jan. 11, 2019 (I)]

Options:

- A. , (C) and (D)
- B. (C) and (D)
- C. (B)
- D. (A) and (C)

Answer: A

Solution:



is aromatic; $(4 \times 0 + 2) = 2\pi e^-$ in conjugation



and



$4\pi e^-$

$8\pi e^-$

$(4 \times 1)\pi e^-$ $(4 \times 2)\pi e^-$

are antiaromatic



is non aromatic due to the presence of sp^3 carbon.

Question146

Which of the following compounds is not aromatic?

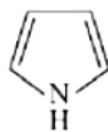
[Jan. 9, 2019 (II)]

Options:

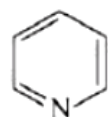
- A.



B.



C.



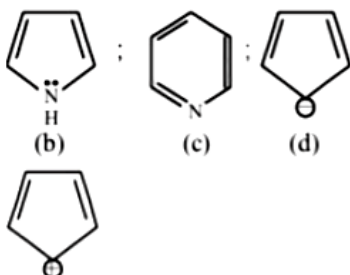
D.



Answer: A

Solution:

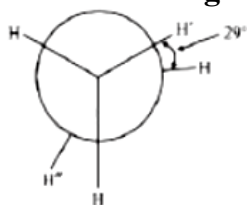
Compounds (b), (c) and (d) are containing $6\pi e^-$ in complete conjugation and are aromatic.



Compound (a) is anti-aromatic as it has $4\pi e^-$ in complete conjugation and doesn't obey Hückel rule.

Question147

In the following skew conformation of ethane, $H' - C - C - H$ 3 dihedral angle is :



[April 12, 2019 (II)]

Options:

A. 58°

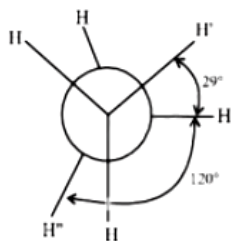
B. 149°

C. 151°

D. 120°

Answer: B

Solution:



$\therefore$ Angle between H' and $H''' = 120^\circ + 29^\circ = 149^\circ$

Question148

25g of an unknown hydrocarbon upon burning produces 88g of CO_2 and 9g of H_2O . This unknown hydrocarbon contains:
[April 12, 2019 (II)]

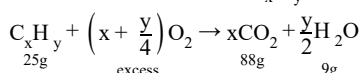
Options:

- A. 20g of carbon and 5g of hydrogen
- B. 22g of carbon and 3g of hydrogen
- C. 24g of carbon and 1g of hydrogen
- D. 18g of carbon and 7g of hydrogen

Answer: C

Solution:

Let the hydrocarbon be C_xH_y



$$2\text{mol} \quad \frac{1}{2}\text{mol}$$

$$x = 2 \quad y = 1$$

$\therefore x = 2$ and $y = 1$, the hydrocarbon will be $(\text{C}_2\text{H})_n$

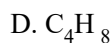
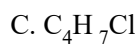
2 mol carbon contains 24g and 1 mol hydrogen contains 1g.

Question149

At 300K and 1 atmospheric pressure, 10mL of a hydrocarbon required 55mL of O_2 for complete combustion, and 40mL of CO_2 is formed. The formula of the hydrocarbon is:
[April 10, 2019, (I)]

Options:

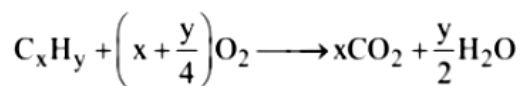
- A. C_4H_{10}
- B. C_4H_6



Answer: B

Solution:

Let the hydrocarbon be C_xH_y



Before combustion:	10 mL	55 mL	0
After combustion:	0	$55 - 10\left(x + \frac{y}{4}\right)$	10 x

Volume of CO_2 , $10x = 40$; $x = 4$

Volume of CO_2 , $10x = 40$; $x = 4$

$55 - 10\left(x + \frac{y}{4}\right) = 0$; $y = 6$

$\therefore$ Hydrocarbon is C_4H_6

Question150

Which of these factors does not govern the stability of a conformation in acyclic compounds?
[April 10,2019, (II)]

Options:

- A. Steric interactions
- B. Torsional strain
- C. Electrostatic forces of interaction
- D. Angle strain

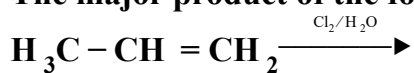
Answer: D

Solution:

In a cyclic or open chain compounds, angle strain is absent.

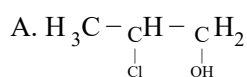
Question151

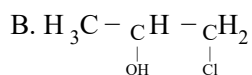
The major product of the following addition reaction is



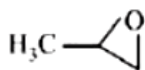
[April 12, 2019(I)]

Options:

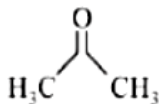




C.

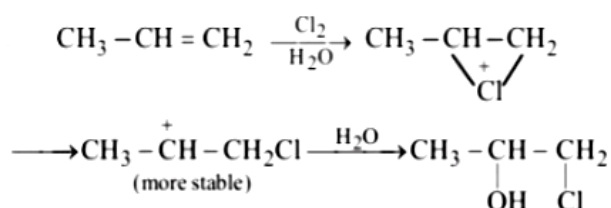


D.



Answer: B

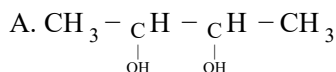
Solution:



Question152

But 2 -ene on reaction with alkaline KMnO_4 at elevated temperature followed by acidification will give :
[April 12, 2019 (I)]

Options:



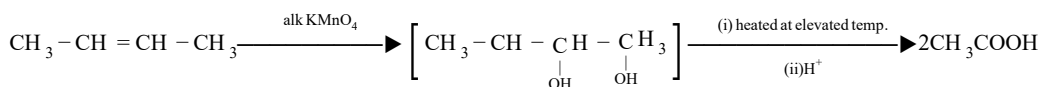
B. one molecule of CH_3CHO and one molecule of CH_3COOH

C. 2 molecules of CH_3COOH

D. 2 molecules of CH_3CHO

Answer: C

Solution:

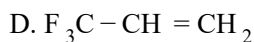
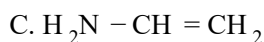
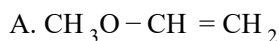


Question153

Which one of the following alkenes when treated with HCl yields majorly an anti Markovnikov product?

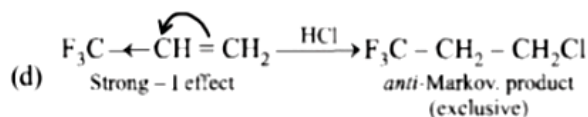
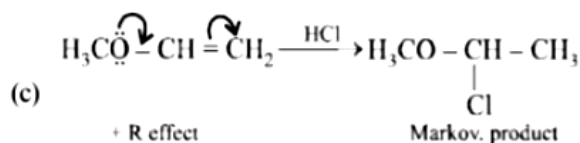
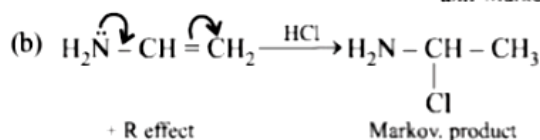
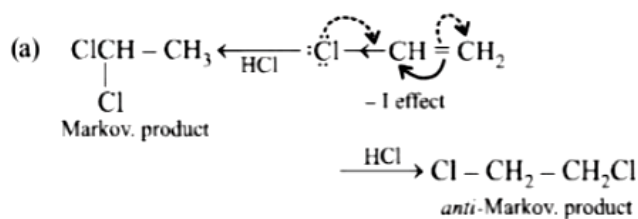
[April 8, 2019 (II)]

Options:



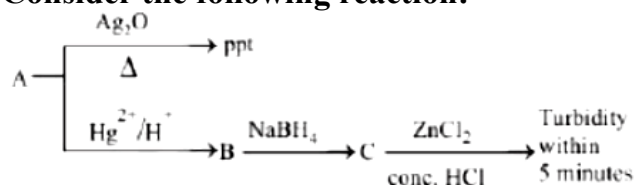
Answer: D

Solution:



Question154

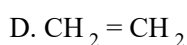
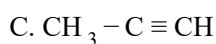
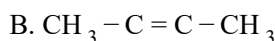
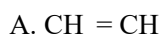
Consider the following reaction:



'A' is:

[April 12, 2019 (II)]

Options:

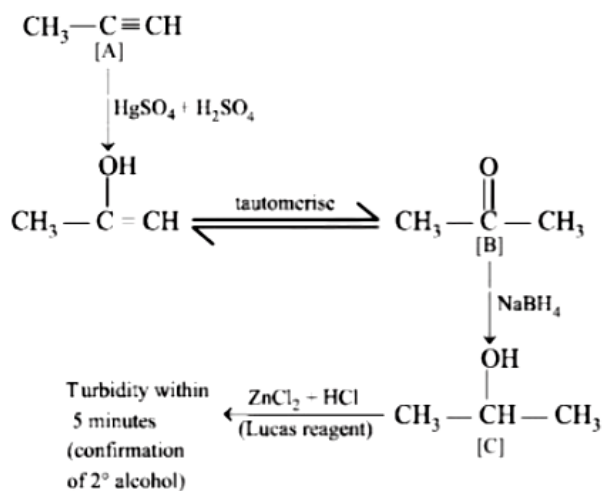


Answer: C

Solution:

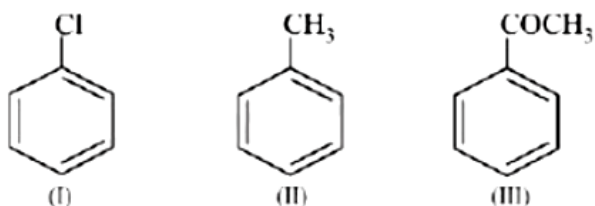
(i) Reaction of 'A' with Ag_2O to give ppt indicates $-\text{C}\equiv\text{CH}$ type of linkage.

(ii) Going backward in the second set, compound 'C' when heated with ZnCl_2 , and conc. HCl gives turbidity within 5 minutes, it indicates that 'C' is a 2° alcohol, (C) hence 'B' is a ketone and 'A' is prop-1-yne.



Question 155

The increasing order of the reactivity of the following compounds towards electrophilic aromatic substitution reactions is :



[April 10, 2019 (I)]

Options:

A. $\text{II} < \text{I} < \text{III}$

B. $\text{III} < \text{II} < \text{I}$

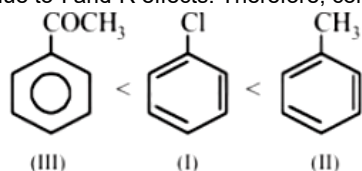
C. $\text{III} < \text{I} < \text{II}$

D. $\text{I} < \text{III} < \text{II}$

Answer: C

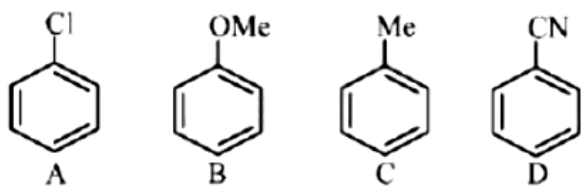
Solution:

CH_3 group when bonded to benzene increases the electron density of benzene ring due to +I and hyper conjugation effects. $-\text{Cl}$ group decreases the electron density of benzene ring due to $-I$ effect, and $-\text{COCH}_3$ group strongly decreases the electron density of benzene ring due to $-I$ and $-R$ effects. Therefore, correct increasing order of the given compounds towards electrophilic aromatic substitution is



Question156

The increasing order of reactivity of the following compounds towards aromatic electrophilic substitution reaction is:



[April 9,2019 (I)]

Options:

A. $D < A < C < B$

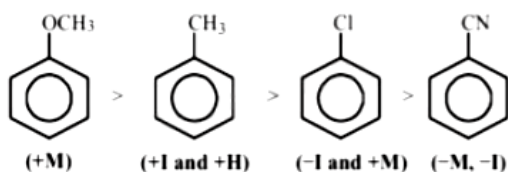
B. $B < C < A < D$

C. $A < B < C < D$

D. $D < B \leq A < C$

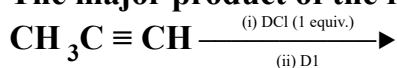
Answer: A

Solution:



Question157

The major product of the following reaction is:



[April 9, 2019 (I)]

Options:

A. $\text{CH}_3\text{CD(I)CH D(Cl)}$

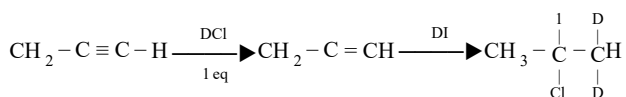
B. $\text{CH}_3\text{CD(Cl)CH D(I)}$

C. $\text{CH}_3\text{CD}_2\text{CH (Cl)(I)}$

D. $\text{CH}_3\text{C(T)(Cl)CH D}_2$

Answer: D

Solution:

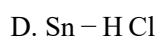
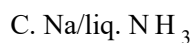
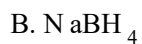
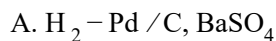


Both additions follow Markovnikov's rule.

Question158

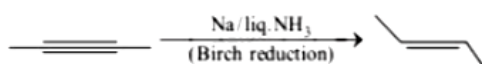
The trans-alkenes are formed by the reduction of alkynes with:
[2018]

Options:



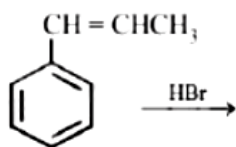
Answer: C

Solution:



Question159

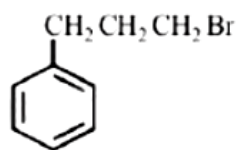
The major product of the following reaction is:



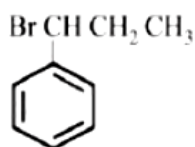
[Online April 16, 2018]

Options:

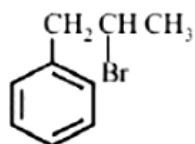
A.



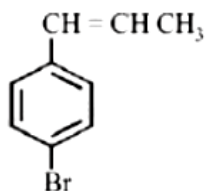
B.



C.

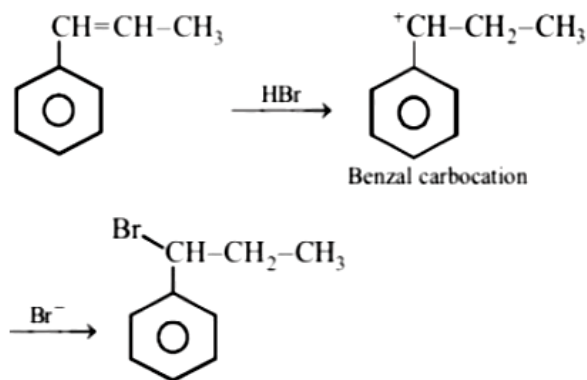


D.



Answer: B

Solution:



Question160

When 2 -butyne is treated with H_2 / Lindlar's catalyst, compound X is produced as the major product and when treated with $\text{Na} / \text{liq. N H}_3$ it produces Y as the major product. Which of the following statements is correct?
[Online April 15, 2018 (II)]

Options:

- A. Y will have higher dipole moment and higher boiling point than X
- B. Y will have higher dipole moment and lower boiling point than X
- C. X will have lower dipole moment and lower boiling point than Y
- D. X will have higher dipole moment and higher boiling point than Y

Answer: D

Solution:

When 2 -butyne is treated with H_2 / Lindlar's catalyst, compound X (cis-2-butene) is produced as the major product; and when treated with $\text{Na} / \text{liq N H}_3$ it produces Y (trans - 2- butene) as the major product. Cis-isomer (X) will have higher dipole moment and higher boiling point than trans(Y).

Question161

3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is :
[2017]

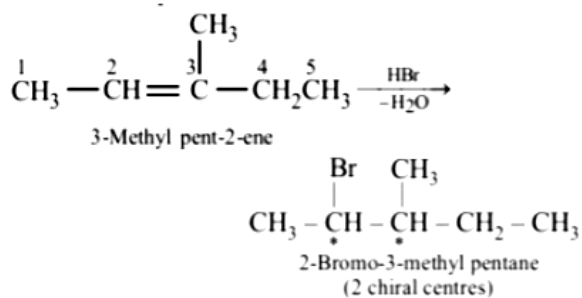
Options:

- A. Six
- B. Zero
- C. Two
- D. Four

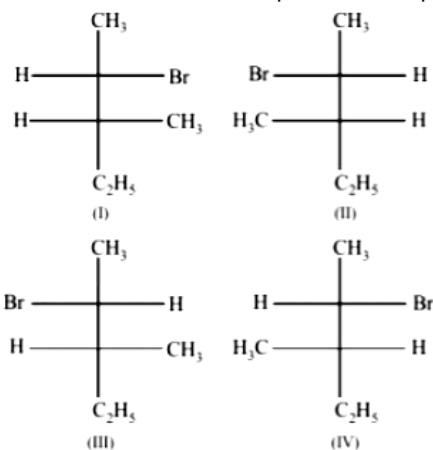
Answer: D

Solution:

Addition of HBr on 3-methylpent-2-ene in presence of peroxide, takes place in anti-Markownikov's rule.



Since two chiral centres are present in the product, four stereoisomers (n^2) are possible.



Question162

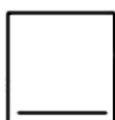
Which of the following compounds is most reactive to an aqueous solution of sodium carbonate?
[Online April 9, 2017]

Options:

A.



B.



C.

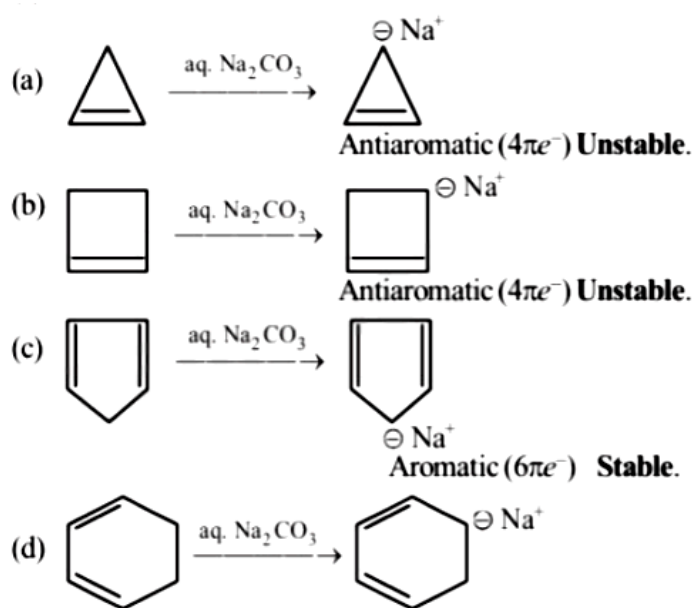


D.



Answer: D

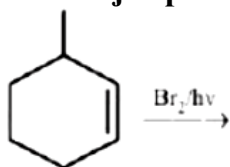
Solution:



No delocalization of electron in whole ring Unstable.

Question163

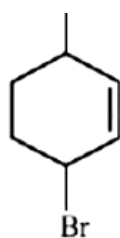
The major product of the following reaction is :



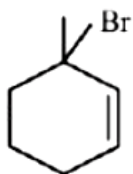
[Online April 9, 2017]

Options:

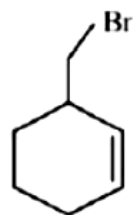
A.



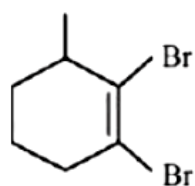
B.



C.

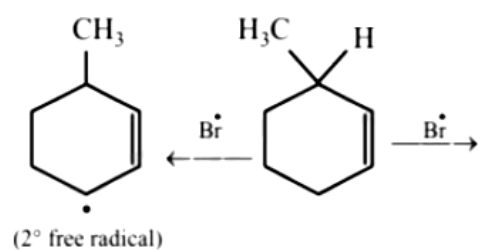
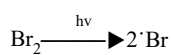


D.



Answer: B

Solution:

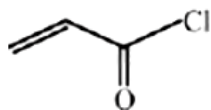


Question164

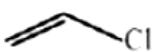
Which of the following compounds will not undergo Friedel Craft's reaction with benzene?
[Online April 8, 2017]

Options:

A.



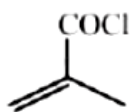
B.



C.

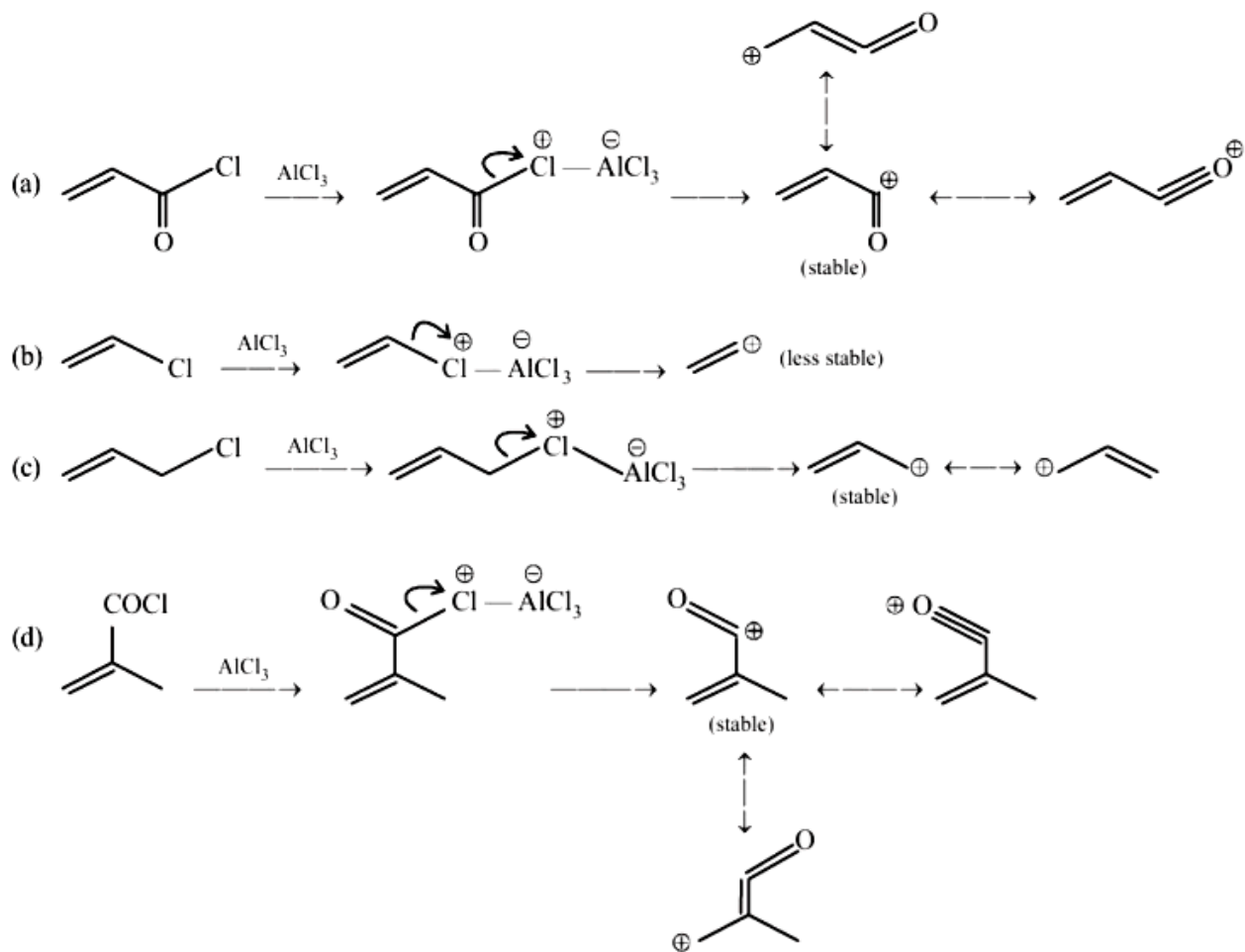


D.



Answer: B

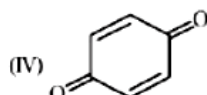
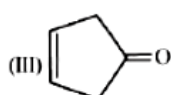
Solution:



Hence formation of carbocation is not possible in the case of vinyl halide.

Question165

Which of the following compounds will show highest dipole moment ?



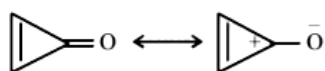
[Online April 9,2017]

Options:

- A. (I)
- B. (II)
- C. (III)
- D. (IV)

Answer: A

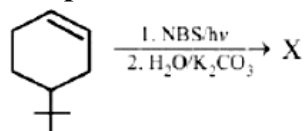
Solution:



A dipolar resonance structure has aromatic character in the ring and would be expected to make a major contribution to the overall structure.

Question166

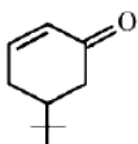
The product of the reaction given below is:



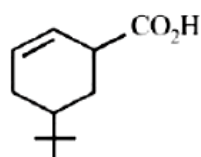
[2016]

Options:

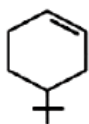
A.



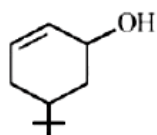
B.



C.



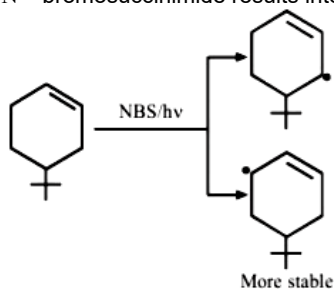
D.



Answer: D

Solution:

N - bromosuccinimide results into bromination at allylic and benzylic positions.



Question167

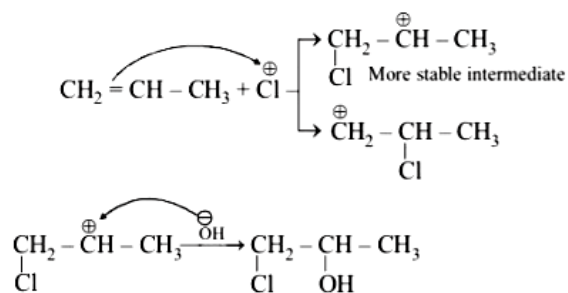
The reaction of propene with H OCl ($\text{Cl}_2 + \text{H}_2\text{O}$) proceeds through the intermediate:
[2016]

Options:

- A. $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_2^+$
- B. $\text{CH}_3 - \text{CH Cl} - \text{CH}_2^+$
- C. $\text{CH}_3 - \text{CH}^+ - \text{CH}_2 - \text{OH}$
- D. $\text{CH}_3 - \text{CH}^+ - \text{CH}_2 - \text{Cl}$

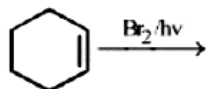
Answer: D

Solution:



Question168

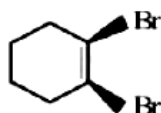
Bromination of cyclohexene under conditions given below yields:



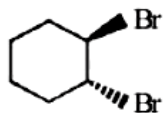
[Online April 10,2016]

Options:

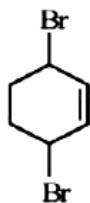
A.



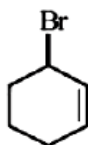
B.



C.



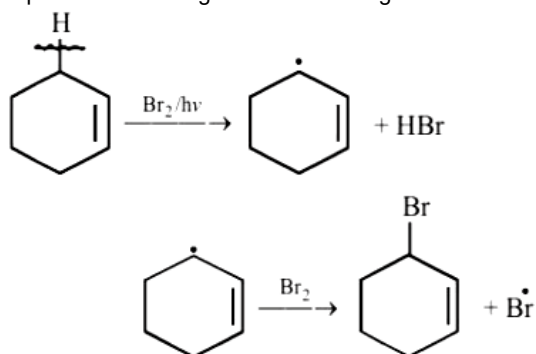
D.



Answer: D

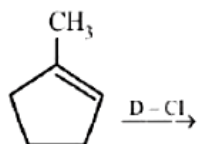
Solution:

In presence of sunlight alkenes undergo free radical substitution.



Question169

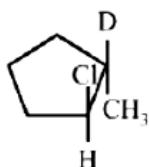
What is the major product expected from the following reaction ?



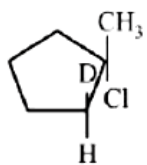
Where D is an isotope of hydrogen
[Online April 11,2015]

Options:

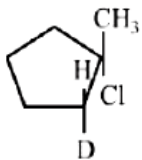
A.



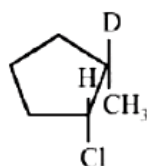
B.



C.

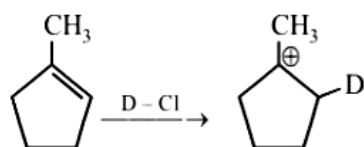


D.

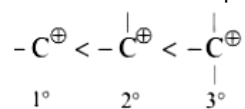


Answer: C

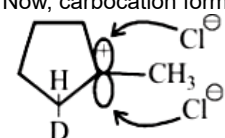
Solution:



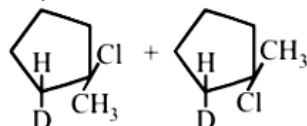
Formation of above species is more favourable. The stability order of carbocations is



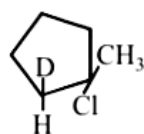
Now, carbocation formed is sp^2 hybridised that is triangular planar as shown below



$\text{Cl}^{\ominus}$ ion can attack either from above or below the plane of the molecule, so there is probability of resultant product as shown below i.e., both product will form.

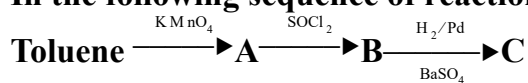


|||



Question170

In the following sequence of reactions:



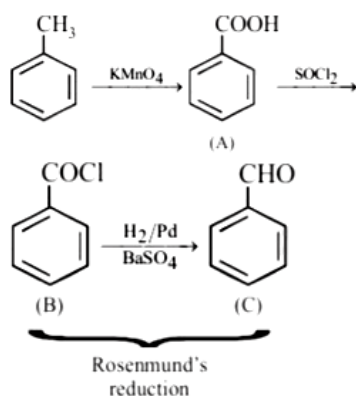
**the product C is :
[2015]**

Options:

- A. $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- B. $\text{C}_6\text{H}_5\text{CHO}$
- C. $\text{C}_6\text{H}_5\text{COOH}$
- D. $\text{C}_6\text{H}_5\text{CH}_3$

Answer: B

Solution:



Question171

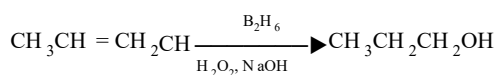
**In the hydroboration - oxidation reaction of propene with diborane, H_2O_2 and NaOH , the organic compound formed is:
[Online April 9, 2014]**

Options:

- A. $\text{CH}_3\text{CH}_2\text{OH}$
- B. $\text{CH}_3\text{CHOHCH}_3$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- D. $(\text{CH}_3)_3\text{COH}$

Answer: C

Solution:



Question172

The gas liberated by the electrolysis of dipotassium succinate solution is:
[Online April 11, 2014]

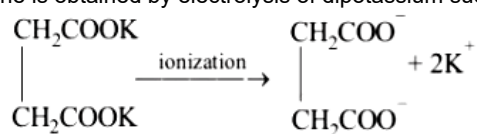
Options:

- A. Ethane
- B. Ethyne
- C. Ethene
- D. Propene

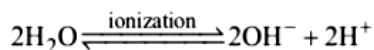
Answer: C

Solution:

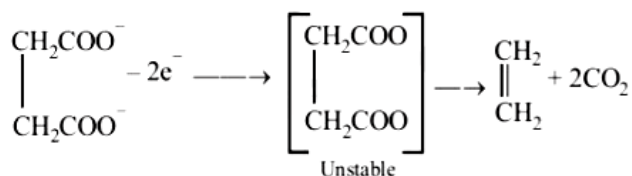
Ethene is obtained by electrolysis of dipotassium succinate as follows



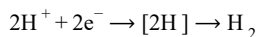
Pot. Succinate



At anode :



At cathode :



Question173

In the presence of peroxide, H Cl and H I do not give anti-Markownikoff's addition of alkenes because:
[Online April 12, 2014]

Options:

- A. One of the steps is endothermic in H Cl and H I
- B. Both HCl and HI are strong acids
- C. H Cl is oxidizing and the H I is reducing
- D. All the steps are exothermic in H Cl and H I

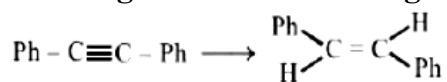
Answer: A

Solution:

Anti-Markownikov addition is possible only in case of HBr and not in HCl and HI. In HBr both the chain initiation and propagation steps are exothermic, while in HCl, first step is exothermic, and second step is endothermic and in HI, no step is exothermic. Hence HCl and HI do not undergo anti-Markownikov's addition.

Question174

The reagent needed for the given conversion is



[Online April 11, 2014]

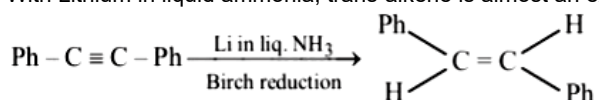
Options:

- A. Cat. hydrogenation
- B. H_2 /Lindlar Cat.
- C. Li/NH_3
- D. LiAlH_4

Answer: C

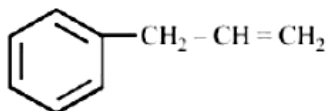
Solution:

With Lithium in liquid ammonia, trans-alkene is almost an exclusive product.



Question175

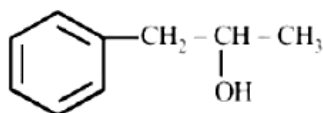
On mercuration-demercuration produces the major product :



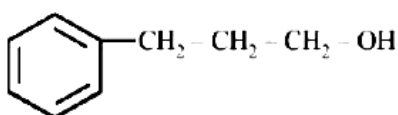
[Online April 12, 2014]

Options:

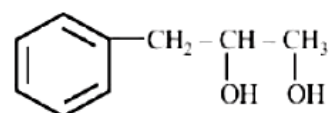
A.



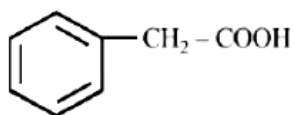
B.



C.

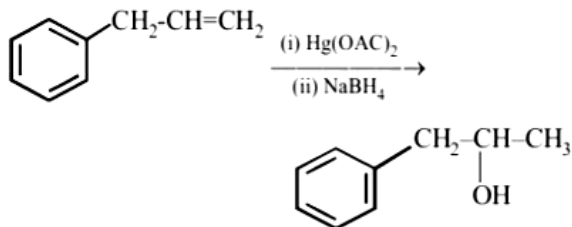


D.



Answer: A

Solution:



Question176

Which one of the following class of compounds is obtained by polymerization of acetylene?
[Online April 9, 2014]

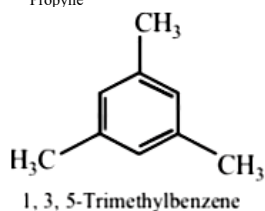
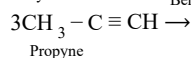
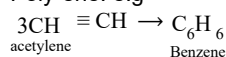
Options:

- A. Poly-yne
- B. Poly-ene
- C. Poly-ester
- D. Poly-amine

Answer: B

Solution:

Poly-ene. e.g



Question177

The addition of HI in the presence of peroxide catalyst does not follow anti-Markovnikov's rule because :
[Online April 9, 2013]

Options:

- A. HI is a strong reducing agent.

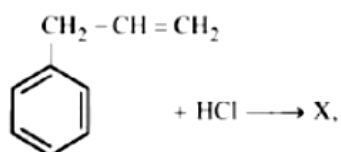
- B. H-I bond is too strong to be broken homolytically.
- C. I atom combines with H atom to give back H I .
- D. Iodine atom is not reactive enough to add across a double bond.

Answer: D

Solution:

HI does not exhibit peroxide effect. HI bond although dissociates easily into iodine radicals, they being bigger in size are not much reactive but recombine together to form iodine molecule.

Question178

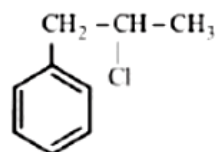


X is:

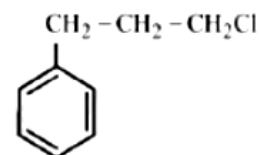
[Online April 9, 2013]

Options:

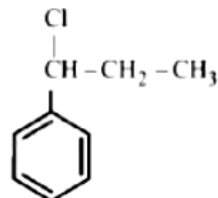
A.



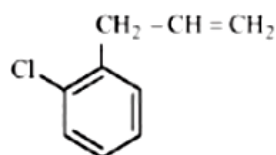
B.



C.

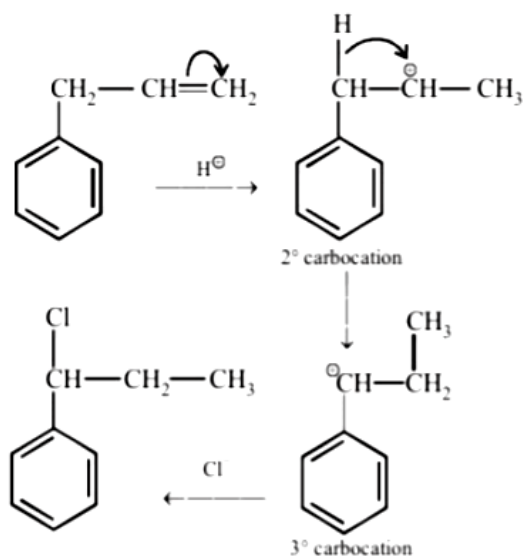


D.



Answer: C

Solution:



Question179

Which of the following would not give 2 -phenyl butane as the major product in a Fricdel-Crafts alkylation reaction?
[Online April 22, 2013]

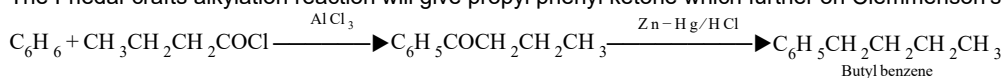
Options:

- A. 1 -butene + H F
- B. 2 -butanol + H_2SO_4
- C. Butanoyl chloride + AlCl_3 then Z n, H Cl
- D. Butyl choloride + AlCl_3

Answer: C

Solution:

The Friedal-crafts alkylation reaction will give propyl phenyl ketone which further on Clemmenson's reduction will give butyl benzene



Question180

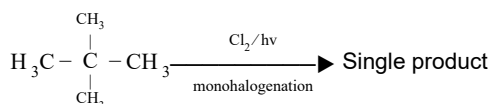
Which branched chain isomer of the hydrocarbon with molecular mass 72u gives only one isomer of mono substituted alkyl halide?
[2012]

Options:

- A. Tertiary butyl chloride
- B. Neopentane
- C. Isohexane
- D. Neohexane

Answer: B

Solution:

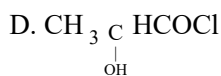
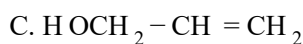
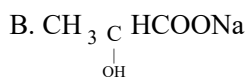
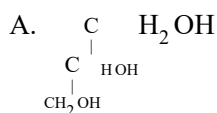


In neopentane, all hydrogen atoms are equivalent.

Question181

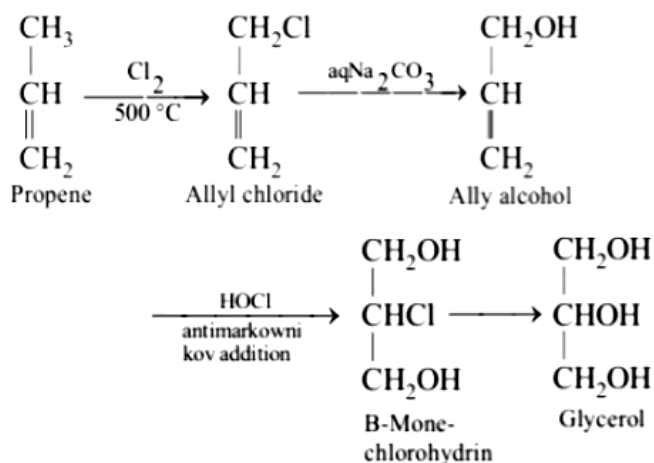
Consider the following sequence of reactions $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow[700\text{K}]{\text{Cl}_2} \text{A} \xrightarrow[420\text{K}, 12\text{atm}]{\text{Na}_2\text{CO}_3} \text{B} \xrightarrow[\text{(ii) NaOH}]{\text{(i) HOCl}} \text{C}$
Compound 'C' is
[Online May 26, 2012]

Options:



Answer: A

Solution:



Question182

2-Hexyne gives trans-2-hexene on treatment with:
[2012]

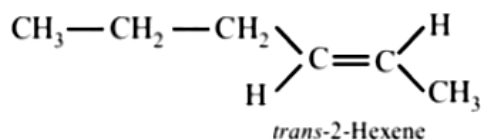
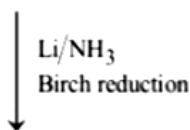
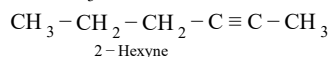
Options:

- A. Pt/H₂
- B. Li/NH₃
- C. Pd /BaSO₄
- D. LiAlH₄

Answer: B

Solution:

Anti addition of hydrogen atoms to the triple bond occurs when alkynes are reduced with sodium (or lithium) metal in liq. ammonia, ethylamine, or alcohol at low temperatures. This reaction called, a dissolving metal reduction, produces an (E)- or trans-alkene. Sodium in liq. NH₃ is used as a source of electrons in the reduction of an alkyne to a trans alkene.



Question183

The hydration of propyne results in formation of
[Online May 26,2012]

Options:

- A. Acetone
- B. Propanol- 1
- C. Propene
- D. Propanal

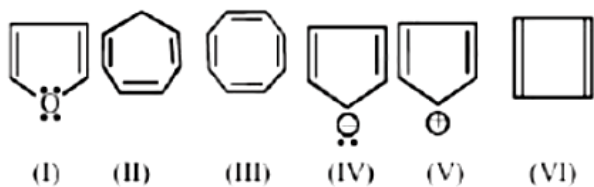
Answer: A

Solution:



Question184

Which of the following compounds are antiaromatic



[Online May 26, 2012]

Options:

- A. (I) and (V)
- B. (II) and (V)
- C. (I) and (IV)
- D. (III) and (VI)

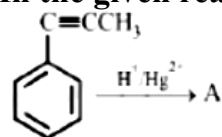
Answer: D

Solution:

For a compound to be aromatic it must have $(4n + 2)\pi e^-$ where n is an integer. (III) have 8 and (VI) have $4e^-$. Hence are antiaromatic

Question185

In the given reaction,

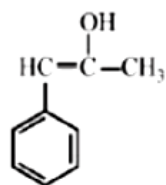


the product ' A ' is

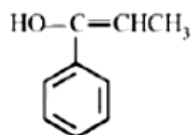
[Online May 12, 2012]

Options:

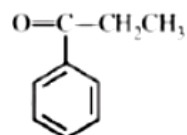
A.



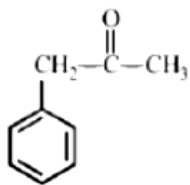
B.



C.

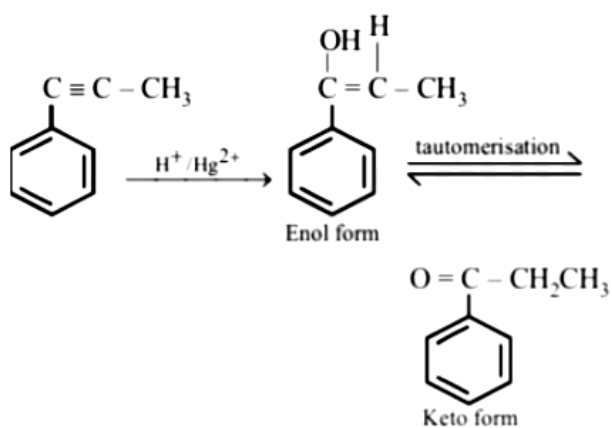


D.



Answer: C

Solution:

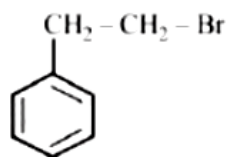


Question186

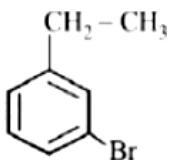
The product of the reaction between ethyl benzene and N-bromosuccinamide is
[Online May 19, 2012]

Options:

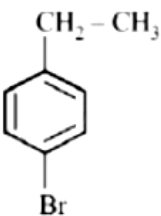
A.



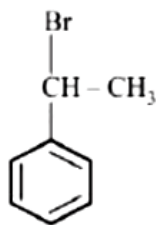
B.



C.

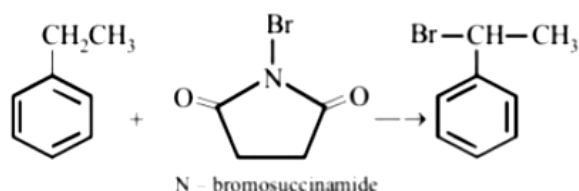


D.



Answer: D

Solution:



Question187

Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds:
[2011RS]

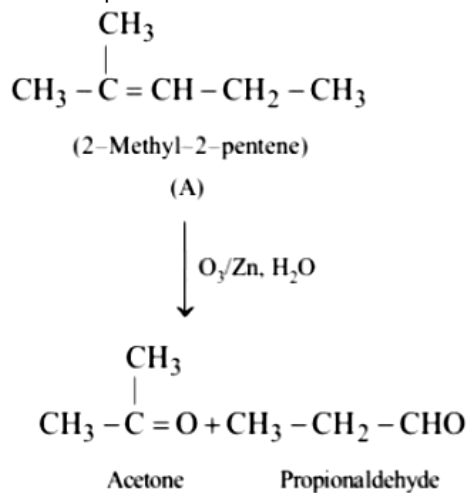
Options:

- A. 1- Pentene
- B. 2 - Pentene
- C. 2- Methyl -2- pentene
- D. 2- Methyl -1 - pentene

Answer: C

Solution:

From the products formed it is clear that the compound has 5 carbon atoms with a double bond and methyl group on 2nd carbon atom.



Question188

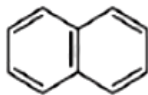
The non aromatic compound among the following is:
[2011RS]

Options:

A.



B.



C.

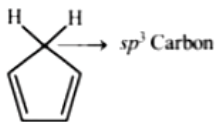


D.



Answer: D

Solution:



Cyclopentadiene does not obey Huckel's Rule, as it has sp^3 carbon in the ring.

Question189

One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44u. The alkene is
[2010]

Options:

A. propene

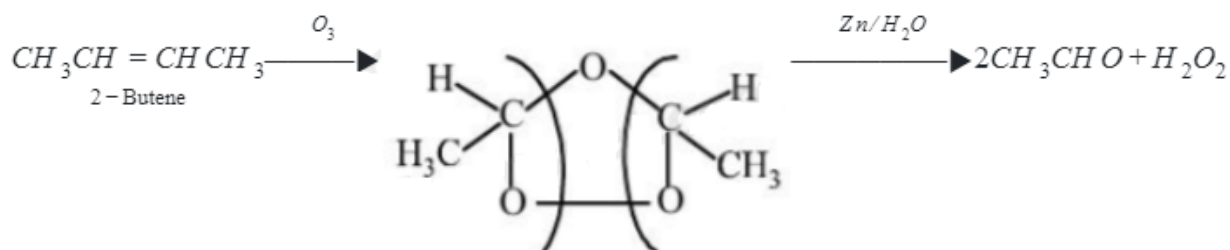
B. 1-butene

C. 2 -butene

D. ethene

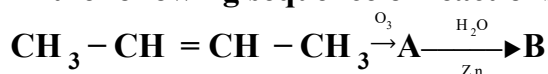
Answer: C

Solution:



Question190

In the following sequence of reactions, the alkene affords the compound ' B '



The compound B is
[2008]

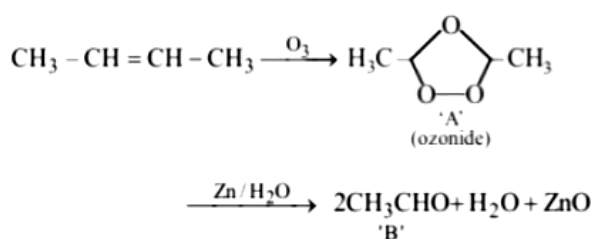
Options:

- A. $\text{CH}_3\text{CH}_2\text{CHO}$
- B. CH_3COCH_3
- C. $\text{CH}_3\text{CH}_2\text{COCH}_3$
- D. CH_3CHO

Answer: D

Solution:

Completing the sequence of given reactions.



Question191

The hydrocarbon which can react with sodium in liquid ammonia is
[2008]

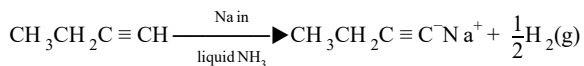
Options:

- A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_2\text{CH}_3$
- B. $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$
- C. $\text{CH}_3\text{CH} = \text{CHCH}_3$
- D. $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_3$

Answer: B

Solution:

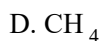
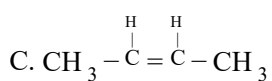
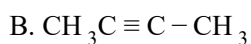
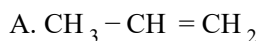
Alkynes having terminal $\text{C} \equiv \text{H}$ react with Na in liquid ammonia to yield H_2 gas



Question 192

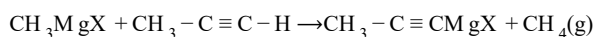
The treatment of CH_3MgX with $\text{CH}_3\text{C} \equiv \text{C}-\text{H}$ produces
[2008]

Options:



Answer: D

Solution:



Question 193

Toluene is nitrated and the resulting product is reduced with tin and hydrochloric acid. The product so obtained is diazotised and then heated with cuprous bromide. The reaction mixture so formed contains
[2008]

Options:

A. mixture of o- and p-bromotoluenes

B. mixture of o- and p-dibromobenzenes

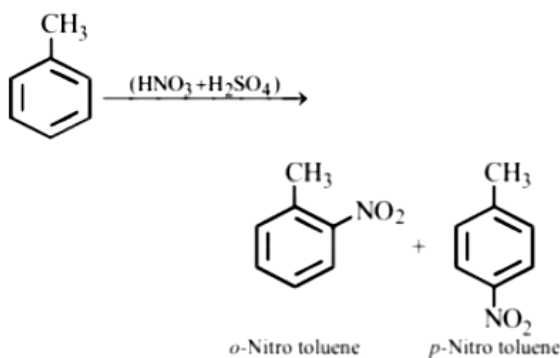
C. mixture of o- and p-bromoanilines

D. mixture of o- and m-bromotoluenes

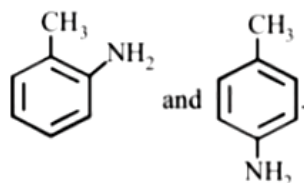
Answer: A

Solution:

Note: Toluene ($\text{C}_6\text{H}_5\text{CH}_3$) contains $-\text{CH}_3$ group which is o-, p- directing group so on nitration of toluene the $-\text{NO}_2$ group will occupy o-, p-positions.



On reduction with Sn/HCl they will form corresponding anilines in which $-\text{NO}_2$ group changes to $-\text{NH}_2$. The mixture now contains

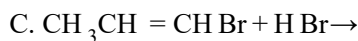
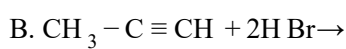
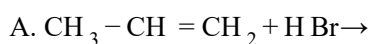


These anilines when diazotized and then treated with CuBr form *o*-, *p*-bromotoluenes (Sandmeyer reductions).

Question194

Which of the following reactions will yield 2, 2-dibromopropane?
[2007]

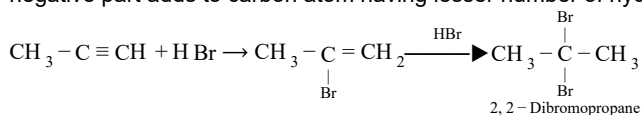
Options:



Answer: B

Solution:

The reaction follows Markownikoff rule which states that when unsymmetrical reagent adds across unsymmetrical double or triple bond, the negative part adds to carbon atom having lesser number of hydrogen atoms.



Question195

The compound formed as a result of oxidation of ethyl benzene by KMnO_4 is
[2007]

Options:

A. benzyl alcohol

B. benzophenone

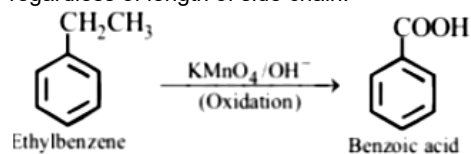
C. acetophenone

D. benzoic acid.

Answer: D

Solution:

When alkyl benzenes are oxidised with alkaline KMnO_4 (strong oxidising agent), the entire alkyl group is oxidised to -COOH group regardless of length of side chain.



Question196

The reaction of toluene with Cl_2 in presence of FeCl_3 gives predominantly [2007]

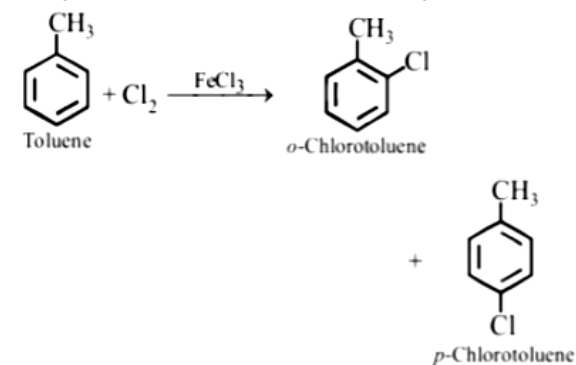
Options:

- A. m -chlorobenzene
- B. benzoyl chloride
- C. benzyl chloride
- D. o - and p -chlorotoluenes.

Answer: D

Solution:

FeCl_3 is Lewis acid. In presence of FeCl_3 and Cl_2 toluene undergoes electrophilic substitution in o - and p positions.



Question197

The major product obtained in the photo catalysed bromination of 2 -methylbutane is: [2005, Online May 19, 2012; Online April 12,2014]

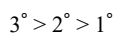
Options:

- A. 1 -bromo-2-methylbutane
- B. 1 -bromo- 3 -methylbutane
- C. 2 -bromo- 3 -methylbutane
- D. 2-bromo-2-methylbutane

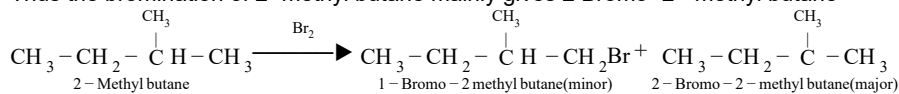
Answer: D

Solution:

The order of substitution in different alkanes is



Thus the bromination of 2-methyl butane mainly gives 2-Bromo-2-methyl butane



Question198

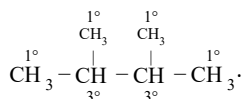
Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is [2005]

Options:

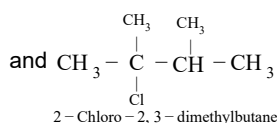
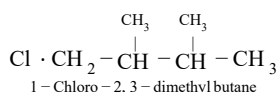
- A. 2-methylpentane
- B. 2,2-dimethylbutane
- C. 2,3-dimethylbutane
- D. n-hexane

Answer: C

Solution:



Since it contains only two types of H-atoms hence it will give only two monochlorinated compounds viz.



Question199

Which one of the following has minimum boiling point? [2004]

Options:

- A. 1-Butene
- B. 1-Butyne
- C. n-Butane
- D. Isobutane

Answer: D

Solution:

Note: Among isomeric alkanes, the straight chain isomer has higher boiling point than the branched chain isomer. The greater the branching of the chain, the lower is the boiling point. Further due to the presence of π electrons, these molecular slightly polar and hence have higher boiling points than the corresponding alkanes.

Thus B.pt. follows the order

alkynes > alkenes > alkanes (straight chain) > branched chain alkanes.

Question200

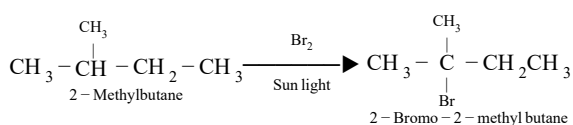
On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be [2003]

Options:

- A. pentane
- B. isopentane
- C. neopentane
- D. propane

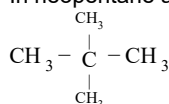
Answer: C

Solution:



Ease of replacement of H-atoms $3^\circ > 2^\circ > 1^\circ$.

In neopentane all H atoms are equivalent (1°).



Question201

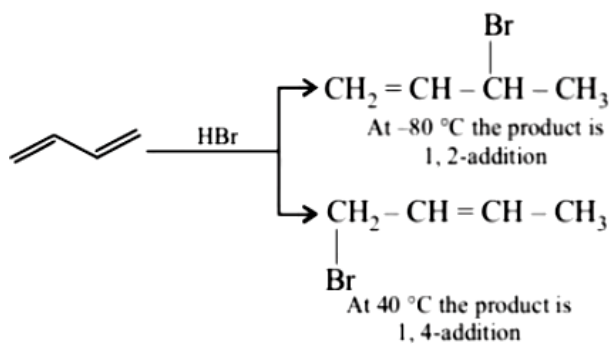
Reaction of one molecule of H Br with one molecule of 1,3-butadiene at 40°C gives predominantly [2003]

Options:

- A. 1-bromo-2-butene under kinetically controlled conditions
- B. 3-bromobutene under thermodynamically controlled conditions
- C. 1-bromo-2-butene under thermodynamically controlled conditions
- D. 3-bromobutene under kinetically controlled conditions

Answer: C

Solution:



Question202

Butene- 1 may be converted to butane by reaction with [2003]

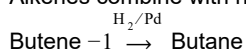
Options:

- A. $\text{Sn} - \text{HCl}$
- B. $\text{Zn} - \text{Hg}$
- C. Pd / H_2
- D. $\text{Zn} - \text{HCl}$

Answer: C

Solution:

Alkenes combine with hydrogen under pressure and in presence of a catalyst (Ni, Pt or Pd) and form alkanes.



Question203

Which of these will not react with acetylene? [2002]

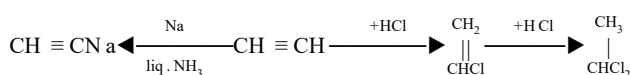
Options:

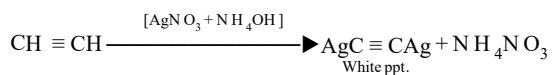
- A. dil. NaOH
- B. ammonical AgNO_3
- C. Na in liq. NH_3
- D. HCl .

Answer: A

Solution:

Acetylene reacts with the other three reagents as:





Question 204

What is the product when acetylene reacts with hypochlorous acid?
[2002]

Options:

- A. CH_3COCl
- B. ClCH_2CHO
- C. Cl_2CHCHO
- D. ClCH_2COOH

Answer: C

Solution:

